

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\105-C.pvdb

Date of Analysis : Apr 21,2026 8:25am

PV Elite 28, January 2026



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FileName : 105-C

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Vessel Design Summary:

Step: 31 8:25am Apr 21,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2480.000 mm ID
Vessel Design Length, Tangent to Tangent 10251.20 mm
Specified Datum Line Distance 50.00 mm
Shell Side Design Temperature 160 °C
Channel Side Design Temperature 245 °C
Shell Side Design Pressure 520.000 KPa.
Channel Side Design Pressure 4950.000 KPa.
Shop Shell Side Test Pressure 676.000 KPa.
Field Shell Side Test Pressure 676.000 KPa.
Shop Channel Side Test Pressure 6435.000 KPa.
Field Channel Side Test Pressure 6435.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-516 70	...	...	K02700	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-266 2	...	...	K03506	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch.shell head	4974.172	0.00	5065.000	0.0000	No	No
Ch.shell	4974.172	0.00	5620.000	0.0000	N/A	No
Shell	544.180	101.30	2745.000	0.8000	N/A	No
shell head	544.172	101.30	977.000	0.0000	No	No

Liquid Level: 2466.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Vessel Design Summary:

Step: 31 8:25am Apr 21,2026

Ellipse	0.0	50.0	54.0	45.9	44.9	...	1.00	1.00
Cylinder	2500.0	2500.0	52.0	52.0	45.7	...	1.00	1.00
Cylinder	10125.0	7200.0	28.0	26.0	5.7	...	1.00	1.00
Ellipse	10201.2	76.2	16.0	9.0	4.9	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	45.9	No Calc	...	2148
20	Tsht	52	No Calc	...	3086
Tsht	30	52	No Calc	...	No Calc
30	50	26	13.5232	101.3	552
50	60	9	6.42337	101.3	199

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2756.64	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	50	7482.9	165502	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N5- Tube inlet	WRC-297	0.891	Shell	Passed
N6- Tube outle	WRC-297	0.891	Shell	Passed
N1- Shell inle	WRC-297	0.670	Nozzle	Passed
N2- Shell inle	WRC-297	0.670	Nozzle	Passed
N3- Shell outl	WRC-297	0.901	Shell	Passed
N4- Shell outl	WRC-297	0.904	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	375.000 mm
Saddle Bearing Angle	140.000 deg.
Centerline Dimension	1910.000 mm
Wear Pad Width	710.000 mm
Wear Pad Thickness	15.000 mm
Wear Pad Bearing Angle	152.000 deg.
Distance from Saddle to Tangent	1440.000 mm
Baseplate Length	2350.000 mm
Baseplate Thickness	40.000 mm
Baseplate Width	400.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	16.000 mm
Web Thickness	16.000 mm
Height of Center Web	680.000 mm
Number of Bolts in Baseplate	8

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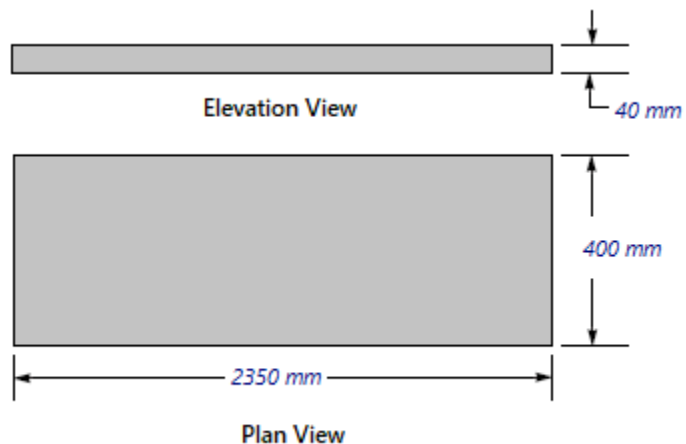
FileName : 105-C

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Vessel Design Summary:

Step: 31 8:25am Apr 21, 2026

Baseplate Sketch



Maximum Tensile Bolt Load 16. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	143825.66	Kgf
Maximum Transverse Saddle Shear Load	5260.62	Kgf
Maximum Longitudinal Saddle Shear Load	8363.24	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	129442.47	Kgf
Maximum Transverse Saddle Shear Load	359.82	Kgf
Maximum Longitudinal Saddle Shear Load	362.69	Kgf

Weights:

Fabricated	- Bare w/o removable internals	92899.6	kg.
Shop Test	- Fabricated + water (full)	140712.9	kg.
Shipping	- Fab. + rem. intls.+ shipping app.	92899.6	kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	96621.7	kg.
Empty	- Fab. + intls. + details + wghts.	96621.7	kg.
Operating	- Empty + operating liquid (no ca)	144676.1	kg.
Field Test	- Empty weight + test medium (full)	144097.2	kg.



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Nozzle Summary:

Step: 29 8:25am Apr 21,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Manway	5065	...	...	...		OK	Passed
N5- Tube inlet	5143	...	...	OK	17.39	OK	Passed
N6- Tube outlet	5143	...	...	OK	17.48	OK	Passed
N1- Shell inlet	2070	OK	...	OK	8.33	OK	Passed
N2- Shell inlet	2070	OK	...	OK	8.33	OK	Passed
N3- Shell outle	1518	OK	...	OK	9.13	OK	Passed
N4- Shell outle	1528	OK	...	OK	8.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1518 Nozzle : N3- Shell outle [Shellside]
 Minimum MAWP Nozzles : 5065 Nozzle : Manway [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Manway	0.000	0.000	1119.200
20	N5- Tube inlet	1300.000	90.000	1828.800
20	N6- Tube outlet	1300.000	270.000	1828.800
30	N1- Shell inlet	5475.000	270.000	611.607
30	N2- Shell inlet	7475.000	270.000	611.607
30	N3- Shell outlet	4475.000	90.000	1832.000
30	N4- Shell outlet	7475.000	90.000	1828.800

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 28 8:25am Apr 21,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1- Shell inlet	12.000 in	40	WNF	12.750	10.312	523.85	16.00	286.62	150
N2- Shell inlet	12.000 in	40	WNF	12.750	10.312	523.85	16.00	286.62	150
Manway	24.000 in	Actual	WNF	24.000	25.000	...	...	416.47	600
N5- Tube inlet	36.000 in	Actual	None	37.535	19.500	...	...	637.32	...
N6- Tube outlet	36.000 in	Actual	None	37.535	19.500	...	...	637.32	...
N3- Shell outlet	36.000 in	Actual	WNF	37.535	19.500	...	...	475.72	150
N4- Shell outlet	36.000 in	Actual	WNF	37.535	19.500	...	...	475.72	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1- Shell i	SA-106 B	15.000	10.000	12.000	16.000	...
N2- Shell i	SA-106 B	15.000	10.000	12.000	16.000	...
Manway	SA-105	45.900	10.000	...	...	...
N5- Tube in	SA-105	48.000	15.000	...	...	...
N6- Tube ou	SA-105	48.000	15.000	...	...	...
N3- Shell o	SA-266 2	18.000	10.000	...	...	...
N4- Shell o	SA-516 70	20.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1- Shell inlet	5425.000	270.0	250.00	0.00	Shell
N2- Shell inlet	7425.000	270.0	250.00	0.00	Shell
Manway	...	0.0	350.00	0.00	Ch.shell head
N5- Tube inlet	1250.000	90.0	415.00	0.00	Ch.shell
N6- Tube outlet	1250.000	270.0	415.00	0.00	Ch.shell
N3- Shell outlet	4425.000	90.0	350.00	0.00	Shell
N4- Shell outlet	7425.000	90.0	350.00	0.00	Shell

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Bill of Material:

Step: 27 8:25am Apr 21,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 54.0mm THK X 2480.0mm ID X 50.0mm	SA-516 70
1	CYLINDER: 52.0mm THK X 2480.0mm ID X 2500.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 2480.0mm ID X 76.2mm	SA-516 70
1	INSULATION: 666mm X 100mm THK	Hot Insulation
1	INSULATION: 2500mm X 100mm THK	Hot Insulation
1	CLADDING: 2500.000mm X 3.000mm THK	...
6	BAFFLES: 2432.0mm DIA X 10.0mm THK	SA-516 70
12	TIE RODS: 25mm DIA X 6500mm	SA-193 B7
1	TUBESHEET: 2584.0mm X 425.0mm THK	SA-266 2
5080	TUBES: 6500.0mm X 25.4mm DIA X 1.6mm THK	SA-213 TP304L
2	SADDLE: 375.0mm X 140 DEG	SA-516 70
1	INSULATION: 7200mm X 100mm THK	Hot Insulation
1	CLADDING: 7200.000mm X 3.000mm THK	...
1	INSULATION: 692mm X 100mm THK	Hot Insulation
1	CLADDING: 692.700mm X 3.000mm THK	...
1	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	CYLINDRICAL SEGMENT 28.0mm THK X 3073.1mm X 7200.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3051.5mm X 7200.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 1666.4mm X 7200.0mm	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 30 8:25am Apr 21,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell	[8]	B	0	-104		0.350	26.000	1.00	No
shell head	[10]	B	-29	-30	-29	0.976	9.000	1.00	No
shell head	[7]	B	-14	-41	-29	0.551	16.000	1.00	No
Left Saddle	[24]	B	-16	-104	-29	0.350	15.000	1.00	No
New Saddle	[24]	B	-16	-104	-29	0.350	15.000	1.00	No
N1- Shell inle[1]		B	-14	-48	-29	0.359	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N2- Shell inle[1]		B	-14	-48	-29	0.359	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N3- Shell outl[1]		B	12	-104		0.174	19.500	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N4- Shell outl[1]		B	12	-104		0.167	19.500	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: SS[13]		B	33	-104		0.119	106.250	1.00	No
Warmest MDMT:			33	-30					
Ch.shell head[10]		B	15	14		0.995	45.900	1.00	Yes
Ch.shell head [7]		B	18	11		0.862	54.000	1.00	Yes
Ch.shell	[8]	B	18	12		0.895	52.000	1.00	Yes
Manway	[1]	A	34	29		0.898	45.900	1.00	Yes
Nozzle Flg	[5]	A	-18	-48					
N5- Tube inlet[1]		A	37	31		0.895	52.000	1.00	No
N6- Tube outle[1]		A	37	32		0.899	52.000	1.00	No
Tubesheet: CS[14]		B	16	6		0.815	106.250	1.00	No
Warmest MDMT:			37	32					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-30.0	-12.0	Pass
Channel/Tube	32.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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MDMT Summary:

Step: 30 8:25am Apr 21,2026

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 8:25am Apr 21,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 8:25am Apr 21,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	520	KPa.
Channel Side Design Pressure	4950	KPa.
Shell Side Design Temperature	160.0	°C
Channel Side Design Temperature	245.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

Step: 1 8:25am Apr 21,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	17000	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	0	mm
Distance Upwind of Crest	0	mm
Distance from Crest to the Vessel	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration	0.147	[Ss]
Max. Eff. Ground Acceleration	0.065	[S]
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	0.000	z/h
Amplification Factor	1.000	Ap
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

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Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Ch.shell head
Distance "FROM" to "TO"	50 mm
Inside Diameter	2480 mm
Element Thickness	45.9 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	54 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	245 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	217 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2466 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation



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Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-616.5 mm
Height/Length of Insulation	666.5 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	10
Detail Type	Nozzle
Detail ID	Manway
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1377.3 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element#	2/4
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Ch.shell
Distance "FROM" to "TO"	2500 mm
Inside Diameter	2480 mm
Element Thickness	52 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	52 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	245 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2466 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Lining
Detail ID	308L Weld overl
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	2500 mm
Thickness of Lining	3 mm
Density	0.00785 kg./cm ³



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Element From Node	20
Detail Type	Nozzle
Detail ID	N5- Tube inlet
Dist. from "FROM" Node / Offset dist	1250 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1033.6 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Nozzle
Detail ID	N6- Tube outlet
Dist. from "FROM" Node / Offset dist	1250 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	996.17 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Weight
Detail ID	Pass partition
Dist. from "FROM" Node / Offset dist	1250 mm
Miscellaneous Weight	750 Kg
Offset from Element Centerline	0 mm

Element#	3/4
Element From Node	30
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	7200 mm
Inside Diameter	2480 mm
Element Thickness	26 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	160 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1440 mm
Width of Saddle	375 mm
Height of Saddle at Bottom	1910 mm
Saddle Contact Angle	140.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	710 mm



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Input Echo:

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Thickness of Wear Plate	15	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	30	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	5760	mm
Width of Saddle	375	mm
Height of Saddle at Bottom	1910	mm
Saddle Contact Angle	140.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	710	mm
Thickness of Wear Plate	15	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2466	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	7200	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Lining	
Detail ID	Weld overlay	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	7200	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1- Shell inlet	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	12	in.
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	35.472	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	



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Detail ID	N2- Shell inlet	
Dist. from "FROM" Node / Offset dist	4500	mm
Nozzle Diameter	12	in.
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.281	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N3- Shell outlet	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	769.41	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N4- Shell outlet	
Dist. from "FROM" Node / Offset dist	4500	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	769.41	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	30	
Detail Type	Weight	
Detail ID	Fireproofing	
Dist. from "FROM" Node / Offset dist	1440	mm
Miscellaneous Weight	410	Kgf
Offset from Element Centerline	1500	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Fireproofing	
Dist. from "FROM" Node / Offset dist	5760	mm
Miscellaneous Weight	410	Kgf
Offset from Element Centerline	1500	mm

Element#	4/4	
Element From Node	50	
Element To Node	60	
Element Type	Elliptical	
Description	shell head	
Distance "FROM" to "TO"	76.2	mm
Inside Diameter	2480	mm
Element Thickness	9	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	520	KPa.



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Design Temperature Internal Pressure	160	°C
Design External Pressure	101.3	KPa.
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2466	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	692.7	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Lining	
Detail ID	Weld oeverlay	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	692.7	mm
Thickness of Lining	3	mm
Density	0.00785	kg./cm ³

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XY Coordinate Calculations: Step: 2 8:25am Apr 21,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Ch.shell head		50	...	50	...
Ch.shell		2550	...	2500	...
Shell		10175	...	7200	...
shell head		10251.2	...	76.2	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Calculations: Step: 3 8:25am Apr 21,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Ch.shell head		4974.2	54	...	2480	138	1
Ch.shell		4974.2	52	...	2480	138	2
Shell		544.18	28	0.8	2480	138	3
shell head		544.17	16	...	2480	138	4

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Ch.shell head		4950	5065	5089	45.9	44.8606	1
Ch.shell		4950	5620	5645	52	45.687	2
Shell		520	2745	2857	26	5.70486	3
shell head		520	977	1001	9	4.89196	4

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 977 KPa.
 Shell Side MAPnc = 1001 KPa.
 Channel Side MAWP = 5065 KPa.
 Channel Side MAPnc = 5089 KPa.

Elements Suitable for Design Internal Pressure.

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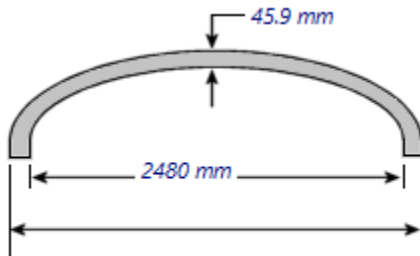
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Element: Ch.shell head

Step: 4 8:25am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Ch.shell head

Elliptical Head Design Information:			
Design Pressure P	4974.172 KPa.	Int. Design Temperature	245 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	134.9 N./mm ²
Spec. Min. Thk t,e, or sa	45.9000 mm	Spec. Nominal Thk.	54.000 mm
Surface Area	76667.8 cm ²	SG of Contents	1.000
Element Volume	2238538.5 cm ³	Weight of Contents	2237.0 Kgf
Empty Weight	3383.0 Kgf	Operating Weight	5620.0 Kgf
		Comp. All. Ope. Stress	113.3 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4974 \cdot 2480 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 4974)$$

$$= 44.8606 + 0.0000 = 44.8606 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.172 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 45.9) / (1 \cdot 2480 + 0.2 \cdot 45.9)$$

$$= 5089 - 24.2 = 5065 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 45.9) / (1 \cdot 2480 + 0.2 \cdot 45.9)$$

$$= 5089 \text{ KPa.}$$



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Element: Ch.shell head

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4974 \cdot (1 \cdot 2480 + 0.2 \cdot 45.9)) / (2 \cdot 1 \cdot 45.9)$$

$$= 134.884 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4974 \cdot 1240) / (138 \cdot 1 - 0.6 \cdot 4974) + 0$$

$$= 45.687 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 24.172 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 54) / (1240 + 0.6 \cdot 54)$$

$$= 5856 - 24.2 = 5832 \text{ KPa.}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.028 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 45.9, tr = 45.7, c = 0 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C

Min Metal Temp. at Required thickness (UCS 66.1) 14 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 54, tr = 46.5, c = 0 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 18 °C

Min Metal Temp. at Required thickness (UCS 66.1) 11 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5064.827 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	54.000 mm	Basic MDMT	18 °C
Stress/Thk. Ratio	0.862	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	11 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.500 mm

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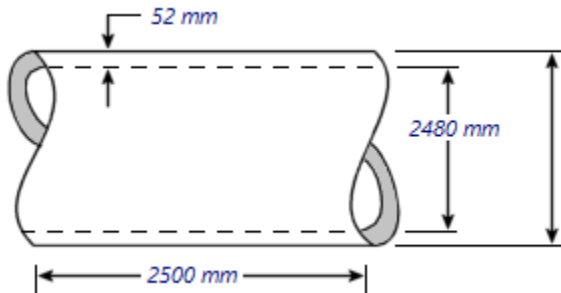
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Element: Ch.shell

Step: 5 8:25am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Ch.shell

Cylindrical Shell Design Information:			
Design Pressure P	4974.172 KPa.	Int. Design Temperature	245 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	2500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	121.6 N./mm ²
Spec. Min. Thk t,e, or sa	52.0000 mm	Spec. Nominal Thk.	52.000 mm
Surface Area	202946.9 cm ²	SG of Contents	1.000
Element Volume	12078445.0 cm ³	Weight of Contents	12062.4 Kg
Empty Weight	8014.7 Kg	Operating Weight	20077.1 Kg
		Comp. All. Ope. Stress	115.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4974 \cdot 1240) / (138 \cdot 1 - 0.6 \cdot 4974) \\
 &= 45.6870 + 0.0000 = 45.6870 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.172 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 52) / (1240 + 0.6 \cdot 52) \\
 &= 5645 - 24.2 = 5620 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 52) / (1240 + 0.6 \cdot 52) \\
 &= 5645 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4974 \cdot (1240 + 0.6 \cdot 52)) / (1 \cdot 52) \\
 &= 121.606 \text{ N./mm}^2
 \end{aligned}$$



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Element: Ch.shell

Step: 5 8:25am Apr 21,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 2.054 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 52$, $t_r = 46.5$, $c = 0$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.89$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 18 °C

Min Metal Temp. at Required thickness (UCS 66.1) 12 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5064.827 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	52.000 mm	Basic MDMT	18 °C
Stress/Thk. Ratio	0.895	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	12 °C
		Joint Efficiency E^*	100 %

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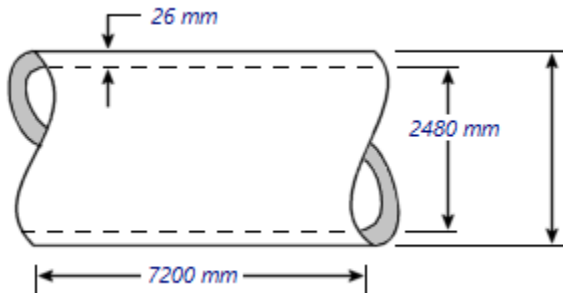
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Element: Shell

Step: 6 8:25am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	544.180 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	160 °C
Length	7200.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	27.1 N./mm ²
Spec. Min. Thk t,e, or sa	26.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	573629.8 cm ²	SG of Contents	1.000
Element Volume	14107923.0 cm ³	Weight of Contents	14099.3 Kg
Empty Weight	12311.2 Kg	Operating Weight	26410.5 Kg
		Comp. All. Ope. Stress	126.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (544 \cdot 1241) / (138 \cdot 1 - 0.6 \cdot 544) \\
 &= 4.9049 + 0.8000 = 5.7049 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.180 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 25.2) / (1241 + 0.6 \cdot 25.2) \\
 &= 2769 - 24.2 = 2745 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 26) / (1240 + 0.6 \cdot 26) \\
 &= 2857 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (544 \cdot (1241 + 0.6 \cdot 25.2)) / (1 \cdot 25.2) \\
 &= 27.122 \text{ N./mm}^2
 \end{aligned}$$



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Element: Shell

Step: 6 8:25am Apr 21,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.116 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 26$, $t_r = 8.82$, $c = 0.8$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.35$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	976.638 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	26.000 mm	Basic MDMT	0 °C
Stress/Thk. Ratio	0.350	Temperature Difference	104 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 30 to 50 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.200	2532.00	7482.90	100.48	2.9553	0.0004192	41.63

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 41.6) / (3 \cdot 100) = 552 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.723	2532.00	7482.90	199.01	2.9553	0.0001523	15.12

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 15.1) / (3 \cdot 199) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.200	2532.00	165502.03	100.48	50.0000	0.0001091	10.83

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 10.8) / (3 \cdot 100) = 144 \text{ KPa.}$$

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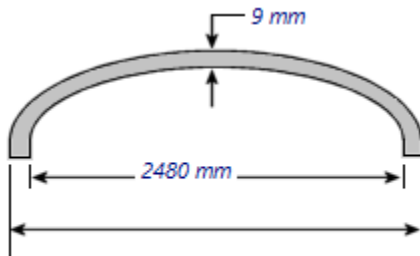
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Element: shell head

Step: 7 8:25am Apr 21,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell head

Elliptical Head Design Information:			
Design Pressure P	544.172 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	160 °C
Length	76.200 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	75.0 N./mm ²
Spec. Min. Thk t,e, or sa	9.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	74414.5 cm ²	SG of Contents	1.000
Element Volume	2365120.5 cm ³	Weight of Contents	2363.5 Kg
Empty Weight	998.8 Kg	Operating Weight	3362.2 Kg
		Comp. All. Ope. Stress	126.9 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (544 \cdot 2480 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 544)$$

$$= 4.8920 + 0.0000 = 4.8920 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.172 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 9) / (1 \cdot 2480 + 0.2 \cdot 9)$$

$$= 1001 - 24.2 = 977 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 9) / (1 \cdot 2480 + 0.2 \cdot 9)$$

$$= 1001 \text{ KPa.}$$



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Element: shell head

Step: 7 8:25am Apr 21,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (544 \cdot (1 \cdot 2480 + 0.2 \cdot 9)) / (2 \cdot 1 \cdot 9)$$

$$= 75.034 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (544 \cdot 1240) / (138 \cdot 1 - 0.6 \cdot 544) + 0$$

$$= 4.902 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 24.172 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1240 + 0.6 \cdot 16)$$

$$= 1767 - 24.2 = 1743 \text{ KPa.}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.793 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 9, tr = 8.78, c = 0 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 8.81, c = 0 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.55$, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	976.638 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	9.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.976	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-30 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.000 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.500 mm

EXTERNAL PRESSURE RESULTS for: shell head

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
9.000	2498.00	277.56	0.0005004	49.68



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Element: shell head

Step: 7 8:25am Apr 21,2026

$$MAEP = B/(K0*Do/t) = 49.7/(0.9 *278) = 199 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.423	2498.00	388.89	0.0003571	35.46

$$MAEP = B/(K0*Do/t) = 35.5/(0.9 *389) = 101 \text{ KPa.}$$

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Shop Pressure Test Results: Step: 8 8:25am Apr 21,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1269.630 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1276.745 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1074.302 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	976.638 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6584.275 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6591.390 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5571.310 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5064.827 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 520 • 1

= 676.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4950 • 1

= 6435.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell head	175.2	234.0	0.749	6459.31
Ch.shell	157.9	234.0	0.675	6459.31
Shell	34.9	234.0	0.149	700.31
shell head	96.6	234.0	0.413	700.31

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manway	Nozzle	138.00	136.20	1.013
N5- Tube inlet	Nozzle	138.00	136.20	1.013
N6- Tube outlet	Nozzle	138.00	136.20	1.013
N1- Shell inlet	Nozzle	118.00	118.00	1.000
N1- Shell inlet	Pad	138.00	138.00	1.000
N2- Shell inlet	Nozzle	118.00	118.00	1.000
N2- Shell inlet	Pad	138.00	138.00	1.000
N3- Shell outlet	Nozzle	138.00	138.00	1.000
N4- Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000



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Shop Pressure Test Results: Step: 8 8:25am Apr 21,2026

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Ch.shell head	138.00	138.00	1.000
Ch.shell	138.00	138.00	1.000
Shell	138.00	138.00	1.000
shell head	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	103.70	1.109
Tubesheet Material	138.00	136.20	1.013

Minimum 1.013

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Manway	76.17	179.40	0.425
N5- Tube inlet	155.33	179.40	0.866
N6- Tube outlet	155.33	179.40	0.866
N1- Shell inlet	12.29	153.40	0.080
N2- Shell inlet	12.29	153.40	0.080
N3- Shell outlet	17.57	225.00	0.078
N4- Shell outlet	16.84	179.40	0.094

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 9 8:25am Apr 21,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1269.630 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1276.745 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1074.302 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	976.638 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6584.275 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6591.390 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5571.310 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5064.827 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 520 • 1

= 676.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4950 • 1

= 6435.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell head	175.2	234.0	0.749	6459.31
Ch.shell	157.9	234.0	0.675	6459.31
Shell	34.9	234.0	0.149	700.31
shell head	96.6	234.0	0.413	700.31

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Manway	Nozzle	138.00	136.20	1.013
N5- Tube inlet	Nozzle	138.00	136.20	1.013
N6- Tube outlet	Nozzle	138.00	136.20	1.013
N1- Shell inlet	Nozzle	118.00	118.00	1.000
N1- Shell inlet	Pad	138.00	138.00	1.000
N2- Shell inlet	Nozzle	118.00	118.00	1.000
N2- Shell inlet	Pad	138.00	138.00	1.000
N3- Shell outlet	Nozzle	138.00	138.00	1.000
N4- Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000



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Field Pressure Test Results: Step: 9 8:25am Apr 21,2026

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Ch.shell head	138.00	138.00	1.000
Ch.shell	138.00	138.00	1.000
Shell	138.00	138.00	1.000
shell head	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	103.70	1.109
Tubesheet Material	138.00	136.20	1.013

Minimum 1.013

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Manway	76.17	179.40	0.425
N5- Tube inlet	155.33	179.40	0.866
N6- Tube outlet	155.33	179.40	0.866
N1- Shell inlet	12.29	153.40	0.080
N2- Shell inlet	12.29	153.40	0.080
N3- Shell outlet	17.57	225.00	0.078
N4- Shell outlet	16.84	179.40	0.094

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 8:25a Apr 21,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		676.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		6435.00 KPa.
Channel Thickness	tc	52.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5080
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6075.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1207.600 mm
Largest Center-to-Center Tube Distance	U1	76.2000 mm
Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	44322.1 cm ²
Perimeter of Tube Layout	Cp	7804.6 mm



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ASME TS Calc: Pressure Test

Case: 1 8:25a Apr 21,2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

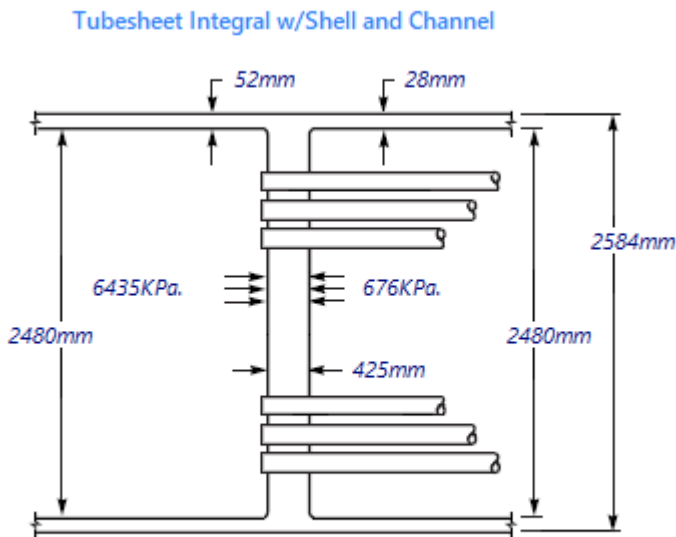
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 425.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 2584.000 mm

Area of the Untubed Lanes AL 1221.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Shell Side Min. Design Pressure Psd,min 100.0002 KPa.

Tubesheet Integral with Both
Tubesheet Extended as Flange No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.



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ASME TS Calc: Pressure Test

Case: 1 8:25a Apr 21,2026

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (6535 \cdot (25.4/2 - 0)) / (115 \cdot 1 + 0.4 \cdot 6535)$
 $= 0.7057 + 0.0000 = 0.7057 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 21.11 °C
 Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7282.60	15.39	50.0000	0.0046419	81.86

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 81.9) / (3 \cdot 15.4) = 7090 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4277	25.40	7282.60	59.38	50.0000	0.0003119	30.11

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 30.1) / (3 \cdot 59.4) = 676 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7057 mm
Allowable Internal Pressure at Corroded thickness	15758.77 KPa.
Required Internal Design Pressure	6535.00 KPa.
Allowable External Pressure at Corroded thickness	7090.01 KPa.
Required External Design Pressure	676.00 KPa.
Required Thickness due to Shell Side pressure	0.4277 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot r_o + dt$
 $= 2 \cdot 1208 + 25.4$
 $= 2440.600 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.8 - 25.4) / 31.8$
 $= 0.200$



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ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 2480/2441$$

$$= 1.0161$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 2480/2441$$

$$= 1.0161$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 50/425$$

$$= 0.118 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (p); dt - 2t_t)$$

$$= \max(25.4 - 2 \cdot 1.65 (195117184/202701232) \cdot$$

$$(115/138) \cdot (0.12); 25.4 - 2 \cdot 1.65)$$

$$= 25.0886 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) }$$

$$= 32.1726 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.2 - 25.1) / 32.2$$

$$= 0.2202$$

$$h/p = 13.385828; \mu^* = 0.220189; E^* = 36395696 \text{ KPa.}$$

$$E^*/E = 0.179553; \nu^* = 0.430867$$

Note: As h/p (13.386) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12 (1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 28) 28 }$$

$$= 0.0069 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6 (1 - \nu_s^2))$$

$$= 0.0069 \cdot 202701232 \cdot 28^3 / (6 (1 - 0.3^2))$$

$$= 5590501376.0000 \text{ KPa.}\cdot\text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6 (D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 (2480 + 28) \cdot 5.59E+09/425^3 \cdot (1 + 425 \cdot 0.0069 + (425 \cdot 0.0069)^2 / 2)$$

$$= 8948154.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2 (D_s + t_s)))$$

$$= 2480^2 / (4 \cdot 202701232 \cdot 28) \cdot (1 - 2480 \cdot 0.3 / (2 (2480 + 28)))$$

$$= 0.23072E+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



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$$= 1.02 \cdot 5.59\text{E}+09 \cdot 0.0069 \cdot 0.23071586 (1 + 425 \cdot 0.0069)$$

$$= 35204.6641 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2480 + 52)52}$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 202701232 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 26151458816.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 26.15\text{E}+09 / 425^3 \cdot (1 + 425 \cdot 0.005 \cdot (1 + 1) / 2 + (425 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 27925762.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 202701232 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.12444\text{E}+00 \text{ mm/N} / \text{mm}^2$$

Channel coefficient [zeta α_c]:

$$zeta\alpha_c = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 26.15\text{E}+09 [0.12443668 (1 + 0.005 \cdot 1 \cdot 425) - 0.000 (1/425 + 1 \cdot 0.005/2)]$$

$$= 51841.3633 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584 / 2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v_s) / E_s \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 36395696 \cdot (8948154 + 27925762 + 202701232 \cdot \ln(1.06))$$

$$= 0.7576$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot Pt]$$

$$= 2441^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 676 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 6435]$$

$$= -70349400.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -70349400 + 51841 \cdot 6435 - 35205 \cdot 676$$

$$= 239451408.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:



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Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((239451408) - 2441^2/32 \cdot 0.76 (676 - 6435)) / (1 + 0.76)$$

$$= 598306432.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= 598306432 + 2441^2/64 \cdot (3 + 0.43) \cdot (676 - 6435)$$

$$= -1240619008.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 598306432 | ; | -1240619008 |)$$

$$= 1240619008.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 1240619008 / (0.22 (425 - 0)^2)$$

$$= 187.1725 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 250$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 1101438464 / (0.23 (248 - 0)^2)$$

$$= 472.9462 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 248 + 0 + 0$$

$$= 248.0621 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(248 ; 86.1)$$

$$= 248.0621 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 676 - 6435 |$$

$$= 5759.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot 0.8 \cdot 0.95 \cdot 250$$

$$= 26.468 \text{ N./mm}^2$$

| Ps - Pt | is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 190) \cdot (4 \cdot 44322 / 7805) \cdot | 676 - 6435 | + 0$$

$$= 86.0731 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:



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Note: For a given Shell thickness of: 28.000mm
Min. Shell len. adjacent to the tubesheet is: 474.327 mm

Note: For a given Channel thickness of: 52.000 mm
Min. Channel len. adjacent to the tubesheet is: 646.398mm

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 161$$

$$= 2027.372 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$$

$$= 1923.2015 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 161/161$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 161$$

$$= 1244.4244 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 2027 is $\geq$ tube strength 2027 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	187.2 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	38.5 $\leq$	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	248.062	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok



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U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	268.468	...	214	475	44	190 Ps+Pt D1 Ok
D2uc	75.769	...	23	475	5	190 Ps+Pt D2 Ok
D3uc	248.062	...	187	475	38	190 Ps+Pt D3 Ok
Max:	268.4677	... mm	0.450	0.230	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	370	...	370	Ok
D2uc	...	...	370	...	370	Ok
D3uc	...	...	370	...	370	Ok
Max	...	...	...	...	...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	268.468	425.000	Ok
Tube thickness :	0.706	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 52.0000 mm

Min Shell length of thk, (28.000) adj. to tubesheet: 474.327 mm

Min Channel length of thk, (52.000) adj. to tubesheet: 646.398 mm

Note: This is a full strength Tube to Tubesheet Joint.



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External Pressure Calculations: Step: 10 8:25am Apr 21,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	2571.8	45.9	0.0024788	108.314
20	Tsht	2756.64	2584	52	0.0036187	115.014
Tsht	30	No Calc	...	52	No Calc	No Calc
30	50	7482.9	2532	25.2	0.00041924	41.6261
50	60	No Calc	2498	9	0.0005004	49.6842

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	45.9	No Calc	...	2148
20	Tsht	52	No Calc	...	3086
Tsht	30	52	No Calc	...	No Calc
30	50	26	13.5232	101.3	552
50	60	9	6.42337	101.3	199

Minimum

199

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	2756.64	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	50	7482.9	165502	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 11

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3383.05	2238538	3383.05	2238538	338.305
20	30	8014.71	12078445	8014.71	12078445	801.471
30	50	12311.2	14107923	11963.2	14152828	1231.12
50	60	998.752	2365120	998.752	2365120	99.8752
Total		24707	30790028.00	24359	30834932.00	2470



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	2236.96	-206.667	7	...	Liquid: 10
10	Insl	354.49	-283.25	...	...	Ins: 10
10	Nozl	1515.06	-620	...	...	Manway
20	Liqd	12062.4	1250	7	...	Liquid: 30
20	Insl	742.888	1250	...	...	Ins: 30
20	Lini	458.685	1250	...	...	308L Weld overl
20	Nozl	1136.96	1250	1697.2	...	N5- Tube inlet
20	Nozl	1095.79	1250	-1697.2	...	N6- Tube outlet
20	Wght	750	1250	...	...	Pass partition
30	Sadl	1222.01	1440	1554.5	...	Left Saddle
30	Sadl	1222.01	5760	1554.5	...	New Saddle
30	Liqd	14099.3	3600	7	...	Liquid: 40
30	Insl	2098.07	3600	...	...	Ins: 30
30	Lini	1321.01	3600	...	...	Weld overlay
30	Nozl	39.0191	2500	-1392.9	...	N1- Shell inlet
30	Nozl	81.7085	4500	-1392.9	...	N2- Shell inlet
30	Nozl	846.348	1500	1697.2	...	N3- Shell outlet
30	Nozl	846.348	4500	1697.2	...	N4- Shell outlet
30	Wght	410	1440	1500	...	Fireproofing
30	Wght	410	5760	-1500	...	Fireproofing
50	Liqd	2363.45	282.867	7	...	Liquid: 50
50	Insl	526.684	346.35	...	...	Ins: 30
50	Lini	187.118	346.35	...	...	Weld oeverlay
20	FTsh	9674.64	2712.5	...	...	Tubesheet
20	Tube	43645.2	5750	...	...	Tubes
30	Bafl	279.626	900	...	...	Baffles
30	Bafl	279.626	1800	...	...	Baffles
30	Bafl	279.626	2700	...	...	Baffles
30	Bafl	279.626	3600	...	...	Baffles
30	Bafl	279.626	4500	...	...	Baffles
30	Bafl	281.123	5400	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	2444.0
Liquid	30762.1
Insulation	3722.1
Cladding & Overlay	1966.8
Nozzles	5561.2



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Element and Detail Weights: Step: 11 8:25am Apr 21,2026

Weights	1570.0
Exchanger Components	54999.1
Liquid in Tubes	16472.2

Sum of the Detail Weights 117497.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	27178.4	27178.4	27178.4	27178.4	27178.4	27178.4
Saddles	2444.0	2444.0	2444.0	2444.0	2444.0	2444.0
Cladding	1966.8	1966.8	1966.8	1966.8	1966.8	1966.8
Nozzles	5561.2	5561.2	5561.2	5561.2	5561.2	5561.2
Wld Weights	750.0	750.0	750.0	750.0	750.0	750.0
Exchanger	54999.1	54999.1	54999.1	54999.1	54999.1	54999.1
Insulation	...	...	...	3722.1	3722.1	3722.1
Test Weights	...	820.0	...	...	...	...
Ope Weights	...	...	...	...	...	820.0
Ope. Liquid	...	...	...	...	...	30762.1
Tube Ope Lqd	...	...	...	...	...	16472.2
Test Liquid	...	30521.1	...	...	...	...
Tube Tst Lqd	...	16472.3	...	...	...	...
Totals	92899.6	140712.9	92899.6	96621.7	96621.7	144676.1

Field Installation Options:

Insulation installed before lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	92899.6 kg.
Shop Test Wt.	- Fabricated weight + water (full)	140712.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	92899.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	96621.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	96621.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	144676.1 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	144293.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	144097.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	16482346.0 cm ³
Weight of Ope Liq in tubes :	16472.2 kg.
Weight of Water in tubes :	16472.3 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	76667.8	70565
20	30	202947	194779
30	50	573630	560963
50	60	74414.5	72606.3

Total	927658.9	898912.8 cm ²
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Element and Detail Weights: Step: 11 8:25am Apr 21,2026

Total	998.5	967.6 ft ²
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Nozzle Flange MAWP:

Step: 12 8:25am Apr 21,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
Manway	8427.1	10200.0	245	600	GR 1.1	...	...	...	...
N1- Shell inle	1540.0	1960.0	160	150	GR 1.1	2214.1	1174.0	1540.0	96.0
N2- Shell inle	1540.0	1960.0	160	150	GR 1.1	2214.1	1174.0	1540.0	96.0
N3- Shell outl	1540.0	1960.0	160	150	GR 1.1	1054.6	1409.5	1540.0	1255.5
N4- Shell outl	1540.0	1960.0	160	150	GR 1.1	1054.6	1409.5	1540.0	1255.5

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1174.010 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 8427.077 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10200.003 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1- Shell inlet	357.251	1.20	1816	2295	976.6	1174.0	Pass
N2- Shell inlet	357.251	1.20	1816	2295	976.6	1174.0	Pass
N3- Shell outlet	947.801	0.60	16348	6884	976.6	1409.5	Pass
N4- Shell outlet	947.801	0.60	16348	6884	976.6	1409.5	Pass

Evaluating Flange on Nozzle [N1- Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) 357.251 \leq \pi \cdot 357.251^3 [(1540.046 - 976.638) + 1.200 \cdot 1540.046]$$

$$= 0.32342E+05 \leq 0.35225E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1- Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1540.046, 1.200 \cdot 1540.046 + 1540.046 - 32342.3 / (\pi \cdot 357.251^3))$$

$$= \min(1540.046, 1174.010)$$

$$= 1174.010 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2- Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) 357.251 \leq \pi \cdot 357.251^3 [(1540.046 - 976.638) + 1.200 \cdot 1540.046]$$

$$= 0.32342E+05 \leq 0.35225E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2- Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1540.046, 1.200 \cdot 1540.046 + 1540.046 - 32342.3 / (\pi \cdot 357.251^3))$$

$$= \min(1540.046, 1174.010)$$

$$= 1174.010 \text{ KPa.}$$



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Nozzle Flange MAWP: Step: 12 8:25am Apr 21, 2026

Evaluating Flange on Nozzle [N3- Shell outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(16348) + 4 \cdot \text{abs}(6884) 947.801 \leq \pi \cdot 947.801^3 [(1540.046 - 976.638) + .600 \cdot 1540.046] \\ &= 0.28767E+06 \leq 0.40574E+06 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3- Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1540.046, 0.600 \cdot 1540.046 + 1540.046 - 287668.0 / (\pi \cdot 947.801^3)) \\ &= \min(1540.046, 1409.481) \\ &= 1409.481 \text{ KPa.} \end{aligned}$$

Evaluating Flange on Nozzle [N4- Shell outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(16348) + 4 \cdot \text{abs}(6884) 947.801 \leq \pi \cdot 947.801^3 [(1540.046 - 976.638) + .600 \cdot 1540.046] \\ &= 0.28767E+06 \leq 0.40574E+06 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4- Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1540.046, 0.600 \cdot 1540.046 + 1540.046 - 287668.0 / (\pi \cdot 947.801^3)) \\ &= \min(1540.046, 1409.481) \\ &= 1409.481 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation:

Step: 13 8:25am Apr 21,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	17000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2 \cdot 0.92)/(1 + 1.7 \cdot 3.4 \cdot 0.2)))$$

$$= \min(0.85, 0.89)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.554
Structure Height to Diameter ratio	4.227

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

= Centroid Height + Vessel Base Elevation
 = 1910 + 17000 = 18910 mm
 = 62 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (62 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (62/900)^{2/9.5}$$

$$= 1.14$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 13 8:25am Apr 21,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.14 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.55 • 19.7

= 149 lbs. [67.4] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Ch.shell head	18910.0	0.000	0.000	0.000	1.145	1.000	78.120
Ch.shell	18910.0	0.000	0.000	0.000	1.145	1.000	78.120
Shell	18910.0	0.000	0.000	0.000	1.145	1.000	78.120
shell head	18910.0	0.000	0.000	0.000	1.145	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass partitio	0.00	18910.00	78.12	0.00
Fireproofing	0.00	20410.00	78.12	0.00
Fireproofing	0.00	17000.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	18910	3326.16	18336.5	78.12	74.1714
20	30	18910	3340.8	83520	78.12	337.84
30	50	18910	3278.4	236045	78.12	954.806
50	60	18910	3237.6	17914.6	78.12	72.4649

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 14 8:25am Apr 21,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(10.30.75)$
 $= 0.115$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.12, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 142232 \cdot 1.25$
 $= 8363.244$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 142232$
 $= 4460.4$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1240	23705.3	1393.87



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Earthquake Load Calculation: Step: 14 8:25am Apr 21,2026

20	30	1240	23705.3	1393.87
30	Sad1	1240	23705.3	1393.87
Sad1	50	1240	23705.3	1393.87
30	50	1240	23705.3	1393.87
50	60	1240	23705.3	1393.87

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 15 8:25am Apr 21,2026

Shop/Field Installation Options :

Insulation is installed in the Field before being lifted.

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	6575.000 mm
Center of Gravity of the Liquid	4311.747 mm
Center of Gravity of the Insulation	5427.421 mm
Center of Gravity of the Lining	5720.254 mm
Center of Gravity of the Nozzles	2319.896 mm
Center of Gravity of the Added Weights (Operating)	4055.095 mm
Center of Gravity of the Added Weights (Empty)	1300.000 mm
Center of Gravity of the Tubesheet(s)	2762.500 mm
Center of Gravity of the Tubes	5800.000 mm
Center of Gravity of the Baffles	6127.005 mm
Center of Gravity of Bare Shell New and Cold	4083.227 mm
Center of Gravity of Bare Shell Corroded	4047.632 mm
Vessel CG in the Operating Condition	4806.759 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4761.372 mm
Vessel CG in the Test Condition	4784.536 mm

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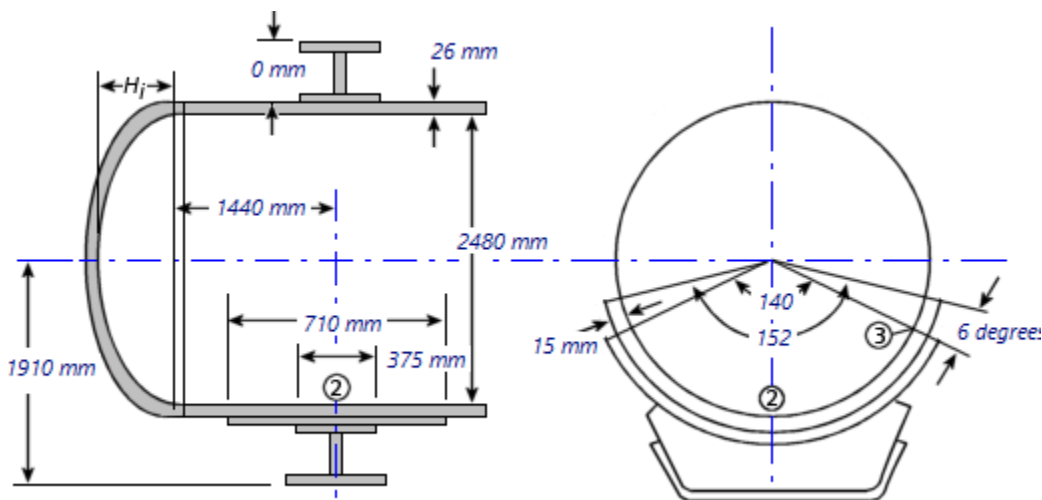
Saddle Calcs: Road Transport: Step: 16 8:25am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.00	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	642.95	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		92716.09	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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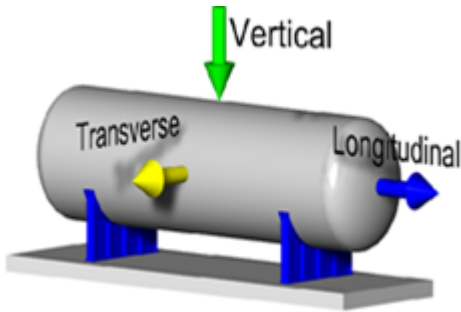
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Saddle Calcs: Road Transport:

Step: 16

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.83	-234.00
Long. Stress at Top of Midspan	1.83	138.00
Long. Stress at Bottom of Midspan	1.83	138.00
Long. Stress at Top of Saddles	16.86	138.00
Long. Stress at Bottom of Saddles	-9.57	-234.00
Long. Stress at Bottom of Saddles	9.57	138.00
Tangential Shear in Shell	13.58	110.40
Circ. Stress at Horn of Saddle	50.79	207.00
Circ. Compressive Stress in Shell	2.35	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 383 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 3253.3 \text{ lbf} \\
 &= 1475.685 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 8.96^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 643.2 \text{ lbf} \\
 &= 291.7 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 90456 \cdot 1 + 1476 \\
 &= 10521.2 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 90456 \cdot 1 + 292 \\
 &= 9337.3 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 10521/2 \cdot 1910/2350 \\
 &= 4275.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 9337 \cdot 1910/4320 \\
 &= 4128.3 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 84229 \cdot 0.05
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 16 8:25am Apr 21,2026

= 4211.450 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(FwL, Fwt)

= 84229 + 4211 + max(4128, 4276)

= 92716.1 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 93938.10 Kgf

Transverse Shear Load 5260.62 Kgf

Longitudinal Shear Load 9337.30 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412 K2 = 0.9003 K3 = 0.5922 K4 = 0.3269

K5 = 0.6971 K6 = 0.0379 K7 = 0.0379 K8 = 0.3167

K9 = 0.2372 K10 = 0.0432 K1* = 0.2486 K6p = 0.0305

K7p = 0.0305

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -92716•1440[1-(1-1440/7276+(1253²-643²)/

(2•1440•7276))/(1+(4•643)/(3•7276))]

= -31118.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 92716•7276/4((1+2(1253²-643²)/(7276²))/(1+(4•643)/

(3•7276))-4•1440/7276)

= 23960.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 1253/(2•26) - 23961/(π•1253²•26)

= -1.83 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 1253/(2 • 26) + 23961/(π • 1253² • 26)

= 1.83 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•1253/(2•26)-31118/(0.14•π•1253²•26)

= 16.86 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•1253/(2•26)+31118/(0.25•π•1253²•26)

= -9.57 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 92716 (7276 - 2 • 1440)/(7276 + (4 • 643/3))

= 50113.7 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 50114 / (1253 \cdot 26)$$

$$= 13.58 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1253 \cdot 26)}$$

$$= 140.785 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1253 \cdot 26)}, 2 \cdot 1440)$$

$$= 656.57 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 92716 \cdot 0.1 / (657(26 + 1 \cdot 15))$$

$$= -2.35 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -92716 / (4(26 + 1 \cdot 15)657) -$$

$$12 \cdot 0.038 \cdot 92716 \cdot 1253 / (7276(26 + 1 \cdot 15)^2)$$

$$= -50.79 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1240 \cdot 26)} + 710)26^3) / 12$$

$$= 145.017 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 145 + 257(231 - 13)^2$$

$$= 122004.336 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	990.1	26.0	13.0	257.4	334655.7	0.122E+06
Wearplate	710.0	15.0	33.5	106.5	356775.0	0.414E+05
Web	16.0	589.0	335.5	94.2	3161752.5	0.376E+05
BasePlate	400.0	40.0	650.0	160.0	10400001.0	0.282E+06
Totals	...	...	...	618.2	14253184.0	0.483E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14253184 / 618$$



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$$= 230.572 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 92716$$

$$= 22291.8105 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 6.0601 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1910 - 1240 \cdot \sin(140/2) / 1.22$$

$$= 956.255 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 21317.0859 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.9858 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(92716 + 1222) \cdot 400) / (4 \cdot 2350 \cdot 173)}$$

$$= 26.048 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2407 / (5 - 1)$$

$$= 601.873 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 602 \cdot 400 / 2$$

$$= 1203.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 93938 / (2350 \cdot 400)$$

$$= 9.993 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1204 \cdot 9.99$$

$$= 12029.534 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 57.4 + 48.1$$



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$$= 105.590 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 12030/106$$

$$= 11.173 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 16$$

$$= 359.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/106$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7031/106)}$$

$$= 81.603 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1910 - \cos(140/2) (1240 + 26 + 15) - 40$$

$$= 1431.872 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (2407 - 2299) / 2$$

$$= 54.146 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(1432^2 + 54.1^2)}$$

$$= 1432.896 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1433/81.6$$

$$= 17.559$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 9337 / (2 \cdot 2350) \cdot 602 \cdot 1433 / 2$$

$$= 856.686 \text{ Kg-m.}$$

Compressive Allowable, KL/r < Cc (17.6 < 123) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (17.6)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (17.6) / (8 \cdot 123) - (17.6^3) / (8 \cdot 123^3))$$

$$= 149.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.2 / 150 + (857 \cdot 188 / 70312504) / 173$$

$$= 0.204$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/154$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1910 - \cos(1281/2) (40 + 0 + 0) - 0$$

$$= 589.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (7051 / 154) }$$

$$= 67.724 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 589/67.7$$

$$= 8.697$$

Unit Force [Force,u]:

$$= Fl / (2 \cdot \text{Baseplate Length})$$

$$= 9337 / (2 \cdot 2350)$$

$$= 1.987 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.99 \cdot 602 \cdot 589$$

$$= 704.291 \text{ Kg-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 704/376067

= 18.366 N./mm²

Compressive Allowable, $KL/r < C_c$ (8.7 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c^2)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c^3)

= (1-(8.7)²/(2 • 123²))260/

(5/3+3•(8.7)/(8• 123)-(8.7³)/(8•123³)

= 153.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 15.4/153 + (704 • 188/7051)/173

= 0.206

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.00	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	624.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		10813.54	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.22	-234.00
Long. Stress at Top of Midspan	0.22	138.00
Long. Stress at Bottom of Midspan	0.22	138.00
Long. Stress at Top of Saddles	1.94	138.00
Long. Stress at Bottom of Saddles	-1.10	-234.00
Long. Stress at Bottom of Saddles	1.10	138.00
Tangential Shear in Shell	1.59	110.40



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Circ. Stress at Horn of Saddle	5.92	207.00
Circ. Compressive Stress in Shell	0.27	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 383 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 3253.3 \text{ lbf} \\
 &= 1475.685 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 8.96^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 643.2 \text{ lbf} \\
 &= 291.7 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 90456 \cdot 1 + 1476 \\
 &= 10521.2 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 90456 \cdot 1 + 292 \\
 &= 9337.3 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 10521/2 \cdot 1910/2350 \\
 &= 4275.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= Fl \cdot B / Ls \\
 &= 9337 \cdot 1910/4320 \\
 &= 4128.3 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 6227 \cdot 0.05 \\
 &= 311.328 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 6227 + 311 + \max(4128, 4276) \\
 &= 10813.5 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	12035.55	Kgf
Transverse Shear Load	5260.62	Kgf
Longitudinal Shear Load	9337.30	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \text{Baseplate Length} \\
 &= 2350.000 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			



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The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 \cdot hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -10814 \cdot 1440 [1 - (1 - 1440/7276 + (1253^2 \cdot 624^2)/(2 \cdot 1440 \cdot 7276))/(1 + (4 \cdot 624)/(3 \cdot 7276))] \\
 &= -3577.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(Rm^2 \cdot hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 10814 \cdot 7276/4 ((1 + 2(1253^2 \cdot 624^2)/(7276^2))/(1 + (4 \cdot 624)/(3 \cdot 7276)) - 4 \cdot 1440/7276) \\
 &= 2865.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\
 &= 0 \cdot 1253/(2 \cdot 26) - 2866/(\pi \cdot 1253^2 \cdot 26) \\
 &= -0.22 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 0 \cdot 1253/(2 \cdot 26) + 2866/(\pi \cdot 1253^2 \cdot 26) \\
 &= 0.22 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t) \\
 &= 0 \cdot 1253/(2 \cdot 26) - 3577/(0.14 \cdot \pi \cdot 1253^2 \cdot 26) \\
 &= 1.94 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 0 \cdot 1253/(2 \cdot 26) + 3577/(0.25 \cdot \pi \cdot 1253^2 \cdot 26) \\
 &= -1.10 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 10814 (7276 - 2 \cdot 1440)/(7276 + (4 \cdot 624/3)) \\
 &= 5862.5 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 0.9 \cdot 5863 / (1253 \cdot 26) \\
 &= 1.59 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1253 \cdot 26)} \\
 &= 140.785 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(375 + 1.56 \cdot \sqrt{(1253 \cdot 26)}, 2 \cdot 1440) \\
 &= 656.57 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7 \cdot 10814 \cdot 0.1 / (657(26 + 1 \cdot 15)) \\
 &= -0.27 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:



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$$\begin{aligned}
 &= -Q/(4(t+\eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm/(L(t+\eta \cdot tr)^2) \\
 &= -10814/(4(26 + 1 \cdot 15)657) - \\
 &\quad 12 \cdot 0.038 \cdot 10814 \cdot 1253/(7276(26+1 \cdot 15)^2) \\
 &= -5.92 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12 \\
 &= ((1.56 \cdot \sqrt{(1240 \cdot 26)} + 710)26^3)/12 \\
 &= 145.017 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 145 + 257(231 - 13)^2 \\
 &= 122004.336 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	990.1	26.0	13.0	257.4	334655.7	0.122E+06
Wearplate	710.0	15.0	33.5	106.5	356775.0	0.414E+05
Web	16.0	589.0	335.5	94.2	3161752.5	0.376E+05
BasePlate	400.0	40.0	650.0	160.0	10400001.0	0.282E+06
Totals	...	...	...	618.2	14253184.0	0.483E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 14253184/618 \\
 &= 230.572 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 140/2 \\
 &= 110.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(110) - 0.5 \cdot \sin(110)^2)/(\pi - 1.92 + \sin(110)\cos(110)) \\
 &= 0.2404
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.24 \cdot 10814 \\
 &= 2599.9094 \text{ Kg}
 \end{aligned}$$

Tension Stress, St = (Fh/As) = 0.7068 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians}) \\
 &= 1910 - 1240 \cdot \sin(140/2)/1.22
 \end{aligned}$$



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= 956.255 mm

Bending Moment, $M = F_h \cdot d = 2486.2268 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 1.1647 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(10814 + 1222) \cdot 400) / (4 \cdot 2350 \cdot 173)}$

= 9.324 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 140/2) (1240 + 26 + 15)$

= 2407.492 mm

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 2407 / (5 - 1)$

= 601.873 mm

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 602 \cdot 400 / 2$

= 1203.746 cm²

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 12036 / (2350 \cdot 400)$

= 1.280 Kg/cm²

Axial Load [P]:

$= A_p \cdot B_p$

$= 1204 \cdot 1.28$

= 1541.250 Kg

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 57.4 + 48.1$

= 105.590 cm²

Compressive Stress [Sc]:

$= P / A_r$

$= 1541 / 106$

= 1.431 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 375 - 16$

= 359.000 mm

Distance to Centroid from Datum [ytot]:

$= AY / A$

= 0/106



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= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375/2

= 187.500 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(7031/106)}$

= 81.603 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1910 - cos(140/2)(1240 + 26 + 15) - 40

= 1431.872 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(140/2)(1240 + 26 + 15)

= 2407.492 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 2350 - 50.8

= 2299.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (2407 - 2299)/2

= 54.146 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$

= $\sqrt{(1432^2 + 54.1^2)}$

= 1432.896 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1433/81.6

= 17.559

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 9337 / (2 • 2350) • 602 • 1433/2

= 856.686 Kg-m.

Compressive Allowable, KL/r < Cc (17.6 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (17.6)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (17.6) / (8 \cdot 123) - (17.6^3) / (8 \cdot 123^3))$

= 149.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 1.43/150 + (857 • 188/70312504) / 173

= 0.139

Check of Inside Ribs:



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Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/154$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1910 - \cos(1281/2)(40 + 0 + 0) - 0$$

$$= 589.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7051/154)}$$

$$= 67.724 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 589/67.7$$

$$= 8.697$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 9337 / (2 \cdot 2350)$$

$$= 1.987 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.99 \cdot 602 \cdot 589$$

$$= 704.291 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 704/376067$$

$$= 18.366 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (8.7) / (8 \cdot 123) - (8.7)^3 / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 1.97/153 + (704 \cdot 188/7051) / 173$$

$$= 0.119$$

Review notes about nozzle loadings and supports in the Warnings report.

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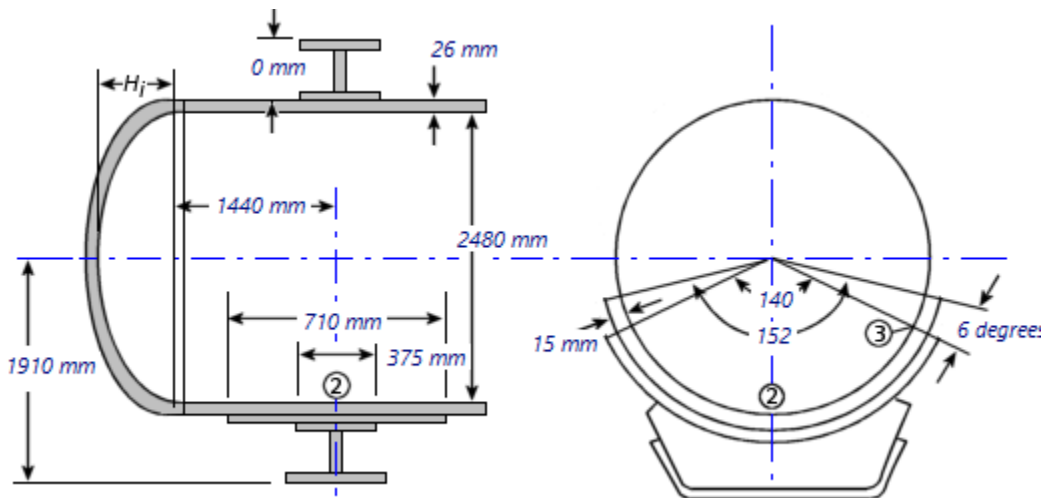
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.40	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	642.95	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	142603.64	Kgf
Pressure used in Analysis	P 532.017	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	10.33	138.00
Long. Stress at Bottom of Midspan	16.14	138.00
Long. Stress at Top of Saddles	39.96	138.00
Long. Stress at Bottom of Saddles	-1.95	-207.36
Long. Stress at Bottom of Saddles	1.95	138.00



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Tangential Shear in Shell	21.55	110.40
Circ. Stress at Horn of Saddle	81.10	172.50
Circ. Compressive Stress in Shell	3.72	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1439/2 + 0) \cdot 1910/2350$$

$$= 1754.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(725, 0) \cdot 1910/4320$$

$$= 320.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8363, 0) \cdot 1910/4320$$

$$= 3697.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (8363/2 + 0) \cdot 1910/2350$$

$$= 10196.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 132408 + \max(321, 1755, 3698, 10196)$$

$$= 142603.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.73 \cdot 1.2)^2)$$

$$= 7113.507 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	143825.66	Kgf
Transverse Shear Load	4181.62	Kgf
Longitudinal Shear Load	8363.24	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2350.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -142604 \cdot 1440 [1 - (1 - 1440/7276 + (1253^2 - 643^2)/$$

$$(2 \cdot 1440 \cdot 7276))/(1 + (4 \cdot 643)/(3 \cdot 7276))]$$

$$= -47852.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:



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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\
 &= 142604 \cdot 7276 / 4 \cdot ((1 + 2(1253^2 - 643^2) / (7276^2)) / (1 + (4 \cdot 643) / (3 \cdot 7276)) - 4 \cdot 1440 / 7276) \\
 &= 36861.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 532 \cdot 1253 / (2 \cdot 25.2) - 36862 / (\pi \cdot 1253^2 \cdot 25.2) \\
 &= 10.33 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 532 \cdot 1253 / (2 \cdot 25.2) + 36862 / (\pi \cdot 1253^2 \cdot 25.2) \\
 &= 16.14 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 532 \cdot 1253 / (2 \cdot 25.2) - 47853 / (0.14 \cdot \pi \cdot 1253^2 \cdot 25.2) \\
 &= 39.96 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 532 \cdot 1253 / (2 \cdot 25.2) + 47853 / (0.25 \cdot \pi \cdot 1253^2 \cdot 25.2) \\
 &= -1.95 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 142604 \cdot (7276 - 2 \cdot 1440) / (7276 + (4 \cdot 643 / 3)) \\
 &= 77078.3 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 0.9 \cdot 77078 / (1253 \cdot 25.2) \\
 &= 21.55 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1253 \cdot 25.2)} \\
 &= 138.624 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(375 + 1.56 \cdot \sqrt{(1253 \cdot 25.2)}, 2 \cdot 1440) \\
 &= 652.25 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr/S, 1) \\
 &= \min(138/138, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7 \cdot 142604 \cdot 0.1 / (652(25.2 + 1 \cdot 15)) \\
 &= -3.72 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -142604 / (4(25.2 + 1 \cdot 15)652) - 12 \cdot 0.038 \cdot 142604 \cdot 1253 / (7276(25.2 + 1 \cdot 15)^2) \\
 &= -81.10 \text{ N./mm}^2
 \end{aligned}$$

Distance between Saddle Supports [Ls]:

$$= 4320.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:



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$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12491\text{E-}04 \cdot 4320 (160 - 21.1)$$

$$= 7.494 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1241 \cdot 25.2) + 710})^3) / 12$$

$$= 131.471 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 131 + 248 (233 - 12.6)^2$$

$$= 120971.883 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	985.9	25.2	12.6	248.4	313027.7	0.121E+06
Wearplate	710.0	15.0	32.7	106.5	348255.0	0.428E+05
Web	16.0	589.0	334.7	94.2	3154212.5	0.370E+05
BasePlate	400.0	40.0	649.2	160.0	10387199.0	0.277E+06
Totals	...	...	...	609.2	14202694.0	0.478E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14202694 / 609$$

$$= 233.147 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 140 / 2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 142604$$

$$= 34286.3164 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 9.3208 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1910 - 1241 \cdot \sin(140 / 2) / 1.22$$

$$= 955.640 \text{ mm}$$



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Bending Moment, $M = F_h \cdot d = 32766.0293 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 15.6754 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(142604 + 1222) \cdot 400) / (4 \cdot 2350 \cdot 154)}$

$= 34.239 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 140/2) \cdot (1240 + 26 + 15)$

$= 2407.492 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 2407 / (5 - 1)$

$= 601.873 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 602 \cdot 400 / 2$

$= 1203.746 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 143826 / (2350 \cdot 400)$

$= 15.301 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 1204 \cdot 15.3$

$= 18418.041 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 57.4 + 48.1$

$= 105.590 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 18418 / 106$

$= 17.106 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 375 - 16$

$= 359.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 106$

$= 0.000 \text{ mm}$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7031 / 106)}$$

$$= 81.603 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1910 - \cos(140 / 2) (1240 + 26 + 15) - 40$$

$$= 1431.872 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140 / 2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (2407 - 2299) / 2$$

$$= 54.146 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$

$$= \sqrt{(1432^2 + 54.1^2)}$$

$$= 1432.896 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 230)}$$

$$= 130.885$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 1433 / 81.6$$

$$= 17.559$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 8363 / (2 \cdot 2350) \cdot 602 \cdot 1433 / 2$$

$$= 767.317 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL} / r < \text{Cc}$ ($17.6 < 131$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL} / r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5 / 3 + 3 \cdot (\text{KL} / r) / (8 \cdot \text{Cc}) - (\text{KL} / r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (17.6)^2 / (2 \cdot 131^2)) 230 /$$

$$(5 / 3 + 3 \cdot (17.6) / (8 \cdot 131) - (17.6^3) / (8 \cdot 131^3))$$

$$= 133 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc} / \text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 17.1 / 133 + (767 \cdot 188 / 70312504) / 154$$

$$= 0.259$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction



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	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/154$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1910 - \cos(1281/2)(40 + 0 + 0) - 0$$

$$= 589.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7051/154)}$$

$$= 67.724 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 589/67.7$$

$$= 8.697$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8363/(2 \cdot 2350)$$

$$= 1.779 \text{ Kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.78 \cdot 602 \cdot 589$$

$$= 630.821 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 631/376067$$

$$= 16.450 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.7 < 131$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.7)^2 / (2 \cdot 131^2)) 230 / (5/3 + 3 \cdot (8.7) / (8 \cdot 131) - (8.7^3) / (8 \cdot 131^3))$$

$$= 135.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 23.5/136 + (631 \cdot 188/7051) / 154$$

$$= 0.280$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm



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Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load QO (Weight)	QO	133629.6 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3697.6 Kgf
Maximum Transverse Force	Ft	4181.6 Kgf
Maximum Longitudinal Force	Fl	8363.2 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 4652.0 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= max(Ff, Fl) / (Bolt Area • Number of Bolts)
= 8363 / (3.04 • 8)
= 33.7 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= Ft / (Bolt Area • Number of Bolts)
= 4182 / (3.04 • 8)
= 16.9 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= Fl / (BoltShearAllowable • Nbolts)
= 8363 / (82.8 • 8)
= 1.2382 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1910 • 4182 + 0
= 7987.06 Kg-m.

Eccentricity (e):
= Rmom / QO
= 7987 / 133630
= 59.77 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 1.24, 0]
= 1.2382 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 15, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E* / (tg - c) = 0.35, Temp. Reduction = 78 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:



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Input and Calculated Values:

Vessel Mean Radius	Rm	1253.40	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	624.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		24480.86	Kgf
Pressure used in Analysis	P	532.017	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	12.72	138.00
Long. Stress at Bottom of Midspan	13.74	138.00
Long. Stress at Top of Saddles	17.75	138.00
Long. Stress at Bottom of Saddles	10.66	138.00
Tangential Shear in Shell	3.71	110.40
Circ. Stress at Horn of Saddle	13.92	172.50
Circ. Compressive Stress in Shell	0.64	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (1439/2 + 0) \cdot 1910/2350$
 $= 1754.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(725, 0) \cdot 1910/4320$
 $= 320.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(8363, 0) \cdot 1910/4320$
 $= 3697.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (8363/2 + 0) \cdot 1910/2350$
 $= 10196.0 \text{ Kgf}$



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Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 14285 + max(321, 1755, 3698, 10196)
 = 24480.9 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (2.73 • 1.2)²)
 = 7113.507 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25702.87	Kgf
Transverse Shear Load	4181.62	Kgf
Longitudinal Shear Load	8363.24	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -24481•1440 [1 - (1-1440/7276 + (1253²-624²) / (2•1440•7276)) / (1 + (4•624) / (3•7276))]
 = -8097.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4 ((1+2(Rm²-hm²)/(L²)) / (1+(4hm²)/(3L)) - 4a/L)
 = 24481•7276/4 ((1+2(1253²-624²) / (7276²)) / (1 + (4•624) / (3•7276)) - 4•1440/7276)
 = 6489.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 532 • 1253/(2•25.2) - 6490/(π•1253²•25.2)
 = 12.72 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 532 • 1253/(2 • 25.2) + 6490/(π • 1253² • 25.2)
 = 13.74 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 532•1253/(2•25.2)--8098/(0.14•π•1253²•25.2)
 = 17.75 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 532•1253/(2•25.2)+8098/(0.25•π•1253²•25.2)
 = 10.66 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:



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$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 24481 (7276 - 2 \cdot 1440)/(7276 + (4 \cdot 624/3))$$

$$= 13272.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 13272 / (1253 \cdot 25.2)$$

$$= 3.71 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (1253 \cdot 25.2) }$$

$$= 138.624 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{ (1253 \cdot 25.2) }, 2 \cdot 1440)$$

$$= 652.25 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot tr))$$

$$= -0.7 \cdot 24481 \cdot 0.1 / (652 (25.2 + 1 \cdot 15))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4 (t + \eta \cdot tr) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L (t + \eta \cdot tr)^2)$$

$$= -24481 / (4 (25.2 + 1 \cdot 15) 652) -$$

$$12 \cdot 0.038 \cdot 24481 \cdot 1253 / (7276 (25.2 + 1 \cdot 15)^2)$$

$$= -13.92 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw, c	680.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (1241 \cdot 25.2) } + 710) 25.2^3) / 12$$

$$= 131.471 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A \text{Shell} (C_1 - Y)^2$$

$$= 131 + 248 (233 - 12.6)^2$$

$$= 120971.883 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	985.9	25.2	12.6	248.4	313027.7	0.121E+06



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Wearplate	710.0	15.0	32.7	106.5	348255.0	0.428E+05
Web	16.0	589.0	334.7	94.2	3154212.5	0.370E+05
BasePlate	400.0	40.0	649.2	160.0	10387199.0	0.277E+06
Totals	...	...	...	609.2	14202694.0	0.478E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14202694/609$$

$$= 233.147 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 24481$$

$$= 5885.9546 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.6001 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1910 - 1241 \cdot \sin(140/2) / 1.22$$

$$= 955.640 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5624.9663 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.6910 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(24481 + 1222)400 / (4 \cdot 2350 \cdot 154)}$$

$$= 14.474 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2407 / (5 - 1)$$

$$= 601.873 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 602 \cdot 400/2$$

$$= 1203.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 25703 / (2350 \cdot 400)$$

$$= 2.734 \text{ Kgf/cm}^2$$



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Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1204 \cdot 2.73$$

$$= 3291.461 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 57.4 + 48.1$$

$$= 105.590 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 3291 / 106$$

$$= 3.057 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 16$$

$$= 359.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 106$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7031 / 106)}$$

$$= 81.603 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1910 - \cos(140/2) (1240 + 26 + 15) - 40$$

$$= 1431.872 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (2407 - 2299) / 2$$

$$= 54.146 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$



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$$= \sqrt{1432^2 + 54.1^2}$$

$$= 1432.896 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{230}}$$

$$= 130.885$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1433/81.6$$

$$= 17.559$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 8363 / (2 \cdot 2350) \cdot 602 \cdot 1433/2$$

$$= 767.317 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($17.6 < 131$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (17.6)^2 / (2 \cdot 131^2)) 230 /$$

$$(5/3 + 3 \cdot (17.6) / (8 \cdot 131) - (17.6)^3 / (8 \cdot 131^3))$$

$$= 133 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.06/133 + (767 \cdot 188/70312504) / 154$$

$$= 0.154$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/154$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{\text{pthk}}$$

$$= 1910 - \cos(1281/2) (40 + 0 + 0) - 0$$

$$= 589.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7051/154}$$

$$= 67.724 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 589/67.7$$

$$= 8.697$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$



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$$= 8363 / (2 \cdot 2350)$$

$$= 1.779 \text{ Kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.78 \cdot 602 \cdot 589$$

$$= 630.821 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 631 / 376067$$

$$= 16.450 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.7 < 131$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.7)^2 / (2 \cdot 131^2)) 230 /$$

$$(5/3 + 3 \cdot (8.7) / (8 \cdot 131) - (8.7^3) / (8 \cdot 131^3))$$

$$= 135.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 4.2 / 136 + (631 \cdot 188 / 7051) / 154$$

$$= 0.138$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	15506.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	3697.6 Kgf
Maximum Transverse Force	Ft	4181.6 Kgf
Maximum Longitudinal Force	F1	8363.2 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8363 / (3.04 \cdot 8)$$

$$= 33.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4182 / (3.04 \cdot 8)$$

$$= 16.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 8363 / (82.8 \cdot 8)$$

$$= 1.2382 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:



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$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 1910 \cdot 4182 + 0 \\
 &= 7987.06 \text{ Kg-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Eccentricity (e):} \\
 &= R_{\text{mom}} / QO \\
 &= 7987/15507 \\
 &= 515.06 \text{ mm} > B_{\text{plen}}/6 \rightarrow \text{Uplift in Transverse direction}
 \end{aligned}$$

$$\begin{aligned}
 f &= B_{\text{plen}} / 2 - \text{Edgedis} \\
 &= 2350/2 - 70 \\
 &= 1105.00 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Modular Ratio Of Steel/Concrete (n1):} \\
 &= E_S / E_C \\
 &= 203402/21526 \\
 &= 9.45
 \end{aligned}$$

$$\begin{aligned}
 K1 &= 3 (e - 0.5 \cdot B_{\text{plen}}) \\
 &= 3 (515 - 0.5 \cdot 2350) \\
 &= -1979.83 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 K2 &= 6 \cdot n1 \cdot A_t / B_{\text{pwid}} \cdot (f + e) \\
 &= 6 \cdot 9.45 \cdot 12.2/400 \cdot (1105 + 515) \\
 &= 279385.14 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 K3 &= -K2 \cdot (0.5 \cdot B_{\text{plen}} + f) \\
 &= -279385 \cdot (0.5 \cdot 2350 + 1105) \\
 &= -636998082.09 \text{ mm}^3
 \end{aligned}$$

$$\begin{aligned}
 &\text{Iteratively Solving for the Effective Bearing Length:} \\
 &Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0 \\
 &Y^3 + -1980 \cdot Y^2 + 279385 \cdot Y + -636998080 = 0 \\
 &Y = 1999.44 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Num} &= (B_{\text{plen}} / 2 - Y / 3 - e) \\
 &= (2350/2 - 1999/3 - 515) \\
 &= -6.54
 \end{aligned}$$

$$\begin{aligned}
 \text{Denom} &= (B_{\text{plen}} / 2 - Y / 3 + f) \\
 &= (2350/2 - 1999/3 + 1105) \\
 &= 1613.52
 \end{aligned}$$

$$\begin{aligned}
 &\text{Total Bolt Tension Force [Tforce]:} \\
 &= -QO \cdot \text{Num} / \text{Denom} \\
 &= -15507 \cdot -6.54/1614 \\
 &= 62.81 \text{ Kg}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\
 &= T_{\text{force}} / (S_{\text{tba}} \cdot N_{\text{bt}}) \\
 &= 62.8 / (138 \cdot 4) \\
 &= 0.0112 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Required Area of a Single Bolt [Bltarear]:} \\
 &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\
 &= \max[0, 1.24, 0.011] \\
 &= 1.2382 \text{ cm}^2
 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}
 &\text{Bearing Pressure (fc)} \\
 &= 2(QO + T_{\text{force}}) / (Y \cdot B_{\text{pwid}}) \\
 &= 2(15507 + 62.8) / (1999 \cdot 400) \\
 &= 381.81 \text{ KPa.}
 \end{aligned}$$



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Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - WebLength) / 2$$

$$= (2350 - 2299) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{force} \cdot A_{dist}$$

$$= 62.8 \cdot 25.4$$

$$= 1.60 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{dist}) / Y$$

$$= 382 \cdot (1999 - 25.4) / 1999$$

$$= 376.96 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{dist}^2 \cdot B_{pwid} / 6) \cdot (f1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 400 / 6) \cdot (377 + 2 \cdot 382)$$

$$= 5.00 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (B_{pwid} \cdot S_{ba}))}$$

$$= \sqrt{(6 \cdot \max(1.6, 5) / (400 \cdot 207))}$$

$$= 1.8855 \text{ mm}$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 15, c = 0.8 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.35, Temp. Reduction = 78 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

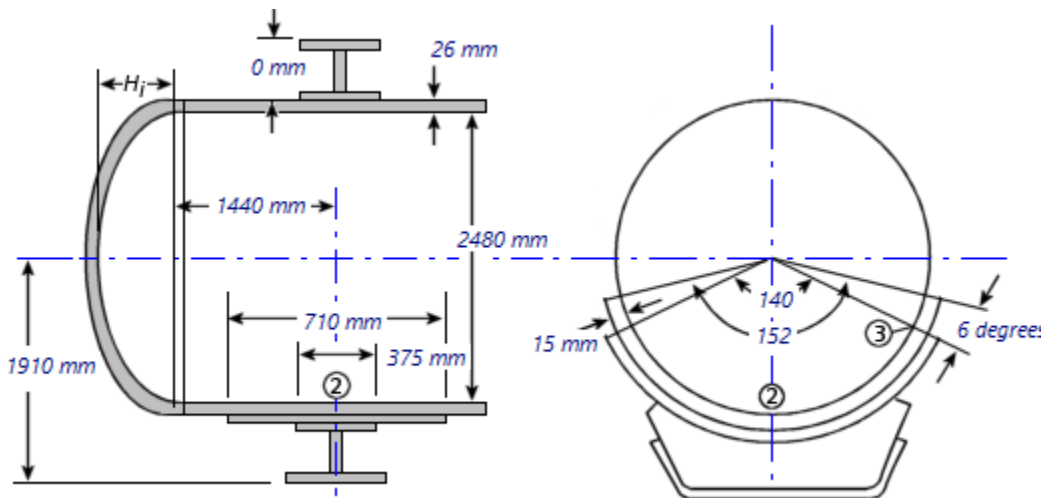
Step: 18 8:25am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.40	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	642.95	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		128220.45	Kgf
Pressure used in Analysis	P	688.286	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	14.50	234.00
Long. Stress at Bottom of Midspan	19.73	234.00
Long. Stress at Top of Saddles	41.15	234.00
Long. Stress at Bottom of Saddles	3.47	234.00
Tangential Shear in Shell	19.37	138.58



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Circ. Stress at Horn of Saddle	72.92	350.99
Circ. Compressive Stress in Shell	3.34	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (720/2 + 0) \cdot 1910/2350$$

$$= 877.4 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(363, 0) \cdot 1910/4320$$

$$= 160.4 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 127343 + \max(160, 877, 0, 0)$$

$$= 128220.5 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.73 \cdot 1.2)^2)$$

$$= 7113.507 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	129442.47	Kgf
Transverse Shear Load	359.82	Kgf
Longitudinal Shear Load	362.69	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2350.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/(3L))]$$

$$= -128220 \cdot 1440 [1 - (1 - 1440/7276 + (1253^2 - 643^2)/(2 \cdot 1440 \cdot 7276)) / (1 + (4 \cdot 643)/(3 \cdot 7276))]$$

$$= -43026.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L)$$

$$= 128220 \cdot 7276/4 ((1 + 2(1253^2 - 643^2)/(7276^2)) / (1 + (4 \cdot 643)/(3 \cdot 7276)) - 4 \cdot 1440/7276)$$

$$= 33143.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 688 \cdot 1253/(2 \cdot 25.2) - 33144/(\pi \cdot 1253^2 \cdot 25.2)$$

$$= 14.50 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1253 / (2 \cdot 25.2) + 33144 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= 19.73 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1253 / (2 \cdot 25.2) - 43026 / (0.14 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= 41.15 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1253 / (2 \cdot 25.2) + 43026 / (0.25 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= 3.47 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 128220 (7276 - 2 \cdot 1440) / (7276 + (4 \cdot 643 / 3))$$

$$= 69304.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 69304 / (1253 \cdot 25.2)$$

$$= 19.37 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1253 \cdot 25.2}$$

$$= 138.624 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{1253 \cdot 25.2}, 2 \cdot 1440)$$

$$= 652.25 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7 \cdot 128220 \cdot 0.1 / (652(25.2 + 1 \cdot 15))$$

$$= -3.34 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -128220 / (4(25.2 + 1 \cdot 15) 652) -$$

$$12 \cdot 0.038 \cdot 128220 \cdot 1253 / (7276(25.2 + 1 \cdot 15)^2)$$

$$= -72.92 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{1241 \cdot 25.2} + 710) 25.2^3) / 12$$



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$$= 131.471 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2$$

$$= 131 + 248 (233 - 12.6)^2$$

$$= 120971.883 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	985.9	25.2	12.6	248.4	313027.7	0.121E+06
Wearplate	710.0	15.0	32.7	106.5	348255.0	0.428E+05
Web	16.0	589.0	334.7	94.2	3154212.5	0.370E+05
BasePlate	400.0	40.0	649.2	160.0	10387199.0	0.277E+06
Totals	...	...	...	609.2	14202694.0	0.478E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14202694/609$$

$$= 233.147 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 128220$$

$$= 30828.1543 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 8.3807 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1910 - 1241 \cdot \sin(140/2) / 1.22$$

$$= 955.640 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 29461.2051 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 14.0943 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(128220 + 1222) \cdot 400) / (4 \cdot 2350 \cdot 173)}$$

$$= 30.577 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2407 / (5 - 1)$$

$$= 601.873 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 602 \cdot 400 / 2$$

$$= 1203.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 129442 / (2350 \cdot 400)$$

$$= 13.770 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1204 \cdot 13.8$$

$$= 16576.158 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 57.4 + 48.1$$

$$= 105.590 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 16576 / 106$$

$$= 15.395 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 375 - 16$$

$$= 359.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 106$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (7031 / 106) }$$

$$= 81.603 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 1910 - \cos(140 / 2)(1240 + 26 + 15) - 40$$

$$= 1431.872 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(140 / 2)(1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:



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= Baseplate length - clearance
 = 2350 - 50.8
 = 2299.200 mm

Length of Outer Rib, step 4, [Dd]:
 = (Lw - Bd)/2
 = (2407 - 2299)/2
 = 54.146 mm

Length of Outer Rib [L]:
 $= \sqrt{L_v^2 + D_d^2}$
 $= \sqrt{1432^2 + 54.1^2}$
 = 1432.896 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008 / 260}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1433/81.6
 = 17.559

Bending Moment [Rm]:
 $= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 363 / (2 \cdot 2350) \cdot 602 \cdot 1433 / 2$
 = 33.277 Kg-m.

Compressive Allowable, $KL/r < C_c$ (17.6 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (17.6)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (17.6) / (8 \cdot 123) - (17.6)^3 / (8 \cdot 123^3))$
 = 149.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 15.4 / 150 + (33.3 \cdot 188 / 70312504) / 173$
 = 0.108

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/154
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 375/2
 = 187.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1910 - cos(1281/2)(40 + 0 + 0) - 0
 = 589.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(7051/154)}$$

$$= 67.724 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 589/67.7$$

$$= 8.697$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 363 / (2 \cdot 2350)$$

$$= 0.077 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 602 \cdot 589$$

$$= 27.357 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 27.4/376067$$

$$= 0.713 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (8.7) / (8 \cdot 123) - (8.7)^3 / (8 \cdot 123^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 21.2/153 + (27.4 \cdot 188/7051) / 173$$

$$= 0.142$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	128565.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	160.4 Kg
Maximum Transverse Force	Ft	359.8 Kg
Maximum Longitudinal Force	F1	362.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 363 / (3.04 \cdot 8)$$

$$= 1.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 360 / (3.04 \cdot 8)$$

$$= 1.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:



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= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (BoltShearAllowable \cdot Nbolts)$

= $363 / (103 \cdot 8)$

= 0.0431 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $1910 \cdot 360 + 0$

= 687.27 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $687 / 128565$

= 5.35 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0, 0.043, 0]$

= 0.0431 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.40	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	624.50	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		12053.29	Kgf
Pressure used in Analysis	P	688.286	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	16.87	234.00
Long. Stress at Bottom of Midspan	17.37	234.00



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Long. Stress at Top of Saddles	19.35	234.00
Long. Stress at Bottom of Saddles	15.85	234.00
Tangential Shear in Shell	1.83	138.58
Circ. Stress at Horn of Saddle	6.85	350.99
Circ. Compressive Stress in Shell	0.31	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (720/2 + 0) \cdot 1910/2350$$

$$= 877.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(363, 0) \cdot 1910/4320$$

$$= 160.4 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 11176 + \max(160, 877, 0, 0)$$

$$= 12053.3 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.73 \cdot 1.2)^2)$$

$$= 7113.507 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 13275.30 Kgf

Transverse Shear Load 359.82 Kgf

Longitudinal Shear Load 362.69 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2350.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1412 \quad K2 = 0.9003 \quad K3 = 0.5922 \quad K4 = 0.3269$$

$$K5 = 0.6971 \quad K6 = 0.0379 \quad K7 = 0.0379 \quad K8 = 0.3167$$

$$K9 = 0.2372 \quad K10 = 0.0432 \quad K1^* = 0.2486 \quad K6p = 0.0305$$

$$K7p = 0.0305$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)]$$

$$= -12053 \cdot 1440 [1 - (1 - 1440/7276 + (1253^2 - 624^2)/$$

$$(2 \cdot 1440 \cdot 7276))/(1 + (4 \cdot 624)/(3 \cdot 7276))]$$

$$= -3986.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 12053 \cdot 7276/4((1 + 2(1253^2 - 624^2)/(7276^2))/(1 + (4 \cdot 624)/$$

$$(3 \cdot 7276)) - 4 \cdot 1440/7276)$$

$$= 3195.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$



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$$= 688 \cdot 1253 / (2 \cdot 25.2) - 3195 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= 16.87 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1253 / (2 \cdot 25.2) + 3195 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= 17.37 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1253 / (2 \cdot 25.2) - 3987 / (0.14 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= 19.35 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1253 / (2 \cdot 25.2) + 3987 / (0.25 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= 15.85 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 12053 (7276 - 2 \cdot 1440) / (7276 + (4 \cdot 624 / 3))$$

$$= 6534.7 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 6535 / (1253 \cdot 25.2)$$

$$= 1.83 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1253 \cdot 25.2)}$$

$$= 138.624 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1253 \cdot 25.2)}, 2 \cdot 1440)$$

$$= 652.25 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 12053 \cdot 0.1 / (652(25.2 + 1 \cdot 15))$$

$$= -0.31 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -12053 / (4(25.2 + 1 \cdot 15)652) -$$

$$12 \cdot 0.038 \cdot 12053 \cdot 1253 / (7276(25.2 + 1 \cdot 15)^2)$$

$$= -6.85 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.300	



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Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1241 \cdot 25.2) + 710})^3) / 12$$

$$= 131.471 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 131 + 248 (233 - 12.6)^2$$

$$= 120971.883 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	985.9	25.2	12.6	248.4	313027.7	0.121E+06
Wearplate	710.0	15.0	32.7	106.5	348255.0	0.428E+05
Web	16.0	589.0	334.7	94.2	3154212.5	0.370E+05
BasePlate	400.0	40.0	649.2	160.0	10387199.0	0.277E+06
Totals	...	...	...	609.2	14202694.0	0.478E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14202694 / 609$$

$$= 233.147 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 140 / 2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 12053$$

$$= 2897.9827 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.7878 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1910 - 1241 \cdot \sin(140 / 2) / 1.22$$

$$= 955.640 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2769.4834 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.3249 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(12053 + 1222)400 / (4 \cdot 2350 \cdot 173))}$$

$$= 9.792 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140 / 2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$



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Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2407 / (5 - 1)$$

$$= 601.873 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 602 \cdot 400 / 2$$

$$= 1203.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 13275 / (2350 \cdot 400)$$

$$= 1.412 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1204 \cdot 1.41$$

$$= 1700.010 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 57.4 + 48.1$$

$$= 105.590 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1700 / 106$$

$$= 1.579 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 16$$

$$= 359.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 106$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (7031 / 106) }$$

$$= 81.603 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1910 - \cos(140 / 2) (1240 + 26 + 15) - 40$$

$$= 1431.872 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:



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$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(140/2) (1240 + 26 + 15) \\
 &= 2407.492 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 2350 - 50.8 \\
 &= 2299.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (2407 - 2299) / 2 \\
 &= 54.146 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{(Lv^2 + Dd^2)} \\
 &= \sqrt{(1432^2 + 54.1^2)} \\
 &= 1432.896 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)} \\
 &= 123.211
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1433 / 81.6 \\
 &= 17.559
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot Bplen) \cdot e \cdot L / 2 \\
 &= 363 / (2 \cdot 2350) \cdot 602 \cdot 1433 / 2 \\
 &= 33.277 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($17.6 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (17.6)^2 / (2 \cdot 123^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (17.6) / (8 \cdot 123) - (17.6^3) / (8 \cdot 123^3)) \\
 &= 149.6 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 1.58 / 150 + (33.3 \cdot 188 / 70312504) / 173 \\
 &= 0.016
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0 / 154 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 375 / 2 \\
 &= 187.500 \text{ mm}
 \end{aligned}$$

Length of Inner Rib [L]:



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= Saddle Height - Outside Radius - Bpthk
 = $1910 - \cos(1281/2)(40 + 0 + 0) - 0$
 = 589.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{7051/154}$
 = 67.724 mm

Slenderness ratio [KL/r]:
 = KL/r
 = $1 \cdot 589/67.7$
 = 8.697

Unit Force [Force,u]:
 = $Fl / (2 \cdot \text{Baseplate Length})$
 = $363 / (2 \cdot 2350)$
 = 0.077 Kg/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force $\cdot e \cdot L$
 = $0.077 \cdot 602 \cdot 589$
 = 27.357 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = $27.4/376067$
 = 0.713 N./mm²

Compressive Allowable, $KL/r < C_c$ ($8.7 < 123$) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (8.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (8.7) / (8 \cdot 123) - (8.7)^3 / (8 \cdot 123^3))$
 = 153.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 = $2.17/153 + (27.4 \cdot 188/7051) / 173$
 = 0.018

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0418 cm ²
Saddle Load Q0 (Weight)	Q0	12397.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	160.4 Kg
Maximum Transverse Force	Ft	359.8 Kg
Maximum Longitudinal Force	F1	362.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = $Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $363 / (3.04 \cdot 8)$
 = 1.5 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $360 / (3.04 \cdot 8)$



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= 1.5 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 363/(103 • 8)
= 0.0431 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 1910 • 360 + 0
= 687.27 Kg-m.

Eccentricity (e):
= Rmom / QO
= 687/12398
= 55.43 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.043, 0]
= 0.0431 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Ext. Pressure:

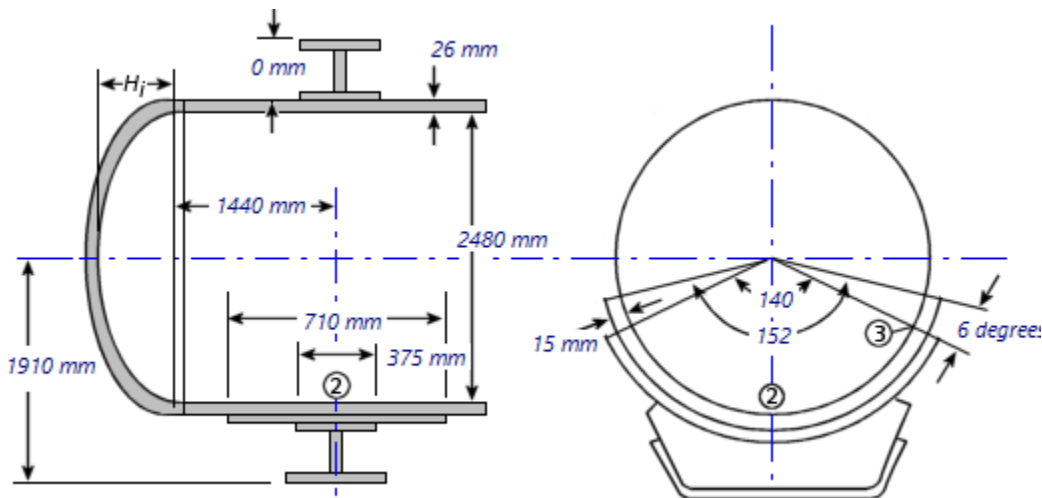
Step: 19 8:25am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.40	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	642.95	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	142603.64	Kgf
Pressure used in Analysis	P	-101.300 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-5.43	-207.36
Long. Stress at Top of Midspan	5.43	138.00
Long. Stress at Bottom of Midspan	0.39	138.00
Long. Stress at Top of Saddles	24.21	138.00
Long. Stress at Bottom of Saddles	-17.70	-207.36
Long. Stress at Bottom of Saddles	17.70	138.00



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

Tangential Shear in Shell	21.55	110.40
Circ. Stress at Horn of Saddle	81.10	172.50
Circ. Compressive Stress in Shell	3.72	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1439/2 + 0) \cdot 1910/2350$$

$$= 1754.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(725, 0) \cdot 1910/4320$$

$$= 320.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(8363, 0) \cdot 1910/4320$$

$$= 3697.6 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (8363/2 + 0) \cdot 1910/2350$$

$$= 10196.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 132408 + \max(321, 1755, 3698, 10196)$$

$$= 142603.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.73 \cdot 1.2)^2)$$

$$= 7113.507 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	143825.66	Kgf
Transverse Shear Load	4181.62	Kgf
Longitudinal Shear Load	8363.24	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2350.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -142604 \cdot 1440 [1 - (1 - 1440/7276 + (1253^2 - 643^2)/(2 \cdot 1440 \cdot 7276))/(1 + (4 \cdot 643)/(3 \cdot 7276))]$$

$$= -47852.8 \text{ Kg-m.}$$



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Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L)$$

$$= 142604 \cdot 7276 / 4 \cdot ((1 + 2(1253^2 - 643^2) / (7276^2)) / (1 + (4 \cdot 643) / (3 \cdot 7276)) - 4 \cdot 1440 / 7276)$$

$$= 36861.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) - 36862 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= -5.43 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) + 36862 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= 0.39 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) - 47853 / (0.14 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= 24.21 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) + 47853 / (0.25 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= -17.70 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 142604 \cdot (7276 - 2 \cdot 1440) / (7276 + (4 \cdot 643 / 3))$$

$$= 77078.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.9 \cdot 77078 / (1253 \cdot 25.2)$$

$$= 21.55 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1253 \cdot 25.2)}$$

$$= 138.624 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1253 \cdot 25.2)}, 2 \cdot 1440)$$

$$= 652.25 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 142604 \cdot 0.1 / (652(25.2 + 1 \cdot 15))$$

$$= -3.72 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -142604 / (4(25.2 + 1 \cdot 15)652) - 12 \cdot 0.038 \cdot 142604 \cdot 1253 / (7276(25.2 + 1 \cdot 15)^2)$$

$$= -81.10 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm



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Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1241 \cdot 25.2)} + 710)25.2^3)/12$$

$$= 131.471 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 131 + 248(233 - 12.6)^2$$

$$= 120971.883 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	985.9	25.2	12.6	248.4	313027.7	0.121E+06
Wearplate	710.0	15.0	32.7	106.5	348255.0	0.428E+05
Web	16.0	589.0	334.7	94.2	3154212.5	0.370E+05
BasePlate	400.0	40.0	649.2	160.0	10387199.0	0.277E+06
Totals	...	...	...	609.2	14202694.0	0.478E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14202694/609$$

$$= 233.147 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110)\cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 142604$$

$$= 34286.3164 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 9.3208 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1910 - 1241 \cdot \sin(140/2) / 1.22$$

$$= 955.640 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 32766.0293 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15.6754 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:



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$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(142604 + 1222) \cdot 400}{4 \cdot 2350 \cdot 154}}$$

$$= 34.239 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2407 / (5 - 1)$$

$$= 601.873 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 602 \cdot 400 / 2$$

$$= 1203.746 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 143826 / (2350 \cdot 400)$$

$$= 15.301 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1204 \cdot 15.3$$

$$= 18418.041 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 57.4 + 48.1$$

$$= 105.590 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 18418 / 106$$

$$= 17.106 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 16$$

$$= 359.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 106$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7031/106}$$

$$= 81.603 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1910 - \cos(140/2) (1240 + 26 + 15) - 40$$

$$= 1431.872 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140/2) (1240 + 26 + 15)$$

$$= 2407.492 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2407 - 2299) / 2$$

$$= 54.146 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1432^2 + 54.1^2}$$

$$= 1432.896 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 230)}$$

$$= 130.885$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1433 / 81.6$$

$$= 17.559$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 8363 / (2 \cdot 2350) \cdot 602 \cdot 1433 / 2$$

$$= 767.317 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($17.6 < 131$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (17.6)^2 / (2 \cdot 131^2)) 230 / (5/3 + 3 \cdot (17.6) / (8 \cdot 131) - (17.6^3) / (8 \cdot 131^3))$$

$$= 133 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 17.1 / 133 + (767 \cdot 188 / 70312504) / 154$$

$$= 0.259$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:



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= AY / A
= 0/154
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 375/2
= 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
= 1910 - cos(1281/2)(40 + 0 + 0) - 0
= 589.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7051/154}$
= 67.724 mm

Slenderness ratio [KL/r]:

= KL/r
= 1 • 589/67.7
= 8.697

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
= 8363/(2 • 2350)
= 1.779 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
= 1.78 • 602 • 589
= 630.821 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
= 631/376067
= 16.450 N./mm²

Compressive Allowable, KL/r < Cc (8.7 < 131) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (8.7)^2 / (2 \cdot 131^2)) 230 /$
 $(5/3 + 3 \cdot (8.7) / (8 \cdot 131) - (8.7^3) / (8 \cdot 131^3))$
= 135.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
= 23.5/136 + (631 • 188/7051)/154
= 0.280

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1253.40	mm
Shell Thickness used in this Case	t	26.000	mm
Stiffened Vessel Length per 4.15.6	L	7276.20	mm
Distance from Saddle to Vessel tangent	a1 or a	1440.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	710.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	15.0	mm



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	624.50	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1910.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		24480.86	Kgf
Pressure used in Analysis	P	-101.300	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.03	-207.36
Long. Stress at Top of Midspan	3.03	138.00
Long. Stress at Bottom of Midspan	-2.01	-207.36
Long. Stress at Bottom of Midspan	2.01	138.00
Long. Stress at Top of Saddles	2.00	138.00
Long. Stress at Bottom of Saddles	-5.09	-207.36
Long. Stress at Bottom of Saddles	5.09	138.00
Tangential Shear in Shell	3.71	110.40
Circ. Stress at Horn of Saddle	13.92	172.50
Circ. Compressive Stress in Shell	0.64	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1439/2 + 0) \cdot 1910/2350$
 $= 1754.7\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(725, 0) \cdot 1910/4320$
 $= 320.7\ Kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(8363, 0) \cdot 1910/4320$
 $= 3697.6\ Kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (8363/2 + 0) \cdot 1910/2350$
 $= 10196.0\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 14285 + \max(321, 1755, 3698, 10196)$
 $= 24480.9\ Kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.73 \cdot 1.2)^2)$
 $= 7113.507\ N$



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21, 2026

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25702.87	Kgf
Transverse Shear Load	4181.62	Kgf
Longitudinal Shear Load	8363.24	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length
= 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0379	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0305			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h_m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h_m)}{3L} \right) \right]$$

$$= -24481 \cdot 1440 \left[1 - \left(1 - \frac{1440}{7276} + \frac{(1253^2 - 624^2)}{(2 \cdot 1440 \cdot 7276)} \right) / \left(1 + \frac{(4 \cdot 624)}{3 \cdot 7276} \right) \right]$$

$$= -8097.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\left(1 + 2 \frac{(R_m^2 - h_m^2)}{(L^2)} \right) / \left(1 + \frac{(4h_m^2)}{3L} \right) - 4a/L \right)$$

$$= 24481 \cdot 7276 / 4 \left(\left(1 + 2 \frac{(1253^2 - 624^2)}{(7276^2)} \right) / \left(1 + \frac{(4 \cdot 624)}{3 \cdot 7276} \right) - 4 \cdot 1440 / 7276 \right)$$

$$= 6489.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) - 6490 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= -3.03 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) + 6490 / (\pi \cdot 1253^2 \cdot 25.2)$$

$$= -2.01 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) - 8098 / (0.14 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= 2.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -101 \cdot 1253 / (2 \cdot 25.2) + 8098 / (0.25 \cdot \pi \cdot 1253^2 \cdot 25.2)$$

$$= -5.09 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 24481 (7276 - 2 \cdot 1440) / (7276 + (4 \cdot 624 / 3))$$

$$= 13272.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 13272 / (1253 \cdot 25.2)$$

$$= 3.71 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

$$= 0.78 \cdot \sqrt{(1253 \cdot 25.2)}$$

$$= 138.624 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1253 \cdot 25.2)}, 2 \cdot 1440)$$

$$= 652.25 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 24481 \cdot 0.1 / (652(25.2 + 1 \cdot 15))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -24481 / (4(25.2 + 1 \cdot 15)652) - 12 \cdot 0.038 \cdot 24481 \cdot 1253 / (7276(25.2 + 1 \cdot 15)^2)$$

$$= -13.92 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	40.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1241 \cdot 25.2)} + 710)25.2^3) / 12$$

$$= 131.471 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 131 + 248(233 - 12.6)^2$$

$$= 120971.883 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	985.9	25.2	12.6	248.4	313027.7	0.121E+06
Wearplate	710.0	15.0	32.7	106.5	348255.0	0.428E+05
Web	16.0	589.0	334.7	94.2	3154212.5	0.370E+05
BasePlate	400.0	40.0	649.2	160.0	10387199.0	0.277E+06
Totals	...	...	...	609.2	14202694.0	0.478E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14202694/609$$

$$= 233.147 \text{ mm}$$

Angle [β]:



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 140/2 \\
 &= 110.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110)) \\
 &= 0.2404
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.24 \cdot 24481 \\
 &= 5885.9546 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 1.6001 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1910 - 1241 \cdot \sin(140/2) / 1.22 \\
 &= 955.640 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5624.9663 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 2.6910 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(24481 + 1222) \cdot 400) / (4 \cdot 2350 \cdot 154)} \\
 &= 14.474 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 140/2) (1240 + 26 + 15) \\
 &= 2407.492 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 2407 / (5 - 1) \\
 &= 601.873 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 602 \cdot 400 / 2 \\
 &= 1203.746 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 25703 / (2350 \cdot 400) \\
 &= 2.734 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 1204 \cdot 2.73 \\
 &= 3291.461 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 57.4 + 48.1 \\
 &= 105.590 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

= P/Ar
= 3291/106
= 3.057 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	375.0	...	60.0	...	0.703E+04

Rib dimension [D]:
= Saddle Width - Web Thickness
= 375 - 16
= 359.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/106
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 375/2
= 187.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7031/106}$
= 81.603 mm

Length of Outer Rib, step 1, [Lv]:
= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
= 1910 - cos(140/2)(1240 + 26 + 15) - 40
= 1431.872 mm

Length of Outer Rib, step 2, [Lw]:
= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
= 2 • sin(140/2)(1240 + 26 + 15)
= 2407.492 mm

Length of Outer Rib, step 3, [Bd]:
= Baseplate length - clearance
= 2350 - 50.8
= 2299.200 mm

Length of Outer Rib, step 4, [Dd]:
= (Lw - Bd)/2
= (2407 - 2299)/2
= 54.146 mm

Length of Outer Rib [L]:
= $\sqrt{Lv^2 + Dd^2}$
= $\sqrt{1432^2 + 54.1^2}$
= 1432.896 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/230}$
= 130.885

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1433/81.6



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

= 17.559

Bending Moment [Rm]:

= $F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

= $8363 / (2 \cdot 2350) \cdot 602 \cdot 1433 / 2$

= 767.317 Kg-m.

Compressive Allowable, $KL/r < C_c$ ($17.6 < 131$) per AISC E2-1 [Sca]:

= $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$

= $(1 - (17.6)^2 / (2 \cdot 131^2)) 230 /$

$(5/3 + 3 \cdot (17.6) / (8 \cdot 131) - (17.6^3) / (8 \cdot 131^3))$

= 133 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$

= $3.06 / 133 + (767 \cdot 188 / 70312504) / 154$

= 0.154

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	359.0	0.0	57.4	0.0	0.703E+04
Web	601.9	16.0	0.0	96.3	0.0	20.5
Totals	...	...	...	153.7	...	0.705E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/154

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375/2

= 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= $1910 - \cos(1281/2) (40 + 0 + 0) - 0$

= 589.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{7051/154}$

= 67.724 mm

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 589 / 67.7$

= 8.697

Unit Force [Force,u]:

= $F_l / (2 \cdot \text{Baseplate Length})$

= $8363 / (2 \cdot 2350)$

= 1.779 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$

= $1.78 \cdot 602 \cdot 589$

= 630.821 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 631/376067



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Saddle Calcs: Ext. Pressure: Step: 19 8:25am Apr 21,2026

= 16.450 N./mm²

Compressive Allowable, $KL/r < C_c$ (8.7 < 131) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(8.7)²/(2 • 131²))230/

(5/3+3•(8.7)/(8• 131)-(8.7³)/(8•131³)

= 135.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_1/I)/S_{ba}$

= 4.2/136 + (631 • 188/7051)/154

= 0.138

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: Manway

Nozl: 15 8:25am Apr 21,2026

Input, Nozzle Desc: Manway

From: 10

Pressure for Reinforcement Calculations	P	5076.845	KPa.
Temperature for Internal Pressure	Temp	245	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	2480.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	45.9000	mm
Head Internal Corrosion Allowance	c	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	43.3013	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		45.9000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



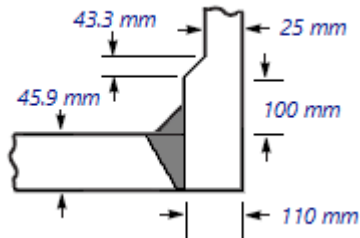
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Nozzle Calcs.: Manway

Nozl: 15 8:25am Apr 21,2026



Reinforcement CALCULATION, Description: Manway

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.984 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K_1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 5077 \cdot 2480 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 5077)$
 $= 41.2110 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5077 \cdot 305 / (136 \cdot 1 + 0.4 \cdot 5077)$
 $= 11.1953 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 85 / 0.5773$
 $= 147.2243 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1119.2000 mm
Parallel to Vessel Wall, opening length	d	559.6000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	114.7500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.987$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.987$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.99, 1)$
 $= 0.987$



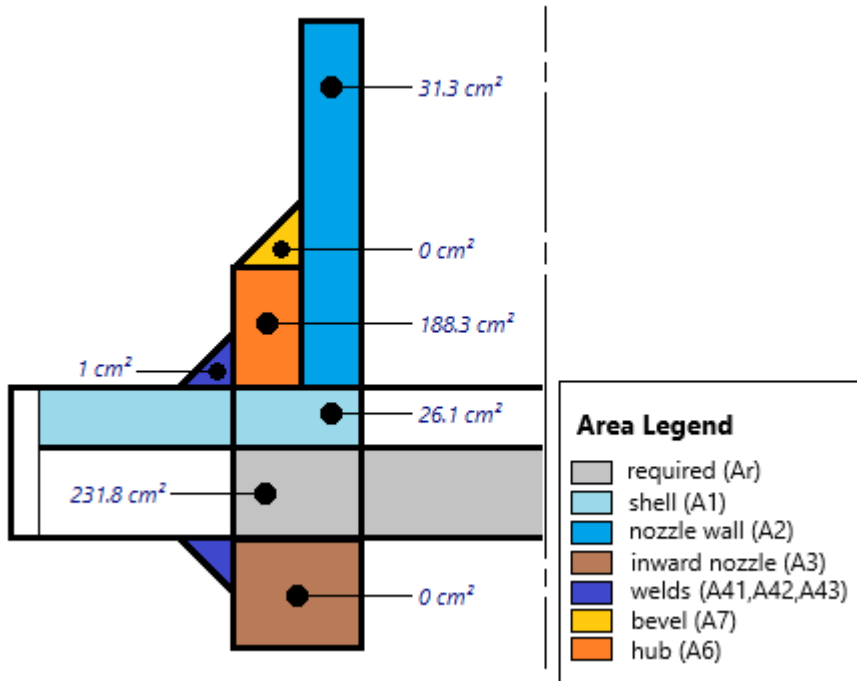
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Nozzle Calcs.: Manway

Noz1: 15 8:25am Apr 21,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	231.799	NA	NA
Area in Shell	A1	26.105	NA	NA
Area in Nozzle Wall	A2	31.269	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		0.987	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	188.317	NA	NA
TOTAL AREA AVAILABLE	Atot	246.678	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (560 \cdot 41.2 \cdot 1 + 2 \cdot 110 \cdot 41.2 \cdot 1 \cdot (1 - 0.99))$$

$$= 231.799 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 560(1 \cdot 45.9 - 1 \cdot 41.2) - 2 \cdot 110$$

$$(1 \cdot 45.9 - 1 \cdot 41.2) \cdot (1 - 0.99)$$

$$= 26.105 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 115)(25 - 11.2)0.99$$

$$= 31.269 \text{ cm}^2$$



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Nozzle Calcs.: Manway

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 0.987 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(115, 350, 100)) \cdot (110 - 25) \cdot 0.99$$

$$= 188.317 \text{ cm}^2 \text{ Includes Bevel Area of } 20.533 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 11.1953 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 11.1953 \text{ mm}$$

Available Nozzle Neck Thickness = 25.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 25$, $t_r = 11.1$, $c = 0 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.44$, Temp. Reduction = 41 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 45.9$, $t_r = 41.2$, $c = 0 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	34 °C
Min Metal Temp. at Required thickness (UCS 66.1)	29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 5076.84 / 10200.00 = 0.498$$

Weld Size Calculations, Description: Manway

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (232 - 26.1 + 2 \cdot 110 \cdot 0.99 \cdot (1 \cdot 45.9 - 41.2)) 138)$$

$$= 303769.50 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.



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Nozzle Calcs.: Manway

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Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot \text{Sv}$$

$$= (220 + 0.99 - 0 \cdot 0.99) \cdot 138$$

$$= 310380.06 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{Sv}$$

$$= (220 + 0 + 0.99 + (99.7)) \cdot 138$$

$$= 450622.28 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{S}$$

$$= (220 + 0 + 0.99 + 0 + (99.7)) \cdot 138$$

$$= 450622.28 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 136$$

$$= 65163. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo}) / 4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn}$$

$$= (3.14 \cdot 292) \cdot (110 - 0) \cdot 0.7 \cdot 136$$

$$= 982003. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot (\text{T} - \text{cas}) \cdot 0.74 \cdot \text{Sng}$$

$$= (3.14/2.0) \cdot 610 \cdot (45.9 - 0) \cdot 0.74 \cdot 138$$

$$= 457669. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (65163 + 982003) = 1047166 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (65163 + 0 + 457669 + 0) = 522833 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (65163 + 457669 + 0) = 522833 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1047166 Kgf, must exceed W = 303769 Kgf	or W1 = 310380 Kgf
Path 2-2 = 522832 Kgf, must exceed W = 303769 Kgf	or W2 = 450622 Kgf
Path 3-3 = 522832 Kgf, must exceed W = 303769 Kgf	or W3 = 450622 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5065 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 20.5679 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 416.4678 mm

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Nozzle Calcs.: N5- Tube inlet

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Input, Nozzle Desc: N5- Tube inlet

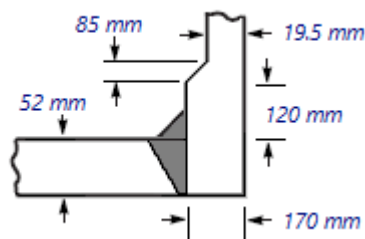
From: 20

Pressure for Reinforcement Calculations	P	5064.827	KPa.
Temperature for Internal Pressure	Temp	245	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Shell Finished (Minimum) Thickness	t	52.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	19.5000	mm
Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	85.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		170.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	415.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N5- Tube inlet



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Nozzle Calcs.: N5- Tube inlet Nozl: 16 8:25am Apr 21,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 0.768 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5065 \cdot 1240 / (138 \cdot 1 - 0.6 \cdot 5065)$
= 46.5384 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5065 \cdot 457 / (136 \cdot 1 - 0.6 \cdot 5065)$
= 17.3910 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $150 / 0.5773$
= 260.6737 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1828.7999	mm
Parallel to Vessel Wall, opening length	d	914.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	130.0000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 136 / 138)$
= 0.987

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 136 / 138)$
= 0.987

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.99, 1)$
= 0.987



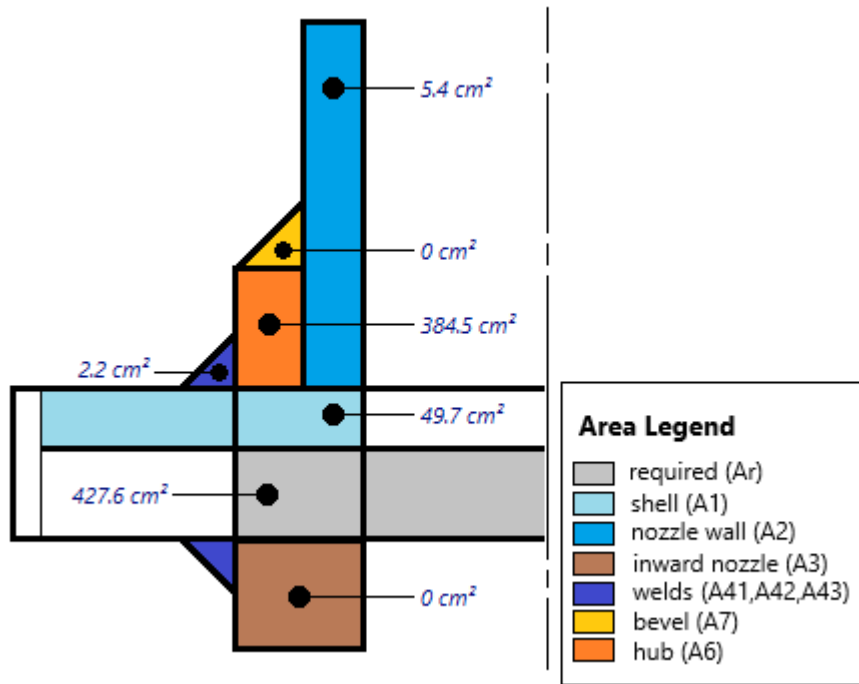
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Nozzle Calcs.: N5- Tube inlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	427.610	NA	NA
Area in Shell	A1	49.699	NA	NA
Area in Nozzle Wall	A2	5.412	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.221	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	384.451	NA	NA
TOTAL AREA AVAILABLE	Atot	441.782	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (914 \cdot 46.5 \cdot 1 + 2 \cdot 170 \cdot 46.5 \cdot 1 \cdot (1 - 0.99))$$

$$= 427.610 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 914(1 \cdot 52 - 1 \cdot 46.5) - 2 \cdot 170$$

$$(1 \cdot 52 - 1 \cdot 46.5) \cdot (1 - 0.99)$$

$$= 49.699 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 130)(19.5 - 17.4)0.99$$

$$= 5.412 \text{ cm}^2$$



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Nozzle Calcs.: N5- Tube inlet Nozl: 16 8:25am Apr 21,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 2.221 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(130, 415, 120)) \cdot (170 - 19.5) \cdot 0.99$$

$$= 384.451 \text{ cm}^2 \text{ Includes Bevel Area of } 27.960 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.3910 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 46.5384 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 46.5384 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.33, \max(46.5, 2.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.4, 8.33)$$

$$= 17.3910 \text{ mm}$$

Available Nozzle Neck Thickness = 19.5000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 74.2,	Allowable	: 136.2 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 268.5 N./mm ²	Passed
Occasional	: 58.1,	Allowable	: 181.1 N./mm ²	Passed
Shear	: 10.2,	Allowable	: 95.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 52, tr = 46.5, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.89, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	31 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 31 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 150 / 0.5773$$

$$= 260.6737 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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Nozzle Calcs.: N5- Tube inlet Nozl: 16 8:25am Apr 21,2026

Parallel to Vessel Wall (Diameter Limit) D1 1371.6000 mm
 Parallel to Vessel Wall, opening length d 685.8000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 130.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		285.088	285.088
Area Available in Shell A1		24.728	24.728
Area Available in Nozzle Wall A2		5.412	5.412
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		2.221	2.221
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		384.451	384.451
TOTAL AREA AVAILABLE Atot		416.812	416.812

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N5- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (428 - 49.7 + 2 \cdot 170 \cdot 0.99 \cdot (1 \cdot 52 - 46.5)) \cdot 138)$
 $= 557570.19 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (390 + 2.22 - 0 \cdot 0.99) \cdot 138$
 $= 551723.19 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (390 + 0 + 2.22 + (174)) \cdot 138$
 $= 797265.06 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (390 + 0 + 2.22 + 0 + (174)) \cdot 138$
 $= 797265.06 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 1254 \cdot 15 \cdot 0.49 \cdot 136$
 $= 201134. \text{ Kgf}$



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Nozzle Calcs.: N5- Tube inlet Nozl: 16 8:25am Apr 21,2026

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 542) \cdot (170 - 0) \cdot 0.7 \cdot 136$$

$$= 2815138. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1254 \cdot (48 - 0) \cdot 0.74 \cdot 138$$

$$= 984853. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (201134 + 2815137) = 3016272 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (201134 + 0 + 984853 + 0) = 1185987 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (201134 + 984853 + 0) = 1185987 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 3016272 Kgf, must exceed W = 557570 Kgf or W1 = 551723 Kgf

Path 2-2 = 1185987 Kgf, must exceed W = 557570 Kgf or W2 = 797265 Kgf

Path 3-3 = 1185987 Kgf, must exceed W = 557570 Kgf or W3 = 797265 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5143 KPa.

The Drop for this Nozzle is : 170.3178 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 637.3178 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N5- Tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	52.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	170.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



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Nozzle Calcs.: N5- Tube inlet

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Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5064.83	KPa.
Radial Load	P	-8260.24	Kgf
Circumferential Shear	Vc	5965.75	Kgf
Longitudinal Shear	Vl	5965.75	Kgf
Circumferential Moment	Mc	16514.19	Kg-m.
Longitudinal Moment	Ml	16514.19	Kg-m.
Torsional Moment	Mt	18991.26	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.379) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1254.4 + 2 \cdot 1.65 \cdot \sqrt{1266.0(52.0 - 0.0)}$$

$$= 2101.106 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2532.000 mm
Nozzle Outside Diameter	(d) =	1254.400 mm
Vessel Thickness used	(T) =	52.000 mm
Nozzle Thickness used	(t) =	170.000 mm
T / t	=	0.306
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.457$$

Nr/P	=	0.056
Mr/P	=	0.092
M0/P	=	0.030
N0/P	=	0.067
MrD/Mc	=	0.260
NrDL/Mc	=	0.094
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.080
NrDL/Ml	=	0.052
M0D/Ml	=	0.023
N0DL/Ml	=	0.075

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-6	6	6	-6
Normal Pressure Stress	118	123	118	123



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Nozzle Calcs.: N5- Tube inlet Nozl: 16 8:25am Apr 21,2026

Outplane Stress Summary	115	123	135	116
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	59	59	59	59

Inplane Stress Summary	52	64	102	23
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	115	123	135	117
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59

Outplane Stress Summary	40	74	92	37
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-74	74	74	-74
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	118	123	118	123

Inplane Stress Summary	57	178	215	38
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Nozzle Calcs.: N5- Tube inlet

Nozl: 16 8:25am Apr 21,2026

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1
Shear Stress Summary		0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		57	178	215	39

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)		2	2	2	2
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-3	-3	3	3
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		13	19	13	19
Hoop Stress Summary		12	18	19	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)		0	0	0	0
Axial Bending (P)		1	-1	1	-1
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-1	-1	1	1
Axial Bending (ML)		-1	1	1	-1
Axial Pressure Stress		6	6	6	6
Axial Stress Summary		5	6	10	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)		Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)		0	0	0	0
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		0	0	0	0
Shear Stress Summary		0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)		Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress		12	18	19	25

Nozzle Stresses (N./mm²):



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Nozzle Calcs.: N5- Tube inlet

Nozl: 16 8:25am Apr 21,2026

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	19	13	19
Hoop Stress Summary	11	17	19	25

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	1	10	14	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11	17	19	25

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	118	123	118	123	118	123	118	123
Circ. Pl (SUS)	-1	-1	5	5	-3	-3	5	5
Circ. Q (SUS)	-1	1	11	-11	-58	58	90	-90
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	-1	-1	3	3	-2	-2	6	6
Long. Q (SUS)	-6	6	38	-38	-17	17	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	118.0	123.0	118.0	123.0	118.0	123.0	118.0	123.0



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Nozzle Calcs.: N5- Tube inlet

Nozl: 16

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Pm+Pl (SUS)	117.0	122.0	123.0	128.0	115.0	120.0	123.0	128.0
Pm+Pl+Q (Total)	116.0	123.0	134.0	117.0	57.1	178.0	213.0	39.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.00	138.00	Passed
Pm+Pl (SUS)	128.00	207.00	Passed
Pm+Pl+Q (TOTAL)	213.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	19	13	19	13	19	13	19
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	19.0	13.0	19.0	13.0	19.0	13.0	19.0
Pm+Pl (SUS)	12.0	18.0	18.0	24.0	11.0	17.0	19.0	25.0
Pm+Pl+Q (Total)	12.0	18.0	18.0	24.0	11.0	17.0	19.0	25.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	136.20	Passed
Pm+Pl (SUS)	25.00	204.30	Passed
Pm+Pl+Q (TOTAL)	25.00	408.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N6- Tube outlet

Nozl: 17 8:25am Apr 21,2026

Input, Nozzle Desc: N6- Tube outlet

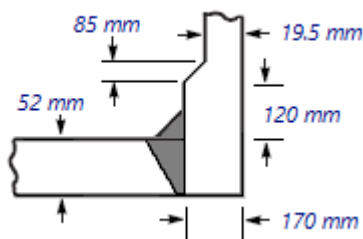
From: 20

Pressure for Reinforcement Calculations	P	5088.999	KPa.
Temperature for Internal Pressure	Temp	245	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Shell Finished (Minimum) Thickness	t	52.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1300.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	19.5000	mm
Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	85.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		170.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	415.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N6- Tube outlet



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 36.000 in.
Actual Thickness Used in Calculation 0.768 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5089 \cdot 1240 / (138 \cdot 1 - 0.6 \cdot 5089)$
= 46.7655 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $5089 \cdot 457 / (136 \cdot 1 - 0.6 \cdot 5089)$
= 17.4759 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $150 / 0.5773$
= 260.6737 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1828.7999	mm
Parallel to Vessel Wall, opening length	d	914.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	130.0000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $136 / 138$)
= 0.987

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $136 / 138$)
= 0.987

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.99, 1)
= 0.987



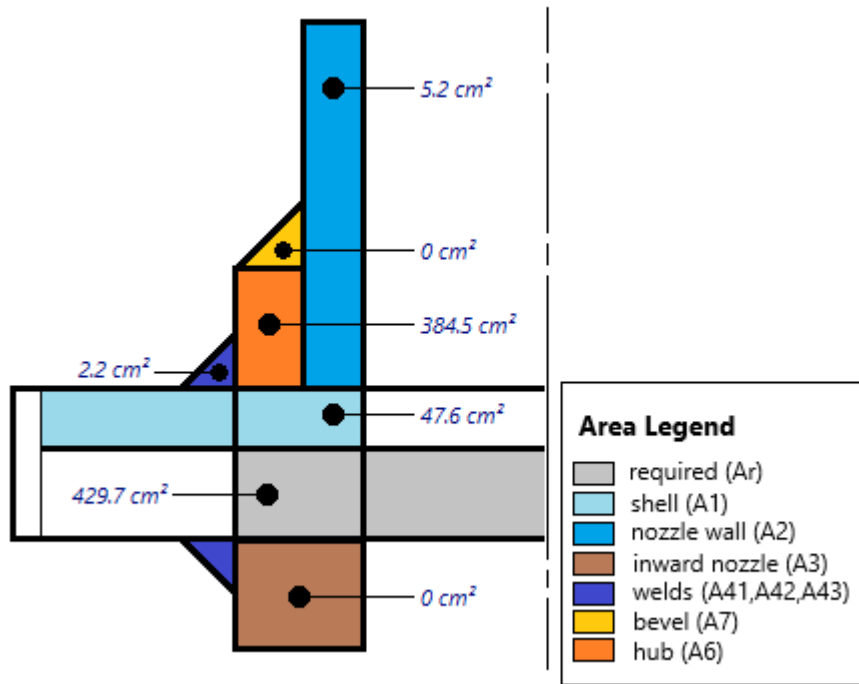
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Nozzle Calcs.: N6- Tube outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	429.697	NA	NA
Area in Shell	A1	47.632	NA	NA
Area in Nozzle Wall	A2	5.194	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.221	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	384.451	NA	NA
TOTAL AREA AVAILABLE	Atot	439.498	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (914 \cdot 46.8 \cdot 1 + 2 \cdot 170 \cdot 46.8 \cdot 1 \cdot (1 - 0.99))$$

$$= 429.697 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 914(1 \cdot 52 - 1 \cdot 46.8) - 2 \cdot 170$$

$$(1 \cdot 52 - 1 \cdot 46.8) \cdot (1 - 0.99)$$

$$= 47.632 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 130)(19.5 - 17.5)0.99$$

$$= 5.194 \text{ cm}^2$$



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 15^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 2.221 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(130, 415, 120)) \cdot (170 - 19.5) \cdot 0.99$$

$$= 384.451 \text{ cm}^2 \text{ Includes Bevel Area of } 27.960 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.4759 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 46.7655 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 46.7655 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.33, \max(46.8, 2.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.5, 8.33)$$

$$= 17.4759 \text{ mm}$$

Available Nozzle Neck Thickness = 19.5000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 76.5,	Allowable	: 136.2 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 266.2 N./mm ²	Passed
Occasional	: 58.4,	Allowable	: 181.1 N./mm ²	Passed
Shear	: 11.2,	Allowable	: 95.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 52, tr = 46.8, c = 0 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 32 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 150 / 0.5773$$

$$= 260.6737 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

Parallel to Vessel Wall (Diameter Limit) D1 1371.6000 mm
 Parallel to Vessel Wall, opening length d 685.8000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 130.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		286.479	286.479
Area Available in Shell A1		23.700	23.700
Area Available in Nozzle Wall A2		5.194	5.194
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		2.221	2.221
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		384.451	384.451
TOTAL AREA AVAILABLE Atot		415.566	415.566

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N6- Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 10.6050 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (430 - 47.6 + 2 \cdot 170 \cdot 0.99 \cdot (1 \cdot 52 - 46.8)) \cdot 138)$
 $= 562342.56 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (390 + 2.22 - 0 \cdot 0.99) \cdot 138$
 $= 551416.62 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (390 + 0 + 2.22 + (174)) \cdot 138$
 $= 796958.44 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (390 + 0 + 2.22 + 0 + (174)) \cdot 138$
 $= 796958.44 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 1254 \cdot 15 \cdot 0.49 \cdot 136$
 $= 201134. \text{ Kgf}$



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 542) \cdot (170 - 0) \cdot 0.7 \cdot 136$$

$$= 2815138. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1254 \cdot (48 - 0) \cdot 0.74 \cdot 138$$

$$= 984853. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (201134 + 2815137) = 3016272 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (201134 + 0 + 984853 + 0) = 1185987 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (201134 + 984853 + 0) = 1185987 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 3016272 Kgf, must exceed W = 562342 Kgf or W1 = 551416 Kgf

Path 2-2 = 1185987 Kgf, must exceed W = 562342 Kgf or W2 = 796958 Kgf

Path 3-3 = 1185987 Kgf, must exceed W = 562342 Kgf or W3 = 796958 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5143 KPa.

The Drop for this Nozzle is : 170.3178 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 637.3178 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N6- Tube outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	52.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	170.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.20	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5089.00	KPa.
Radial Load	P	-8260.24	Kgf
Circumferential Shear	Vc	6297.15	Kgf
Longitudinal Shear	Vl	6297.15	Kgf
Circumferential Moment	Mc	18400.06	Kg-m.
Longitudinal Moment	Ml	18400.06	Kg-m.
Torsional Moment	Mt	21160.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (7.379) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1254.4 + 2 \cdot 1.65 \cdot \sqrt{1266.0(52.0 - 0.0)}$$

$$= 2101.106 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2532.000 mm
Nozzle Outside Diameter	(d) =	1254.400 mm
Vessel Thickness used	(T) =	52.000 mm
Nozzle Thickness used	(t) =	170.000 mm
T / t	=	0.306
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.457$$

Nr/P	=	0.056
Mr/P	=	0.092
M0/P	=	0.030
N0/P	=	0.067
MrD/Mc	=	0.260
NrDL/Mc	=	0.094
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.080
NrDL/Ml	=	0.052
M0D/Ml	=	0.023
N0DL/Ml	=	0.075

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-7	7	7	-7
Normal Pressure Stress	118	123	118	123



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

Outplane Stress Summary	114	123	137	117
-------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	59	59	59	59

Inplane Stress Summary	49	67	105	21
------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	114	124	137	117
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-24	24	24	-24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59

Outplane Stress Summary	37	76	96	35
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-83	83	83	-83
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	118	123	118	123

Inplane Stress Summary	49	187	225	30
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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1
Shear Stress Summary		1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		49	187	225	36

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		2	2	2	2
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-3	-3	3	3
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		13	19	13	19
Hoop Stress Summary		11	17	19	25

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		0	0	0	0
Axial Bending (P)		1	-1	1	-1
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-1	-1	1	1
Axial Bending (ML)		-1	1	1	-1
Axial Pressure Stress		6	6	6	6
Axial Stress Summary		5	6	11	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		0	0	0	0
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		0	0	0	0
Shear Stress Summary		0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		11	18	19	25

Nozzle Stresses (N./mm²):



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Nozzle Calcs.: N6- Tube outlet Nozl: 17 8:25am Apr 21,2026

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	19	13	19
Hoop Stress Summary	11	17	20	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	0	11	15	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	11	17	20	26

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	118	123	118	123	118	123	118	123
Circ. Pl (SUS)	-1	-1	5	5	-3	-3	5	5
Circ. Q (SUS)	-2	2	12	-12	-67	67	99	-99
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	-1	-1	3	3	-2	-2	6	6
Long. Q (SUS)	-9	9	41	-41	-19	19	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	118.0	123.0	118.0	123.0	118.0	123.0	118.0	123.0



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Nozzle Calcs.: N6- Tube outlet

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Pm+Pl (SUS)	117.0	122.0	123.0	128.0	115.0	120.0	123.0	128.0
Pm+Pl+Q (Total)	115.0	124.0	135.0	116.0	48.1	187.0	222.0	36.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.00	138.00	Passed
Pm+Pl (SUS)	128.00	207.00	Passed
Pm+Pl+Q (TOTAL)	222.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	19	13	19	13	19	13	19
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-5	5	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	13.0	19.0	13.0	19.0	13.0	19.0	13.0	19.0
Pm+Pl (SUS)	12.0	18.0	18.0	24.0	11.0	17.0	19.0	25.0
Pm+Pl+Q (Total)	12.0	18.0	18.0	24.0	11.0	17.0	19.0	25.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	136.20	Passed
Pm+Pl (SUS)	25.00	204.30	Passed
Pm+Pl+Q (TOTAL)	25.00	408.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N1- Shell inlet Nozl: 18 8:25am Apr 21,2026

Input, Nozzle Desc: N1- Shell inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.810	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	7482.8965	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5475.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	523.8500	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



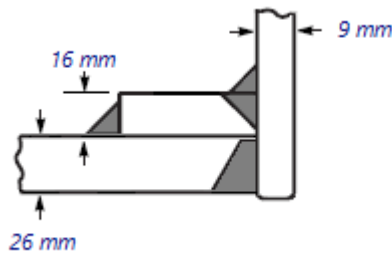
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Nozzle Calcs.: N1- Shell inlet

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Reinforcement CALCULATION, Description: N1- Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.040 in.
Actual Thickness Used in Calculation	0.355 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1001 \cdot 1241 / (138 \cdot 1 - 0.6 \cdot 1001)$
 $= 9.0386 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1001 \cdot 153 / (118 \cdot 1 - 0.6 \cdot 1001)$
 $= 1.3036 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9241 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	611.6066 mm
Parallel to Vessel Wall, opening length	d	305.8033 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		38.5584 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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Nozzle Calcs.: N1- Shell inlet

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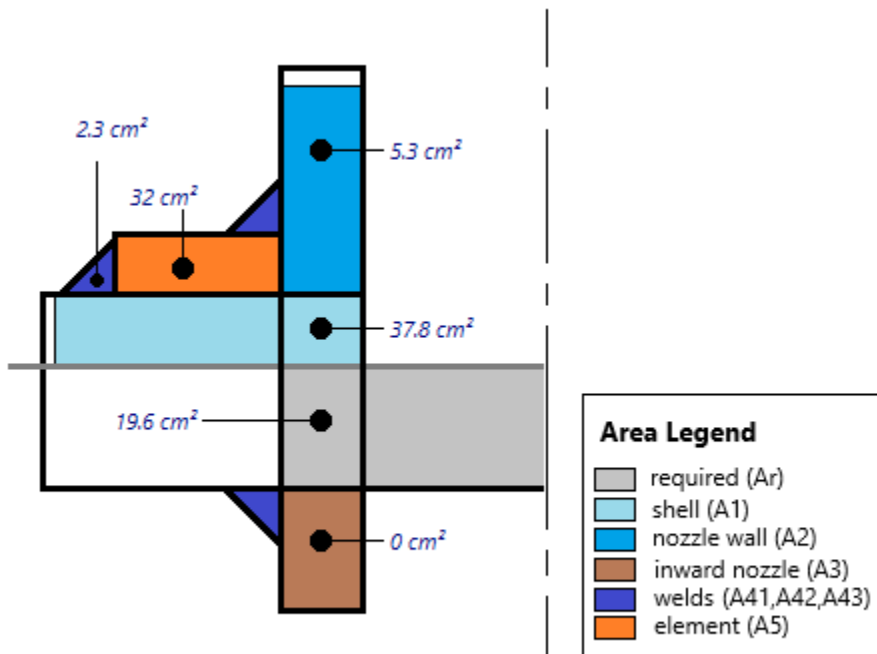


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	27.877	19.620	NA
Area in Shell	A1	48.999	37.828	NA
Area in Nozzle Wall	A2	5.090	5.341	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	32.000	32.000	NA
TOTAL AREA AVAILABLE	Atot	88.385	77.464	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

= $0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)

= $0.5(306 \cdot 12.7 \cdot 1 + 2 \cdot 9.02 \cdot 12.7 \cdot 1(1 - 0.86))$

= 19.620 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= $d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$

= $306(1 \cdot 25.2 - 1 \cdot 12.7) - 2 \cdot 9.02$

$(1 \cdot 25.2 - 1 \cdot 12.7) \cdot (1 - 0.86)$

= 37.828 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:

= $(2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$

= $(2 \cdot 38.6) \cdot (9.02 - 0.92) \cdot 0.86$



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Nozzle Calcs.: N1- Shell inlet Nozl: 18 8:25am Apr 21,2026

= 5.341 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$

= 2.295 cm²

Area Available in Element [A5]:

= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$

= $(524 - 324) \cdot 16 \cdot 1$

= 32.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.3036$ mm

Wall Thickness per UG16(b), $tr_{16b} = 3.3000$ mm

Wall Thickness, shell/head, internal pressure $trb_1 = 9.8386$ mm

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 9.8386$ mm

Wall Thickness, shell/head, external pressure $trb_2 = 1.7113$ mm

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.3000$ mm

Wall Thickness per table UG-45 $tb_3 = 8.3312$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[8.33, \max(9.84, 3.3)]$

= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(ta, tb)$

= $\max(1.3, 8.33)$

= 8.3312 mm

Available Nozzle Neck Thickness = 9.0233 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.2 N./mm ²	Passed
Occasional	: 8.2,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 18.5,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: N1- Shell inlet Nozl: 18 8:25am Apr 21,2026

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $c = 0.8$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 71 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.81 / 1960.01 = 0.511$

Weld Size Calculations, Description: N1- Shell inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 9.0233 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$8.0000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (19.6 - 37.8 + 2 \cdot 9.02 \cdot 0.86 \cdot (1 \cdot 25.2 - 12.7)) \cdot 138)$
 $= \max(0, -22912.06) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5.34 + 32 + 2.3 - 0 \cdot 0.86) \cdot 138$
 $= 55773.72 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5.34 + 0 + 0.86 + (3.89)) \cdot 138$
 $= 14190.36 \text{ Kg}$



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Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (5.34 + 0 + 2.3 + 32 + (3.89)) \cdot 138$$

$$= 61245.70 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 524 \cdot 12 \cdot 0.49 \cdot 138$$

$$= 68084. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 157) \cdot (9.02 - 0) \cdot 0.7 \cdot 118$$

$$= 37584. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 324 \cdot 16 \cdot 0.74 \cdot 138$$

$$= 84754. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 324 \cdot (15 - 0) \cdot 0.74 \cdot 138$$

$$= 79456. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (68084 + 37584) = 105668 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (29992 + 84754 + 79456 + 0) = 194202 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (68084 + 79456 + 0) = 147541 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 105668 Kgf, must exceed W = 0 Kgf	or W1 = 55773 Kgf
Path 2-2 = 194202 Kgf, must exceed W = 0 Kgf	or W2 = 14190 Kgf
Path 3-3 = 147540 Kgf, must exceed W = 0 Kgf	or W3 = 61245 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2070 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 10.6180 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 286.6179 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 281.52 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 305.80 mm



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Input Echo, WRC297 Item 1, Description: N1- Shell inle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.8000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	305.803	mm
Nozzle Thickness	9.0233	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	523.850	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1000.81	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 323.85 + 2 \cdot 1.65 \cdot \sqrt{1261.4(26.0 - 0.8)} \\
 &= 912.207 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2522.800	mm
Nozzle Outside Diameter	(d) =	323.850	mm
Vessel Thickness used	(T) =	41.200	mm
Nozzle Thickness used	(t) =	9.023	mm
T / t	=	4.566	
d / t	=	35.890	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.005$$



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Nr/P = 0.060
 Mr/P = 0.058
 MØ/P = 0.103
 NØ/P = 0.291
 MrD/Mc = 0.106
 NrDL/Mc = 0.098
 MØd/Mc = 0.248
 NØDL/Mc = 0.230
 MrD/Ml = 0.091
 NrDL/Ml = 0.086
 MØD/Ml = 0.185
 NØDL/Ml = 0.445

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	29	30	29	30
Outplane Stress Summary	6	41	77	11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	14	14	14	14
Inplane Stress Summary	5	21	34	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	9	42	77	11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14

Outplane Stress Summary	-12	39	66	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	29	30	29	30

Inplane Stress Summary	18	39	51	14
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	31	41	67	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-10	-10	10	10
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20

Hoop Stress Summary	13	14	33	35
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
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Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-136	136	136	-136
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-96	84	212	-151

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	114	87	213	186

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20
Hoop Stress Summary	18	19	28	30

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-162	162	162	-162
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-123	111	239	-177

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12
Shear Stress Summary	8	8	15	15



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	142	111	240	209

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{1253.4(26.0 - 0.8)}$$

$$= 910.338 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2506.800 mm
Nozzle Outside Diameter	(d) =	523.850 mm
Vessel Thickness used	(T) =	25.200 mm
Nozzle Thickness used	(t) =	109.023 mm
T / t	=	0.231
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.084$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.068
Mr/P	=	0.141
M0/P	=	0.041
N0/P	=	0.079
MrD/Mc	=	0.286
NrDL/Mc	=	0.091
M0d/Mc	=	0.088
N0DL/Mc	=	0.078
MrD/Ml	=	0.108
NrDL/Ml	=	0.083
M0D/Ml	=	0.033
N0DL/Ml	=	0.102

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	48	49	48	49
Outplane Stress Summary	48	47	71	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3



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Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	24	24	24	24
Inplane Stress Summary	29	18	84	-24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	49	47	84	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24
Outplane Stress Summary	13	35	58	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-65	65	65	-65
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	49	48	49
Inplane Stress Summary	12	83	149	-39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	13	83	149	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	-7	-7	13	13	-2	-2	2	2
Circ. Q (SUS)	-17	17	33	-33	-10	10	18	-18
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-1	-1	1	1	-2	-2	8	8
Long. Q (SUS)	-8	8	16	-16	-26	26	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	22.0	23.0	42.0	43.0	27.0	28.0	31.0	32.0
Pm+Pl+Q (Total)	7.0	40.2	75.1	11.7	31.3	40.0	64.3	34.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	138.00	Passed
Pm+Pl (SUS)	43.00	207.00	Passed
Pm+Pl+Q (TOTAL)	75.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	49	48	49	48	49	48	49
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Circ. Q (SUS)	1	-1	15	-15	-36	36	94	-94
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-1	-1	5	5	0	0	4	4
Long. Q (SUS)	5	-5	53	-53	-11	11	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	48.0	49.0	48.0	49.0	48.0	49.0	48.0	49.0



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Pm+Pl (SUS)	47.0	48.0	53.0	54.0	47.0	48.0	53.0	54.0
Pm+Pl+Q (Total)	48.0	47.0	82.1	63.0	13.4	84.0	147.0	41.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+Pl (SUS)	54.00	207.00	Passed
Pm+Pl+Q (TOTAL)	147.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-7	-7	13	13	-2	-2	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-91	91	181	-181	-117	117	207	-207
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	19.9	20.9	33.9	34.6	24.7	25.7	31.2	31.5
Pm+Pl+Q (Total)	114.0	87.0	210.5	185.9	142.1	110.9	237.1	208.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	118.00	Passed
Pm+Pl (SUS)	34.61	177.00	Passed
Pm+Pl+Q (TOTAL)	237.07	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Input, Nozzle Desc: N2- Shell inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.810	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	7482.8965	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7475.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	15.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	523.8500	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



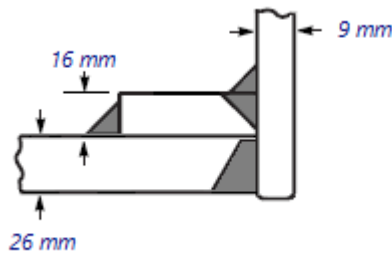
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Nozzle Calcs.: N2- Shell inlet

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Reinforcement CALCULATION, Description: N2- Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.040 in.
Actual Thickness Used in Calculation	0.355 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1001 \cdot 1241 / (138 \cdot 1 - 0.6 \cdot 1001)$
 $= 9.0386 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1001 \cdot 153 / (118 \cdot 1 - 0.6 \cdot 1001)$
 $= 1.3036 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9241 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	611.6066 mm
Parallel to Vessel Wall, opening length	d	305.8033 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		38.5584 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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Nozzle Calcs.: N2- Shell inlet

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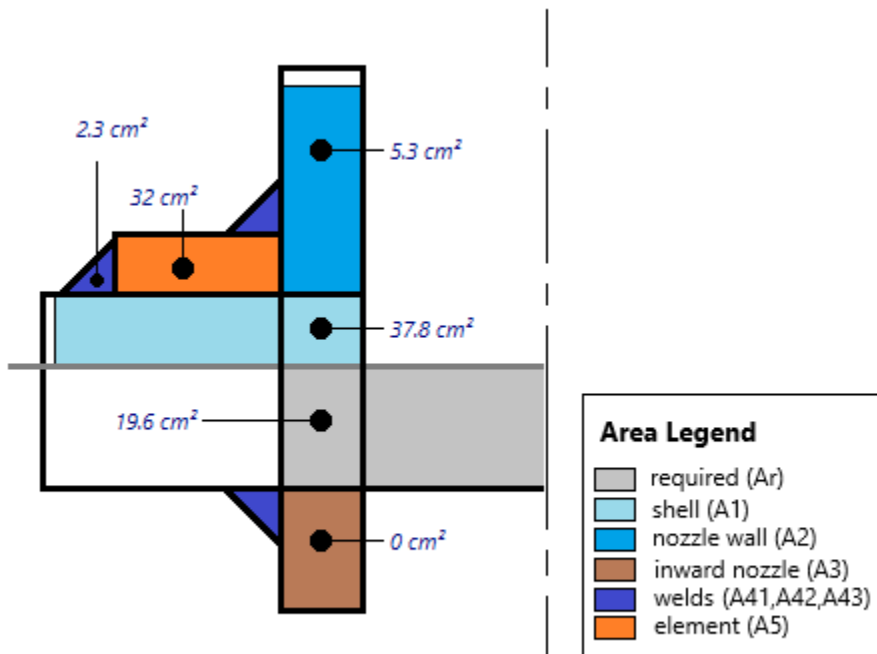


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	27.877	19.620	NA
Area in Shell	A1	48.999	37.828	NA
Area in Nozzle Wall	A2	5.090	5.341	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	32.000	32.000	NA
TOTAL AREA AVAILABLE	Atot	88.385	77.464	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(306 \cdot 12.7 \cdot 1 + 2 \cdot 9.02 \cdot 12.7 \cdot 1(1 - 0.86))$$

$$= 19.620 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 306(1 \cdot 25.2 - 1 \cdot 12.7) - 2 \cdot 9.02$$

$$(1 \cdot 25.2 - 1 \cdot 12.7) \cdot (1 - 0.86)$$

$$= 37.828 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 38.6) \cdot (9.02 - 0.92) \cdot 0.86$$



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

= 5.341 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 12^2 \cdot 1$

= 2.295 cm²

Area Available in Element [A5]:

= $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$

= $(524 - 324) \cdot 16 \cdot 1$

= 32.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.3036$ mm

Wall Thickness per UG16(b), $tr_{16b} = 3.3000$ mm

Wall Thickness, shell/head, internal pressure $trb_1 = 9.8386$ mm

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 9.8386$ mm

Wall Thickness, shell/head, external pressure $trb_2 = 1.7113$ mm

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.3000$ mm

Wall Thickness per table UG-45 $tb_3 = 8.3312$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[8.33, \max(9.84, 3.3)]$

= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(ta, tb)$

= $\max(1.3, 8.33)$

= 8.3312 mm

Available Nozzle Neck Thickness = 9.0233 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.2 N./mm ²	Passed
Occasional	: 8.2,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 18.5,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $c = 0.8$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 71 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.02$, $t_r = 1.3$, $c = 0$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.14$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.81 / 1960.01 = 0.511$

Weld Size Calculations, Description: N2- Shell inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 9.0233 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$8.0000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (19.6 - 37.8 + 2 \cdot 9.02 \cdot 0.86 \cdot (1 \cdot 25.2 - 12.7)) \cdot 138)$
 $= \max(0, -22912.06) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5.34 + 32 + 2.3 - 0 \cdot 0.86) \cdot 138$
 $= 55773.72 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5.34 + 0 + 0.86 + (3.89)) \cdot 138$
 $= 14190.36 \text{ Kg}$



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (5.34 + 0 + 2.3 + 32 + (3.89)) \cdot 138$$

$$= 61245.70 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 524 \cdot 12 \cdot 0.49 \cdot 138$$

$$= 68084. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 157) \cdot (9.02 - 0) \cdot 0.7 \cdot 118$$

$$= 37584. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 324 \cdot 16 \cdot 0.74 \cdot 138$$

$$= 84754. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 324 \cdot (15 - 0) \cdot 0.74 \cdot 138$$

$$= 79456. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (68084 + 37584) = 105668 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (29992 + 84754 + 79456 + 0) = 194202 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (68084 + 79456 + 0) = 147541 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 105668 Kg, must exceed W = 0 Kg	or W1 = 55773 Kg
Path 2-2 = 194202 Kg, must exceed W = 0 Kg	or W2 = 14190 Kg
Path 3-3 = 147540 Kg, must exceed W = 0 Kg	or W3 = 61245 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2070 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 10.6180 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 286.6179 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 281.52 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 305.80 mm



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Input Echo, WRC297 Item 1, Description: N2- Shell inle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.8000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	305.803	mm
Nozzle Thickness	9.0233	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	523.850	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1000.81	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{1261.4(26.0 - 0.8)}$$

$$= 912.207 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2522.800	mm
Nozzle Outside Diameter	(d) =	323.850	mm
Vessel Thickness used	(T) =	41.200	mm
Nozzle Thickness used	(t) =	9.023	mm
T / t	=	4.566	
d / t	=	35.890	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.005$$



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Nr/P = 0.060
 Mr/P = 0.058
 MØ/P = 0.103
 NØ/P = 0.291
 MrD/Mc = 0.106
 NrDL/Mc = 0.098
 MØd/Mc = 0.248
 NØDL/Mc = 0.230
 MrD/Ml = 0.091
 NrDL/Ml = 0.086
 MØD/Ml = 0.185
 NØDL/Ml = 0.445

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	29	30	29	30
Outplane Stress Summary	6	41	77	11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	14	14	14	14
Inplane Stress Summary	5	21	34	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	9	42	77	11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14

Outplane Stress Summary	-12	39	66	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	29	30	29	30

Inplane Stress Summary	18	39	51	14
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	31	41	67	33

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-10	-10	10	10
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20

Hoop Stress Summary	13	14	33	35
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-136	136	136	-136
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-96	84	212	-151

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	114	87	213	186

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	19	20	19	20
Hoop Stress Summary	18	19	28	30

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-162	162	162	-162
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9
Axial Stress Summary	-123	111	239	-177

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12
Shear Stress Summary	8	8	15	15



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	142	111	240	209

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{1253.4(26.0 - 0.8)}$$

$$= 910.338 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2506.800 mm
Nozzle Outside Diameter	(d) =	523.850 mm
Vessel Thickness used	(T) =	25.200 mm
Nozzle Thickness used	(t) =	109.023 mm
T / t	=	0.231
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.084$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.068
Mr/P	=	0.141
M0/P	=	0.041
N0/P	=	0.079
MrD/Mc	=	0.286
NrDL/Mc	=	0.091
M0d/Mc	=	0.088
N0DL/Mc	=	0.078
MrD/Ml	=	0.108
NrDL/Ml	=	0.083
M0D/Ml	=	0.033
N0DL/Ml	=	0.102

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	48	49	48	49
Outplane Stress Summary	48	47	71	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	24	24	24	24
Inplane Stress Summary	29	18	84	-24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	49	47	84	64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24
Outplane Stress Summary	13	35	58	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-65	65	65	-65
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	49	48	49
Inplane Stress Summary	12	83	149	-39

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	13	83	149	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	-7	-7	13	13	-2	-2	2	2
Circ. Q (SUS)	-17	17	33	-33	-10	10	18	-18
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-1	-1	1	1	-2	-2	8	8
Long. Q (SUS)	-8	8	16	-16	-26	26	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	22.0	23.0	42.0	43.0	27.0	28.0	31.0	32.0
Pm+Pl+Q (Total)	7.0	40.2	75.1	11.7	31.3	40.0	64.3	34.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	138.00	Passed
Pm+Pl (SUS)	43.00	207.00	Passed
Pm+Pl+Q (TOTAL)	75.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	49	48	49	48	49	48	49
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Circ. Q (SUS)	1	-1	15	-15	-36	36	94	-94
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-1	-1	5	5	0	0	4	4
Long. Q (SUS)	5	-5	53	-53	-11	11	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	48.0	49.0	48.0	49.0	48.0	49.0	48.0	49.0



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Nozzle Calcs.: N2- Shell inlet Nozl: 19 8:25am Apr 21,2026

Pm+Pl (SUS)	47.0	48.0	53.0	54.0	47.0	48.0	53.0	54.0
Pm+Pl+Q (Total)	48.0	47.0	82.1	63.0	13.4	84.0	147.0	41.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+Pl (SUS)	54.00	207.00	Passed
Pm+Pl+Q (TOTAL)	147.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	19	20	19	20	19	20	19	20
Circ. Pl (SUS)	-7	-7	13	13	-2	-2	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-91	91	181	-181	-117	117	207	-207
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	19.0	20.0	19.0	20.0	19.0	20.0	19.0	20.0
Pm+Pl (SUS)	19.9	20.9	33.9	34.6	24.7	25.7	31.2	31.5
Pm+Pl+Q (Total)	114.0	87.0	210.5	185.9	142.1	110.9	237.1	208.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	118.00	Passed
Pm+Pl (SUS)	34.61	177.00	Passed
Pm+Pl+Q (TOTAL)	237.07	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N3- Shell outlet

Nozl: 20 8:25am Apr 21,2026

Input, Nozzle Desc: N3- Shell outlet

From: 30

Pressure for Reinforcement Calculations	P	976.638	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	7482.8965	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4475.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	19.5000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	0.8000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



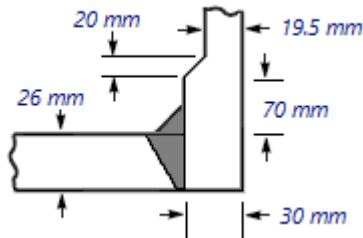
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Nozzle Calcs.: N3- Shell outlet

Noz1: 20 8:25am Apr 21,2026



Reinforcement CALCULATION, Description: N3- Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.768 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 977 \cdot 1241 / (138 \cdot 1 - 0.6 \cdot 977)$
 $= 8.8194 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 977 \cdot 458 / (138 \cdot 1 - 0.6 \cdot 977)$
 $= 3.2554 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.0033 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.5 / 0.5773$
 $= 18.1865 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1832.0000 mm
Parallel to Vessel Wall, opening length	d	916.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	63.0000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



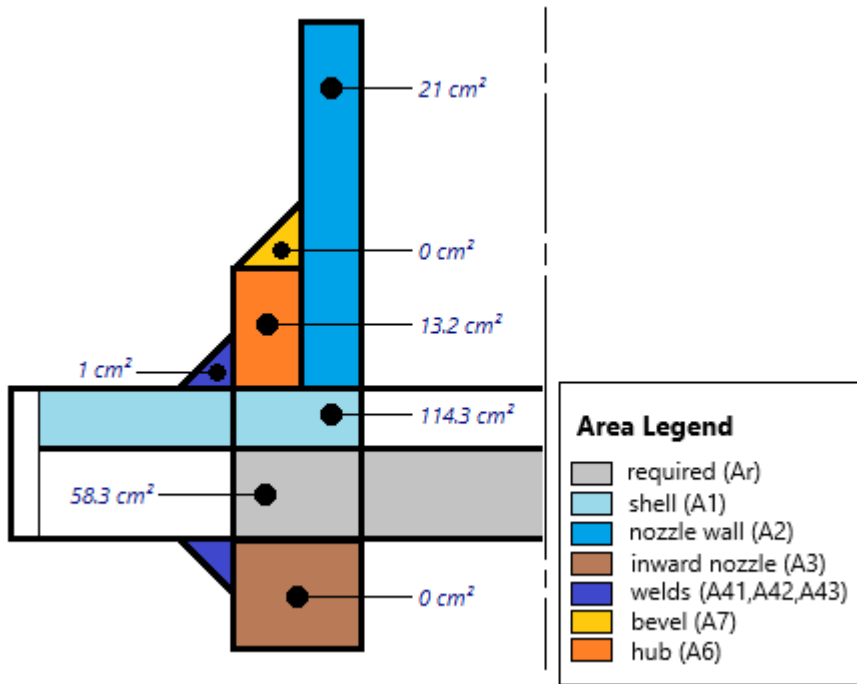
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Nozzle Calcs.: N3- Shell outle

Noz1: 20 8:25am Apr 21,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	80.785	58.272	NA
Area in Shell	A1	150.047	114.288	NA
Area in Nozzle Wall	A2	19.460	21.038	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	13.230	13.230	NA
TOTAL AREA AVAILABLE	Atot	183.737	149.556	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(916 \cdot 12.7 \cdot 1 + 2 \cdot 29.2 \cdot 12.7 \cdot 1(1 - 1))$
 $= 58.272 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 916 (1 \cdot 25.2 - 1 \cdot 12.7) - 2 \cdot 29.2$
 $(1 \cdot 25.2 - 1 \cdot 12.7) \cdot (1 - 1)$
 $= 114.288 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 63)(18.7 - 2) 1$
 $= 21.038 \text{ cm}^2$



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Nozzle Calcs.: N3- Shell outle Nozl: 20 8:25am Apr 21,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(63, 350, 70)) \cdot (30 - 19.5) \cdot 1$$

$$= 13.230 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0554 mm
Wall Thickness per UG16(b),	tr16b = 3.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.6194 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.6194 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.7113 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.3000 mm
Wall Thickness per table UG-45	tb3 = 9.1312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.13, \max(9.62, 3.3)]$$

$$= 9.1312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.06, 9.13)$$

$$= 9.1312 \text{ mm}$$

Available Nozzle Neck Thickness = 19.5000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 23.2,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 321.8 N./mm ²	Passed
Occasional	: 11.7,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 8.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 19.5, tr = 3.26, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.17, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26, tr = 8.82, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.35, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
--	--------



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Nozzle Calcs.: N3- Shell outle Nozl: 20 8:25am Apr 21,2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 976.64/1960.01 = 0.498

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 10.5/0.5773
= 18.1865 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1374.0000 mm
Parallel to Vessel Wall, opening length d 687.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 63.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure] AR		38.850	38.850
Area Available in Shell A1		57.144	57.144
Area Available in Nozzle Wall A2		21.038	21.038
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.000	1.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		13.230	13.230
TOTAL AREA AVAILABLE Atot		92.412	92.412

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N3- Shell outle

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (58.3 - 114 + 2 * 29.2 * 1 *
(1 * 25.2 - 12.7))138)
= max(0, -68569.34) Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²*fr2)*Sv
= (34.3 + 1 - 0 * 1) * 138
= 49627.48 Kgf



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Nozzle Calcs.: N3- Shell outle Nozl: 20 8:25am Apr 21,2026

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (34.3 + 0 + 1 + (14.7)) \cdot 138$$

$$= 70336.32 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (34.3 + 0 + 1 + 0 + (14.7)) \cdot 138$$

$$= 70336.32 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 974 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 105535. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 473) \cdot (30 - 0.8) \cdot 0.7 \cdot 138$$

$$= 427039. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 974 \cdot (18 - 0) \cdot 0.74 \cdot 138$$

$$= 286882. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (105535 + 427039) = 532573 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (105535 + 0 + 286882 + 0) = 392417 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (105535 + 286882 + 0) = 392417 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 532573 Kg, must exceed W = 0 Kg	or W1 = 49627 Kg
Path 2-2 = 392417 Kg, must exceed W = 0 Kg	or W2 = 70336 Kg
Path 3-3 = 392417 Kg, must exceed W = 0 Kg	or W3 = 70336 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1518 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 99.7210 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 475.7210 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N3- Shell outl:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	0.8000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	2480.000 mm



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Nozzle Calcs.: N3- Shell outle Nozl: 20 8:25am Apr 21,2026

Shell Thickness		26.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		0.8000	mm
Nozzle Diameter		914.400	mm
Nozzle Thickness		30.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	976.64	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.89	Kg-m.
Longitudinal Moment	Ml	11559.89	Kg-m.
Torsional Moment	Mt	16514.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 974.4 + 2 \cdot 1.65 \cdot \sqrt{1253.4(26.0 - 0.8)}$$

$$= 1560.888 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2506.800	mm
Nozzle Outside Diameter	(d) =	974.400	mm
Vessel Thickness used	(T) =	25.200	mm
Nozzle Thickness used	(t) =	29.200	mm
T / t	=	0.863	
d / t	=	33.370	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.877$$

Nr/P	=	0.038
Mr/P	=	0.092
M0/P	=	0.029
N0/P	=	0.120
MrD/Mc	=	0.229
NrDL/Mc	=	0.070
M0d/Mc	=	0.079
N0DL/Mc	=	0.121
MrD/Ml	=	0.079



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Nozzle Calcs.: N3- Shell outle

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NrDL/M1 = 0.032
MØD/M1 = 0.025
NØDL/M1 = 0.133

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	47	48	47	48
Outplane Stress Summary	26	46	131	38

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-86	86	86	-86
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-6	50	179	-111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	35	54	179	150

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-22	-22	22	22
Outplane Bending (Mc)	-86	86	86	-86
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23



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Nozzle Calcs.: N3- Shell outle Nozl: 20 8:25am Apr 21,2026

Outplane Stress Summary	-53	82	163	-46
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-252	252	252	-252
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	48	47	48

Inplane Stress Summary	-154	233	375	-245
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	154	233	375	245
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	12	12	12	12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-24	-24	24	24
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18

Hoop Stress Summary	6	7	54	56
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	34	-34	34	-34
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-51	51	51	-51
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-12	20	101	-70
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Nozzle Calcs.: N3- Shell outle

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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	21	22	101	126

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	12	12	12	12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-22	-22	22	22
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18
Hoop Stress Summary	8	9	52	53

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	34	-34	34	-34
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-159	159	159	-159
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-120	128	209	-178

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	129	128	209	232

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Nozzle Calcs.: N3- Shell outle

Nozl: 20 8:25am Apr 21,2026

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	48	47	48	47	48	47	48
Circ. Pl (SUS)	-12	-12	36	36	-8	-8	16	16
Circ. Q (SUS)	-9	9	45	-45	-194	194	310	-310
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-1	-1	9	9	-10	-10	34	34
Long. Q (SUS)	-28	28	144	-144	-68	68	104	-104
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	47.0	48.0	47.0	48.0	47.0	48.0	47.0	48.0
Pm+Pl (SUS)	35.1	36.1	83.0	84.0	39.0	40.0	63.2	64.1
Pm+Pl+Q (Total)	33.5	53.1	176.2	151.1	155.1	234.1	373.1	246.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	48.00	138.00	Passed
Pm+Pl (SUS)	84.02	207.00	Passed
Pm+Pl+Q (TOTAL)	373.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	18	17	18	17	18	17	18
Circ. Pl (SUS)	-12	-12	36	36	-10	-10	34	34
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-17	17	85	-85	-125	125	193	-193
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	17.0	18.0	17.0	18.0	17.0	18.0	17.0	18.0
Pm+Pl (SUS)	5.4	6.3	53.0	54.0	7.2	8.2	51.0	52.0
Pm+Pl+Q (Total)	20.6	21.1	98.1	126.1	129.1	128.0	206.1	232.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Nozzle Calcs.: N3- Shell outle

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Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	54.02	207.00	Passed
Pm+Pl+Q (TOTAL)	232.14	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N4- Shell outlet

Nozl: 21 8:25am Apr 21,2026

Input, Nozzle Desc: N4- Shell outlet

From: 30

Pressure for Reinforcement Calculations	P	976.638	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	7482.8965	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7475.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	19.5000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



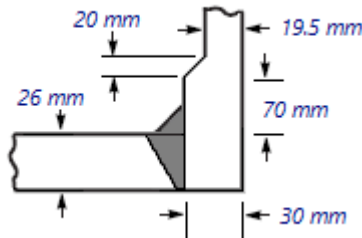
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Nozzle Calcs.: N4- Shell outlet

Noz1: 21 8:25am Apr 21,2026



Reinforcement CALCULATION, Description: N4- Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.768 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 977 \cdot 1241 / (138 \cdot 1 - 0.6 \cdot 977)$
 $= 8.8194 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 977 \cdot 457 / (138 \cdot 1 - 0.6 \cdot 977)$
 $= 3.2497 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.0033 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 10.5 / 0.5773$
 $= 18.1865 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1828.7999 mm
Parallel to Vessel Wall, opening length	d	914.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	63.0000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



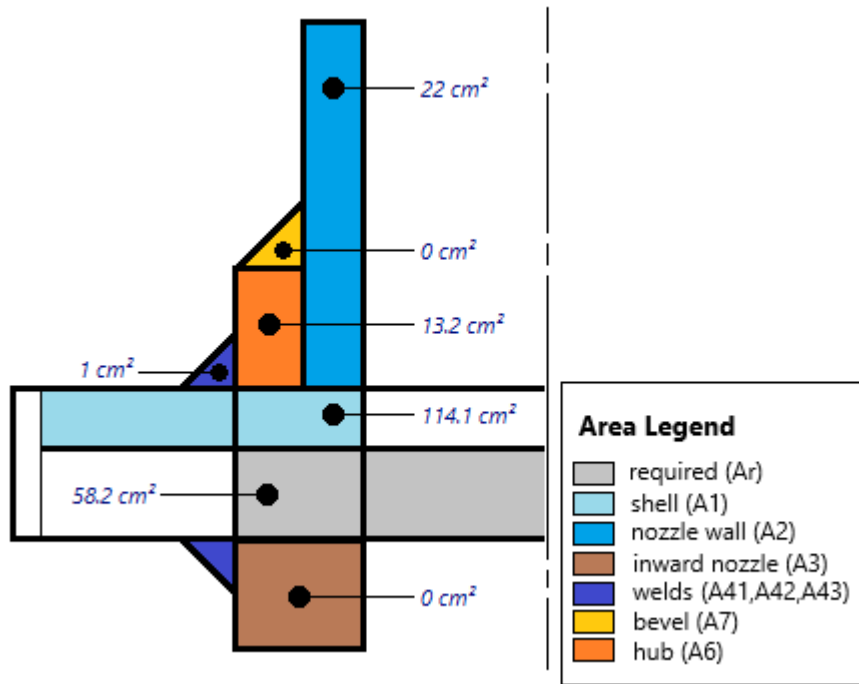
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Nozzle Calcs.: N4- Shell outle

Noz1: 21 8:25am Apr 21,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	80.644	58.170	NA
Area in Shell	A1	149.784	114.088	NA
Area in Nozzle Wall	A2	20.475	22.046	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	13.230	13.230	NA
TOTAL AREA AVAILABLE	Atot	184.490	150.364	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(914 \cdot 12.7 \cdot 1 + 2 \cdot 30 \cdot 12.7 \cdot 1(1 - 1))$
 $= 58.170 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 914 (1 \cdot 25.2 - 1 \cdot 12.7) - 2 \cdot 30$
 $(1 \cdot 25.2 - 1 \cdot 12.7) \cdot (1 - 1)$
 $= 114.088 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 63)(19.5 - 2) 1$
 $= 22.046 \text{ cm}^2$



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Nozzle Calcs.: N4- Shell outle Nozl: 21 8:25am Apr 21,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(63, 350, 70)) \cdot (30 - 19.5) \cdot 1$$

$$= 13.230 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.2497 mm
Wall Thickness per UG16(b),	tr16b = 3.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.6194 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.6194 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.7113 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.3000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.33, \max(9.62, 3.3)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.25, 8.33)$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 19.5000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 22.3,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 322.7 N./mm ²	Passed
Occasional	: 11.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 8.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 19.5, tr = 3.25, c = 0 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.17, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 26, tr = 8.82, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.35, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
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Nozzle Calcs.: N4- Shell outle Nozl: 21 8:25am Apr 21,2026

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 976.64/1960.01 = 0.498

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 10.5/0.5773
= 18.1865 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1371.6000 mm
Parallel to Vessel Wall, opening length d 685.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 63.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure] AR		38.782	38.782
Area Available in Shell A1		57.044	57.044
Area Available in Nozzle Wall A2		22.046	22.046
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.000	1.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		13.230	13.230
TOTAL AREA AVAILABLE Atot		93.320	93.320

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N4- Shell outle

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (58.2 - 114 + 2 * 30 * 1 *
(1 * 25.2 - 12.7))138)
= max(0, -68150.75) Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²*fr2)*Sv
= (35.3 + 1 - 0 * 1) * 138
= 51045.89 Kgf



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Nozzle Calcs.: N4- Shell outle Nozl: 21 8:25am Apr 21,2026

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (35.3 + 0 + 1 + (15.1)) \cdot 138$$

$$= 72322.10 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (35.3 + 0 + 1 + 0 + (15.1)) \cdot 138$$

$$= 72322.10 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 974 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 105535. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 472) \cdot (30 - 0) \cdot 0.7 \cdot 138$$

$$= 438367. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 974 \cdot (20 - 0) \cdot 0.74 \cdot 138$$

$$= 318758. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (105535 + 438367) = 543902 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (105535 + 0 + 318758 + 0) = 424293 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (105535 + 318758 + 0) = 424293 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 543901 Kg, must exceed W = 0 Kg	or W1 = 51045 Kg
Path 2-2 = 424293 Kg, must exceed W = 0 Kg	or W2 = 72322 Kg
Path 3-3 = 424293 Kg, must exceed W = 0 Kg	or W3 = 72322 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1528 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 99.7210 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 475.7210 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N4- Shell outl:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	0.8000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	2480.000 mm



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Nozzle Calcs.: N4- Shell outle Nozl: 21 8:25am Apr 21,2026

Shell Thickness		26.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		0.0000	mm
Nozzle Diameter		914.400	mm
Nozzle Thickness		30.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	976.64	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.89	Kg-m.
Longitudinal Moment	Ml	11559.89	Kg-m.
Torsional Moment	Mt	16514.19	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 974.4 + 2 \cdot 1.65 \cdot \sqrt{1253.4(26.0 - 0.8)}$$

$$= 1560.888 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2506.800	mm
Nozzle Outside Diameter	(d) =	974.400	mm
Vessel Thickness used	(T) =	25.200	mm
Nozzle Thickness used	(t) =	30.000	mm
T / t	=	0.840	
d / t	=	32.480	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.877$$

Nr/P	=	0.038
Mr/P	=	0.092
M0/P	=	0.029
N0/P	=	0.119
MrD/Mc	=	0.230
NrDL/Mc	=	0.069
M0d/Mc	=	0.079
N0DL/Mc	=	0.120
MrD/Ml	=	0.079



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Nozzle Calcs.: N4- Shell outle

Nozl: 21 8:25am Apr 21,2026

NrDL/M1 = 0.032
MØD/M1 = 0.025
NØDL/M1 = 0.131

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-24	-24	24	24
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	47	48	47	48
Outplane Stress Summary	27	46	130	38

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-87	87	87	-87
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-6	50	179	-111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	35	54	179	150

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	12	12	12	12
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-21	-21	21	21
Outplane Bending (Mc)	-86	86	86	-86
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23



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Nozzle Calcs.: N4- Shell outle Nozl: 21 8:25am Apr 21,2026

Outplane Stress Summary	-53	82	163	-47
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-253	253	253	-253
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	48	47	48

Inplane Stress Summary	-155	234	376	-246
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	155	234	376	246
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	12	12	12	12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-24	-24	24	24
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18

Hoop Stress Summary	5	7	53	55
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	32	-32	32	-32
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-48	48	48	-48
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-12	19	96	-66
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Nozzle Calcs.: N4- Shell outle Nozl: 21 8:25am Apr 21,2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	20	21	96	121

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	12	12	12	12
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-21	-21	21	21
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18
Hoop Stress Summary	8	9	51	53

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	32	-32	32	-32
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-151	151	151	-151
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-115	122	199	-169

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	123	123	199	222

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Nozzle Calcs.: N4- Shell outle

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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	48	47	48	47	48	47	48
Circ. Pl (SUS)	-12	-12	36	36	-8	-8	16	16
Circ. Q (SUS)	-9	9	45	-45	-195	195	311	-311
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-1	-1	9	9	-9	-9	33	33
Long. Q (SUS)	-29	29	145	-145	-68	68	104	-104
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	47.0	48.0	47.0	48.0	47.0	48.0	47.0	48.0
Pm+Pl (SUS)	35.1	36.1	83.0	84.0	39.0	40.0	63.1	64.1
Pm+Pl+Q (Total)	34.5	53.8	177.2	152.1	156.1	235.1	374.1	247.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	48.00	138.00	Passed
Pm+Pl (SUS)	84.02	207.00	Passed
Pm+Pl+Q (TOTAL)	374.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	18	17	18	17	18	17	18
Circ. Pl (SUS)	-12	-12	36	36	-9	-9	33	33
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-16	16	80	-80	-119	119	183	-183
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	17.0	18.0	17.0	18.0	17.0	18.0	17.0	18.0
Pm+Pl (SUS)	5.0	6.0	53.0	54.0	8.0	9.0	50.0	51.0
Pm+Pl+Q (Total)	19.0	19.7	93.2	121.1	124.1	122.1	196.1	221.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Nozzle Calcs.: N4- Shell outle

Nozl: 21

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Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	54.00	207.00	Passed
Pm+Pl+Q (TOTAL)	221.08	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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ASME TS Calc: CASE-1

Case: 1 8:25a Apr 21,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	177.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	52.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	245.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5080
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6075.00 mm
Design Temperature of the Tubes		245.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		103.70 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	114.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.600 mm
Largest Center-to-Center Tube Distance	U1	76.2000 mm



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ASME TS Calc: CASE-1

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Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	44322.1 cm ²
Perimeter of Tube Layout	Cp	7804.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	245.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

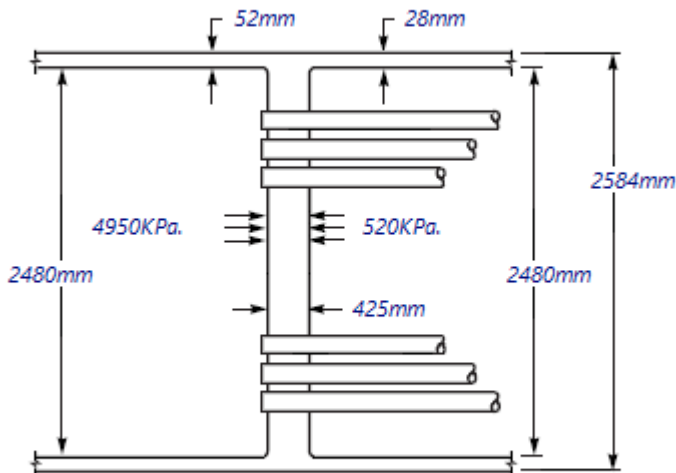
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	136.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	425.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2584.000 mm

Area of the Untubed Lanes	AL	1221.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 100.0002 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	245.0 °C	0.17950E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5050 \cdot (25.4/2 - 0)) / (104 \cdot 1 + 0.4 \cdot 5050)$
 $= 0.6067 + 0.0000 = 0.6067 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 245.00 °C
 Elastic Modulus for Material 175048832.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7282.60	15.39	50.0000	0.0046419	46.94

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.9) / (3 \cdot 15.4) = 4065 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4049	25.40	7282.60	62.73	50.0000	0.0002795	24.47

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.5) / (3 \cdot 62.7) = 520 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6067 mm
Allowable Internal Pressure at Corroded thickness	14210.70 KPa.
Required Internal Design Pressure	5050.00 KPa.
Allowable External Pressure at Corroded thickness	4065.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.4049 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 1208 + 25.4 \\
 &= 2440.600 \text{ mm}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.200
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2480/2441 \\
 &= 1.0161
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 2480/2441 \\
 &= 1.0161
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 50/425 \\
 &= 0.118 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t_t(E_t T/E) \cdot (S_t T/S) \cdot (\rho); dt - 2t_t) \\
 &= \max(25.4 - 2 \cdot 1.65 (179502208/189341056) \cdot \\
 &\quad (104/136) \cdot (0.12); 25.4 - 2 \cdot 1.65) \\
 &= 25.1198 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\
 &= 31.8 / \sqrt{ (1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\
 &= 32.1726 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (32.2 - 25.1) / 32.2 \\
 &= 0.2192
 \end{aligned}$$

$h/p = 13.385828$; $\mu^* = 0.219219$; $E^* = 33773356 \text{ KPa}$.

$E^*/E = 0.178373$; $v^* = 0.432112$

Note: As h/p (13.386) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2480 + 28) 28) } \\
 &= 0.0069 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.0069 \cdot 194566320 \cdot 28^3 / (6(1 - 0.3^2)) \\
 &= 5366140416.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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$$= 6(2480 + 28) \cdot 5.37E+09/425^3 \cdot (1 + 425 \cdot 0.0069 + (425 \cdot 0.0069)^2/2)$$

$$= 8589042.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2480^2 / (4 \cdot 194566320 \cdot 28) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 28)))$$

$$= 0.24036E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 5.37E+09 \cdot 0.0069 \cdot 0.2403622 (1 + 425 \cdot 0.0069)$$

$$= 35204.6680 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2480 + 52)52)$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09/425^3 \cdot (1 + 425 \cdot 0.005 \cdot (1 + 1)/2 + (425 \cdot 0.005)^2/2 \cdot 1)$$

$$= 26085156.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 425) - 0.000 (1/425 + 1 \cdot 0.005/2)]$$

$$= 51841.3750 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584/2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33773356 \cdot (8589042 + 26085156 + 189341056 \cdot \ln(1.06))$$

$$= 0.7648$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$



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$$= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 520 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -54114960.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -54114960 + 51841 \cdot 4950 - 35205 \cdot 520$$

$$= 184193600.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((184193600) - 2441^2/32 \cdot 0.76 (520 - 4950)) / (1 + 0.76)$$

$$= 461727296.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$$

$$= 461727296 + 2441^2/64 \cdot (3 + 0.43) \cdot (520 - 4950)$$

$$= -953345344.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 461727296 | ; | -953345344 |)$$

$$= 953345344.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 953345344 / (0.22 (425 - 0)^2)$$

$$= 144.4674 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 890372992 / (0.22 (297 - 0)^2)$$

$$= 271.9282 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 297 + 0 + 0$$

$$= 296.8341 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(297 ; 115)$$

$$= 296.8341 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | 520 - 4950 |$$

$$= 4430.003 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.2 \cdot 425/2441 \cdot \min(0.8 \cdot 136 ; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$



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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | 520 - 4950 | + 0$$

$$= 115.4538 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot 520$$

$$= 11.3864 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 1206383 / 28^2 \cdot [0.0069 \cdot 0.12 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0.0069 / 2) \cdot (461727296 + 2441^2 / 32 \cdot (520 - 4950))]$$

$$= -82.271 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_s, m | + | \sigma_s, b | \text{ (Should be } \leq \text{SPS, s)}$$

$$= 93.657 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= Dc^2 / (4 \cdot tc (Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\Theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0.005 \cdot 425 / 2 \cdot 1) \cdot (461727296 + 2441^2 / 32 \cdot (520 - 4950))]$$

$$= 309.351 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_c, m | + | \sigma_c, b | \text{ (Should be } \leq \text{SPS, c)}$$

$$= 367.161 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, ks, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, kc, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E * (\lambda_s + \lambda_c + E * \ln(K)) \\
 &= (1 - 0.43) / 33773356 * (0 + 0 + 189341056 * \ln(1.06)) \\
 &= 0.1818
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 * [(p_s - 1) * (p_s^2 + 1) * P_s - (p_c - 1) * (p_c^2 + 1) * P_t] \\
 &= 2441^2 / 16 * [(1.02 - 1) * (1.02^2 + 1) * 520 - (1.02 - 1) * (1.02^2 + 1) * 4950] \\
 &= -54114960.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + \omega_c * P_t - \omega_s * P_s \\
 &= -54114960 + 0 * 4950 - 0 * 520 \\
 &= -54114960.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 * (3 + \nu) * (P_s - P_t) \\
 &= 81044472 + 2441^2 / 64 * (3 + 0.43) * (520 - 4950) \\
 &= -1334028160.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |M_o| \\
 &= |-1334028160| \\
 &= 1334028160.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1334028160 / (0.22 * (425 - 0)^2) \\
 &= 202.1550 \text{ N/mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 * S \\
 &= 2 * 136 \\
 &= 272.40 \text{ N/mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1334012672 / (0.22 * (365 - 0)^2) \\
 &= 272.3606 \text{ N/mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 365 + 0 + 0 \\
 &= 364.9322 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(365; 124) \\
 &= 364.9322 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |520 - 4950| \\
 &= 4430.003 \text{ KPa}
 \end{aligned}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |520 - 4950| + 0$$

$$= 115.4538 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot 520$$

$$= 11.3864 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 28^2 \cdot [0 \cdot 0.12 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0 / 2) \cdot (81044472 + 2441^2 / 32 \cdot (520 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS}, s)$$

$$= 11.386 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0 \cdot 425 / 2 \cdot 1) \cdot (81044472 + 2441^2 / 32 \cdot (520 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS}, c)$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1) - 0.75 \cdot 25.4} \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	202.2 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	29.6 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	11.4 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	364.932	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:
The Tubesheet stress reduction option was Simply Supported. Performing



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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1208 + 25.4 \\ &= 2440.600 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2482 / 2441 \\ &= 1.0168 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 399 \\ &= 0.125 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot \\ &\quad (104 / 136) \cdot (0.13); 25.4 - 2 \cdot 1.65) \\ &= 25.1015 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\ &= 32.1726 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.2 - 25.1) / 32.2 \\ &= 0.2198 \end{aligned}$$

$h/p = 12.566931$; $\mu^* = 0.219787$; $E^* = 33904192 \text{ KPa}$.

$E^*/E = 0.179064$; $v^* = 0.431382$

Note: As h/p (12.567) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2482 + 27.2) 27.2 } \\ &= 0.0070 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.007 \cdot 194566320 \cdot 27.2^3 / (6(1 - 0.3^2)) \end{aligned}$$



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$$= 4990222848.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2482 + 27.2) \cdot 4.99\text{E}+09 / 399^3 \cdot (1 + 399 \cdot 0.007 + (399 \cdot 0.007)^2 / 2)$$

$$= 9024370.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2482^2 / (4 \cdot 194566320 \cdot 27.2) \cdot (1 - 2482 \cdot 0.3 / (2(2482 + 27.2)))$$

$$= 0.24774\text{E}+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 4.99\text{E}+09 \cdot 0.007 \cdot 0.24773699 (1 + 399 \cdot 0.007)$$

$$= 33037.6133 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 52) 52 }$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43\text{E}+09 / 399^3 \cdot (1 + 399 \cdot 0.005 \cdot (1 + 1) / 2 + (399 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 29192318.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43\text{E}+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 399) - 0.000 (1 / 399 + 1 \cdot 0.005 / 2)]$$

$$= 49683.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584 / 2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (9024370 + 29192318 + 189341056 \cdot \ln(1.06))$$

$$= 0.8222$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 520 -$
 $(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$
 $= -53852644.0000 \text{ KPa} \cdot \text{mm}^2$

Moment acting on unperforated tubesheet rim [M*]
 $= Mts + wc \cdot Pt - ws \cdot Ps$
 $= -53852644 + 49683 \cdot 4950 - 33038 \cdot 520$
 $= 174900880.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((174900880) - 2441^2/32 \cdot 0.82 (520 - 4950)) / (1 + 0.82)$
 $= 468066432.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$
 $= 468066432 + 2441^2/64 \cdot (3 + 0.43) \cdot (520 - 4950)$
 $= -946705024.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 468066432 | ; | -946705024 |)$
 $= 946705024.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 946705024 / (0.22 (399 - 0)^2)$
 $= 162.3466 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S$
 $= 2 \cdot 136$
 $= 272.40 \text{ N./mm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 892215360 / (0.22 (297 - 0)^2)$
 $= 271.9947 \text{ N./mm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 297 + 13 + 13$
 $= 323.1125 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(323 ; 141)$
 $= 323.1125 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:
 $= | Ps - Pt |$
 $= | 520 - 4950 |$
 $= 4430.003 \text{ KPa}$

Shear Stress check [τ_{limit}]:



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$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205) \\
 &= 14.250 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |520 - 4950| + 26 \\
 &= 141.4538 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\
 &= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot 520 \\
 &= 11.7327 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot Ps + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 1121872 / 27.2^2 \cdot [0.007 \cdot 0.13 + 6(1 - 0.43) / 33904192 \cdot 2441 / 399^3 \cdot (1 + 399 \cdot 0.007 / 2) \cdot (468066432 + 2441^2 / 32 \cdot (520 - 4950))] \\
 &= -96.961 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned}
 &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)} \\
 &= 108.694 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)
 \end{aligned}$$

Axial Channel Membrane Stress [σ_c, m]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt \\
 &= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950 \\
 &= 57.811 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σ_c, b]:

$$\begin{aligned}
 &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 + 0.005 \cdot 399 / 2 \cdot 1) \cdot (468066432 + 2441^2 / 32 \cdot (520 - 4950))] \\
 &= 328.496 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)} \\
 &= 386.306 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)
 \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$\beta_s, ks, \lambda_s, \delta_s$ are all = 0



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Intermediate Channel side parameters for the Simply Supported condition:
 β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (0 + 0 + 189341056 \cdot \ln(1.06))$$

$$= 0.1813$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2441^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot 520 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -53852644.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -53852644 + 0 \cdot 4950 - 0 \cdot 520$$

$$= -53852644.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 80971584 + 2441^2 / 64 \cdot (3 + 0.43) \cdot (520 - 4950)$$

$$= -1333799808.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= | -1333799808 |$$

$$= 1333799808.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1333799808 / (0.22 (399 - 0)^2)$$

$$= 228.7279 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1333790464 / (0.22 (365 - 0)^2)$$

$$= 272.3968 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 365 + 13 + 13$$

$$= 390.9011 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(391; 150)$$

$$= 390.9011 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | 520 - 4950 |$$

$$= 4430.003 \text{ KPa.}$$

Shear Stress check [τlimit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 14.250 \text{ N./mm}^2$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | 520 - 4950 | + 26$$

$$= 141.4538 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot 520$$

$$= 11.7327 \text{ N./mm}^2$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 27.2^2 \cdot [0 \cdot 0.13 + 6(1 - 0.43) / 33904192 \cdot$$

$$2441 / 399^3 \cdot (1 + 399 \cdot 0 / 2) \cdot (80971584 + 2441^2 / 32 \cdot$$

$$(520 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)}$$

$$= 11.733 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σc,m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 +$$

$$0 \cdot 399 / 2 \cdot 1) \cdot (80971584 + 2441^2 / 32 \cdot (520 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σc]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)}$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$



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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	228.7 <=	272.4 N./mm ²	Ok
Tubesheet shear stress	31.5 <=	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	11.7 <=	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 <=	434.0 N./mm ²	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	390.901	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	390.216 ...	230	272 34 109	Ps+Pt D1	Ok



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ASME TS Calc: CASE-1

Case: 1 8:25a Apr 21, 2026

D2uc	119.509	...	24	272	3	109	Ps+Pt D2	Ok
D3uc	364.932	...	202	272	30	109	Ps+Pt D3	Ok
D4uc	48.212	...	5	272	1	109	Ps+Pt D4	Ok
D1c	416.221	...	261	272	36	109	Ps+Pt-c D1	Ok
D2c	145.600	...	27	272	4	109	Ps+Pt-c D2	Ok
D3c	390.901	...	229	272	32	109	Ps+Pt-c D3	Ok
D4c	74.248	...	5	272	1	109	Ps+Pt-c D4	Ok

Max: 416.2215 ... mm 0.957 0.330 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.			Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	2	461	58	434	Ok
D2uc	...	...	11	461	...	434	Ok
D3uc	...	...	11	461	58	434	Ok
D4uc	...	...	2	461	...	434	Ok
D1c	...	...	2	461	58	434	Ok
D2c	...	...	12	461	...	434	Ok
D3c	...	...	12	461	58	434	Ok
D4c	...	...	2	461	...	434	Ok
Max	...	...	0.025		0.133		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	416.221	425.000	Ok
Tube thickness :	0.607	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 52.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5274.89	1.000	5267.42	1.000
Tubesheet Shear Stress	15309.77	1.000	15309.77	1.000
Shell Stress (Axial, Junction)	15113.33	1.000	3647.20	1.000



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ASME TS Calc: CASE-1

Case: 1 8:25a Apr 21, 2026

Tubesheet-Channel Junction	5289.86	1.000	11381.71	1.000
Tube Pressure Stress	14210.69	1.000	4065.19	1.000
Minimum MAWP	5274.89		3647.20	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6074.90	1.000	6074.90	1.000
Tubesheet Shear Stress	16522.81	1.000	16522.81	1.000
Shell Stress (Axial, Junction)	16522.81	1.000	4379.35	1.000
Tubesheet-Channel Junction	6696.49	1.000	15673.79	1.000
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	6074.90		4379.35	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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ASME TS Calc: CASE-2 -Q-7

Case: 2 8:25a Apr 21,2026

Input Echo, Tubesheet Number 2, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	177.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	52.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	245.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5080
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6075.00 mm
Design Temperature of the Tubes		245.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		103.70 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	114.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.600 mm
Largest Center-to-Center Tube Distance	Ul	76.2000 mm



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ASME TS Calc: CASE-2 -Q-7

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Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	44322.1 cm ²
Perimeter of Tube Layout	Cp	7804.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	245.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

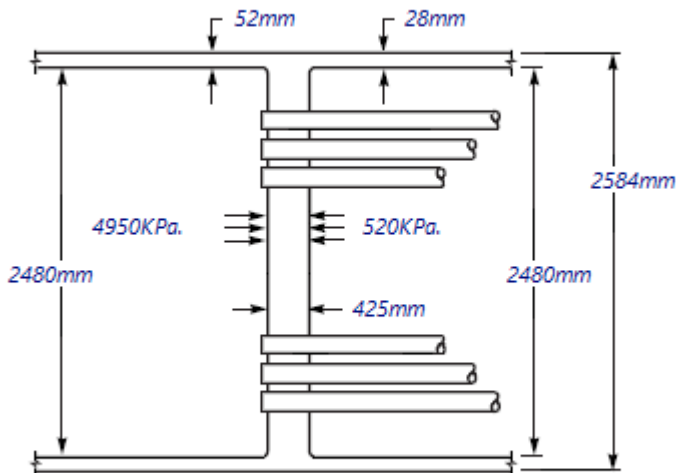
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	136.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	425.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2584.000 mm

Area of the Untubed Lanes	AL	1221.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 100.0002 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: CASE-2 -Q-7

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Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	245.0 °C	0.17950E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(5050 \cdot (25.4/2 - 0)) / (104 \cdot 1 + 0.4 \cdot 5050)$

= 0.6067 + 0.0000 = 0.6067 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	245.00 °C
Elastic Modulus for Material			175048832.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7282.60	15.39	50.0000	0.0046419	46.94

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.9) / (3 \cdot 15.4) = 4065$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4049	25.40	7282.60	62.73	50.0000	0.0002795	24.47

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.5) / (3 \cdot 62.7) = 520$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6067 mm
Allowable Internal Pressure at Corroded thickness	14210.70 KPa.
Required Internal Design Pressure	5050.00 KPa.
Allowable External Pressure at Corroded thickness	4065.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.4049 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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ASME TS Calc: CASE-2 -Q-7

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Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 1208 + 25.4 \\
 &= 2440.600 \text{ mm}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.200
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2480/2441 \\
 &= 1.0161
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 2480/2441 \\
 &= 1.0161
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 50/425 \\
 &= 0.118 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t_t(E_t T/E) \cdot (S_t T/S) \cdot (\rho); dt - 2t_t) \\
 &= \max(25.4 - 2 \cdot 1.65 (179502208/189341056) \cdot \\
 &\quad (104/136) \cdot (0.12); 25.4 - 2 \cdot 1.65) \\
 &= 25.1198 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\
 &= 31.8 / \sqrt{ (1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\
 &= 32.1726 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (32.2 - 25.1) / 32.2 \\
 &= 0.2192
 \end{aligned}$$

$h/p = 13.385828$; $\mu^* = 0.219219$; $E^* = 33773356 \text{ KPa}$.

$E^*/E = 0.178373$; $v^* = 0.432112$

Note: As h/p (13.386) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2480 + 28) 28) } \\
 &= 0.0069 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.0069 \cdot 194566320 \cdot 28^3 / (6(1 - 0.3^2)) \\
 &= 5366140416.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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$$= 6(2480 + 28) \cdot 5.37E+09/425^3 \cdot (1 + 425 \cdot 0.0069 + (425 \cdot 0.0069)^2/2)$$

$$= 8589042.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2480^2 / (4 \cdot 194566320 \cdot 28) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 28)))$$

$$= 0.24036E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 5.37E+09 \cdot 0.0069 \cdot 0.2403622 (1 + 425 \cdot 0.0069)$$

$$= 35204.6680 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(2480 + 52)52})$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09/425^3 \cdot (1 + 425 \cdot 0.005 \cdot (1 + 1)/2 + (425 \cdot 0.005)^2/2 \cdot 1)$$

$$= 26085156.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 425) - 0.000 (1/425 + 1 \cdot 0.005/2)]$$

$$= 51841.3750 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584/2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33773356 \cdot (8589042 + 26085156 + 189341056 \cdot \ln(1.06))$$

$$= 0.7648$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$



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ASME TS Calc: CASE-2 -Q-7

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$$= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61688608.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -61688608 + 51841 \cdot 4950 - 35205 \cdot -100$$

$$= 198446832.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2/32 \cdot F(P_s - P_t) / (1 + F)$$

$$= (198446832) - 2441^2/32 \cdot 0.76 (-100 - 4950) / (1 + 0.76)$$

$$= 519817472.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 519817472 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|519817472|; |-1093301376|)$$

$$= 1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1093301376 / (0.22 (425 - 0)^2)$$

$$= 165.6760 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1039441152 / (0.22 (321 - 0)^2)$$

$$= 272.1014 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 321 + 0 + 0$$

$$= 321.3042 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(321; 132)$$

$$= 321.3042 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= |P_s - P_t|$$

$$= |-100 - 4950|$$

$$= 5050.003 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$



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ASME TS Calc: CASE-2 -Q-7

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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1206383 / 28^2 \cdot [0.0069 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0.0069 / 2) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -142.815 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s, m} | + | \sigma_{s, b} | \text{ (Should be } \leq \text{SPS, s)}$$

$$= 145.005 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0.005 \cdot 425 / 2 \cdot 1) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 329.926 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c, m} | + | \sigma_{c, b} | \text{ (Should be } \leq \text{SPS, c)}$$

$$= 387.737 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, k_s, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, k_c, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E * (\lambda_s + \lambda_c + E * \ln(K)) \\
 &= (1 - 0.43) / 33773356 * (0 + 0 + 189341056 * \ln(1.06)) \\
 &= 0.1818
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 * [(p_s - 1) * (p_s^2 + 1) * P_s - (p_c - 1) * (p_c^2 + 1) * P_t] \\
 &= 2441^2 / 16 * [(1.02 - 1) * (1.02^2 + 1) * -100 - (1.02 - 1) * (1.02^2 + 1) * 4950] \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + \omega_c * P_t - \omega_s * P_s \\
 &= -61688608 + 0 * 4950 - 0 * -100 \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 * (3 + \nu) * (P_s - P_t) \\
 &= 92387032 + 2441^2 / 64 * (3 + 0.43) * (-100 - 4950) \\
 &= -1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |M_o| \\
 &= |-1520731776| \\
 &= 1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520731776 / (0.22 * (425 - 0)^2) \\
 &= 230.4476 \text{ N/mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 * S \\
 &= 2 * 136 \\
 &= 272.40 \text{ N/mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520722048 / (0.22 * (390 - 0)^2) \\
 &= 272.3968 \text{ N/mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 390 + 0 + 0 \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(390; 141) \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |-100 - 4950| \\
 &= 5050.003 \text{ KPa}
 \end{aligned}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 28^2 \cdot [0 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0 / 2) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS}, s)$$

$$= 2.190 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0 \cdot 425 / 2 \cdot 1) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS}, c)$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 1.65 (25.4 - 1.65) 104 \\
 &= 1301.823 \text{ Kgf}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104 \\
 &= 1234.9329 \text{ Kgf}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 104/104 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104 \\
 &= 799.0742 \text{ Kgf}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right) \\
 &= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1} - 0.75 \cdot 25.4 \right) \\
 &= 2.141 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	230.4 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	33.7 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.2 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	390.216	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:
The Tubesheet stress reduction option was Simply Supported. Performing



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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1208 + 25.4 \\ &= 2440.600 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2482 / 2441 \\ &= 1.0168 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 399 \\ &= 0.125 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot \\ &\quad (104 / 136) \cdot (0.13); 25.4 - 2 \cdot 1.65) \\ &= 25.1015 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ 1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\ &= 32.1726 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.2 - 25.1) / 32.2 \\ &= 0.2198 \end{aligned}$$

$h/p = 12.566931$; $\mu^* = 0.219787$; $E^* = 33904192 \text{ KPa}$.

$E^*/E = 0.179064$; $v^* = 0.431382$

Note: As h/p (12.567) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2482 + 27.2) 27.2 } \\ &= 0.0070 \text{ 1/mm} \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.007 \cdot 194566320 \cdot 27.2^3 / (6(1 - 0.3^2)) \end{aligned}$$



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$$= 4990222848.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2482 + 27.2) \cdot 4.99\text{E}+09 / 399^3 \cdot (1 + 399 \cdot 0.007 + (399 \cdot 0.007)^2 / 2)$$

$$= 9024370.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2482^2 / (4 \cdot 194566320 \cdot 27.2) \cdot (1 - 2482 \cdot 0.3 / (2(2482 + 27.2)))$$

$$= 0.24774\text{E}+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 4.99\text{E}+09 \cdot 0.007 \cdot 0.24773699 (1 + 399 \cdot 0.007)$$

$$= 33037.6133 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 52) 52 }$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43\text{E}+09 / 399^3 \cdot (1 + 399 \cdot 0.005 \cdot (1 + 1) / 2 + (399 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 29192318.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43\text{E}+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 399) - 0.000 (1 / 399 + 1 \cdot 0.005 / 2)]$$

$$= 49683.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584 / 2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (9024370 + 29192318 + 189341056 \cdot \ln(1.06))$$

$$= 0.8222$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 -$
 $(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$
 $= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$

Moment acting on unperforated tubesheet rim [M*]
 $= Mts + wc \cdot Pt - ws \cdot Ps$
 $= -61739052 + 49683 \cdot 4950 - 33038 \cdot -100$
 $= 187497808.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((187497808) - 2441^2/32 \cdot 0.82 (-100 - 4950)) / (1 + 0.82)$
 $= 527054432.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$
 $= 527054432 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$
 $= -1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 527054432 | ; | -1085721088 |)$
 $= 1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1085721088 / (0.22 (399 - 0)^2)$
 $= 186.1859 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S$
 $= 2 \cdot 136$
 $= 272.40 \text{ N./mm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1041836800 / (0.22 (322 - 0)^2)$
 $= 272.1585 \text{ N./mm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 322 + 13 + 13$
 $= 347.6495 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(348 ; 158)$
 $= 347.6495 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:
 $= | Ps - Pt |$
 $= | -100 - 4950 |$
 $= 5050.003 \text{ KPa}$

Shear Stress check [τ_{limit}]:



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$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205) \\
 &= 14.250 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 26 \\
 &= 157.6121 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\
 &= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100 \\
 &= -2.2563 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σs,b]:

$$\begin{aligned}
 &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot Ps + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 1121872 / 27.2^2 \cdot [0.007 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot 2441 / 399^3 \cdot (1 + 399 \cdot 0.007 / 2) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= -161.308 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σs]:

$$\begin{aligned}
 &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)} \\
 &= 163.564 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)
 \end{aligned}$$

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt \\
 &= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950 \\
 &= 57.811 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σc,b]:

$$\begin{aligned}
 &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 + 0.005 \cdot 399 / 2 \cdot 1) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= 352.138 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σc]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)} \\
 &= 409.949 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)
 \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

βs, ks, λs, δs are all = 0



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Intermediate Channel side parameters for the Simply Supported condition:
 β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (0 + 0 + 189341056 \cdot \ln(1.06))$$

$$= 0.1813$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2441^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -61739052 + 0 \cdot 4950 - 0 \cdot -100$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 92008104 + 2441^2 / 64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= | -1520767360 |$$

$$= 1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520767360 / (0.22 (399 - 0)^2)$$

$$= 260.7902 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520764928 / (0.22 (390 - 0)^2)$$

$$= 272.3629 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 390 + 13 + 13$$

$$= 416.2215 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(416; 167)$$

$$= 416.2215 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | -100 - 4950 |$$

$$= 5050.003 \text{ KPa.}$$

Shear Stress check [rlimit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 14.250 \text{ N./mm}^2$$

| Ps - Pt | is ≤ the Shear Stress limit (rlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 26$$

$$= 157.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100$$

$$= -2.2563 \text{ N./mm}^2$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot$$

$$Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 27.2^2 \cdot [0 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot$$

$$2441 / 399^3 \cdot (1 + 399 \cdot 0 / 2) \cdot (92008104 + 2441^2 / 32 \cdot$$

$$(-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)}$$

$$= 2.256 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σc,m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 +$$

$$0 \cdot 399 / 2 \cdot 1) \cdot (92008104 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σc]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)}$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$



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ASME TS Calc: CASE-2 -Q-7

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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	260.8 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	35.9 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.3 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	416.221	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	390.216 ...	230	272	34 109	Ps+Pt D1	Ok



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ASME TS Calc: CASE-2 -Q-7

Case: 2 8:25a Apr 21, 2026

D2uc	119.509	...	24	272	3	109	Ps+Pt D2	Ok
D3uc	364.932	...	202	272	30	109	Ps+Pt D3	Ok
D4uc	48.212	...	5	272	1	109	Ps+Pt D4	Ok
D1c	416.221	...	261	272	36	109	Ps+Pt-c D1	Ok
D2c	145.600	...	27	272	4	109	Ps+Pt-c D2	Ok
D3c	390.901	...	229	272	32	109	Ps+Pt-c D3	Ok
D4c	74.248	...	5	272	1	109	Ps+Pt-c D4	Ok

Max: 416.2215 ... mm 0.957 0.330 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.			Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	2	461	58	434	Ok
D2uc	...	...	11	461	...	434	Ok
D3uc	...	...	11	461	58	434	Ok
D4uc	...	...	2	461	...	434	Ok
D1c	...	...	2	461	58	434	Ok
D2c	...	...	12	461	...	434	Ok
D3c	...	...	12	461	58	434	Ok
D4c	...	...	2	461	...	434	Ok
Max	...	...		0.025		0.133	

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	416.221	425.000	Ok
Tube thickness :	0.607	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 52.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5274.89	1.000	5267.42	1.000
Tubesheet Shear Stress	15309.77	1.000	15309.77	1.000
Shell Stress (Axial, Junction)	15113.33	1.000	3647.20	1.000



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ASME TS Calc: CASE-2 -Q-7

Case: 2 8:25a Apr 21, 2026

Tubesheet-Channel Junction	5289.86	1.000	11381.71	1.000
Tube Pressure Stress	14210.69	1.000	4065.19	1.000
Minimum MAWP	5274.89		3647.20	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6074.90	1.000	6074.90	1.000
Tubesheet Shear Stress	16522.81	1.000	16522.81	1.000
Shell Stress (Axial, Junction)	16522.81	1.000	4379.35	1.000
Tubesheet-Channel Junction	6696.49	1.000	15673.79	1.000
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	6074.90		4379.35	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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ASME TS Calc: CASE-3 -Q-7

Case: 3 8:25a Apr 21,2026

Input Echo, Tubesheet Number 3, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	175.99 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	52.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	245.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5080
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6500.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	6075.00 mm
Design Temperature of the Tubes		245.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		103.70 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	114.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.600 mm
Largest Center-to-Center Tube Distance	U1	76.2000 mm



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ASME TS Calc: CASE-3 -Q-7

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Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	44322.1 cm ²
Perimeter of Tube Layout	Cp	7804.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	245.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

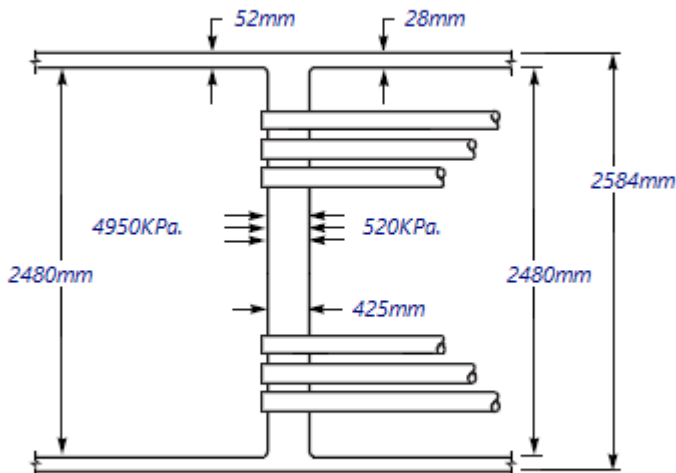
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	136.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	425.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2584.000 mm

Area of the Untubed Lanes	AL	1221.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 100.0002 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: CASE-3 -Q-7

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Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	245.0 °C	0.17950E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5050 \cdot (25.4/2 - 0)) / (104 \cdot 1 + 0.4 \cdot 5050)$
 $= 0.6067 + 0.0000 = 0.6067 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 245.00 °C
 Elastic Modulus for Material 175048832.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7282.60	15.39	50.0000	0.0046419	46.94

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.9) / (3 \cdot 15.4) = 4065 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4049	25.40	7282.60	62.73	50.0000	0.0002795	24.47

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.5) / (3 \cdot 62.7) = 520 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6067 mm
Allowable Internal Pressure at Corroded thickness	14210.70 KPa.
Required Internal Design Pressure	5050.00 KPa.
Allowable External Pressure at Corroded thickness	4065.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.4049 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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ASME TS Calc: CASE-3 -Q-7

Case: 3 8:25a Apr 21, 2026

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1208 + 25.4 \\ &= 2440.600 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 425 \\ &= 0.118 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\ &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot \\ &\quad (104 / 136) \cdot (0.12); 25.4 - 2 \cdot 1.65) \\ &= 25.1198 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ (1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\ &= 32.1726 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.2 - 25.1) / 32.2 \\ &= 0.2192 \end{aligned}$$

$h/p = 13.385828$; $\mu^* = 0.219219$; $E^* = 33773356 \text{ KPa}$.

$E^*/E = 0.178373$; $v^* = 0.432112$

Note: As h/p (13.386) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2480 + 28) 28) } \\ &= 0.0069 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.0069 \cdot 194566320 \cdot 28^3 / (6(1 - 0.3^2)) \\ &= 5366140416.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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$$= 6(2480 + 28) \cdot 5.37E+09/425^3 \cdot (1 + 425 \cdot 0.0069 + (425 \cdot 0.0069)^2/2)$$

$$= 8589042.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2480^2 / (4 \cdot 194566320 \cdot 28) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 28)))$$

$$= 0.24036E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 5.37E+09 \cdot 0.0069 \cdot 0.2403622 (1 + 425 \cdot 0.0069)$$

$$= 35204.6680 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2480 + 52)52)$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa.} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09/425^3 \cdot (1 + 425 \cdot 0.005 \cdot (1 + 1)/2 + (425 \cdot 0.005)^2/2 \cdot 1)$$

$$= 26085156.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 425) - 0.000 (1/425 + 1 \cdot 0.005/2)]$$

$$= 51841.3750 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584/2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33773356 \cdot (8589042 + 26085156 + 189341056 \cdot \ln(1.06))$$

$$= 0.7648$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$



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$$= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61688608.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -61688608 + 51841 \cdot 4950 - 35205 \cdot -100$$

$$= 198446832.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((198446832) - 2441^2/32 \cdot 0.76 (-100 - 4950)) / (1 + 0.76)$$

$$= 519817472.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$$

$$= 519817472 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 519817472 | ; | -1093301376 |)$$

$$= 1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 1093301376 / (0.22 (425 - 0)^2)$$

$$= 165.6760 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 1039441152 / (0.22 (321 - 0)^2)$$

$$= 272.1014 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 321 + 0 + 0$$

$$= 321.3042 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(321 ; 132)$$

$$= 321.3042 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | -100 - 4950 |$$

$$= 5050.003 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.2 \cdot 425/2441 \cdot \min(0.8 \cdot 136 ; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$



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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 1206383 / 28^2 \cdot [0.0069 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0.0069 / 2) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -142.815 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_s, m | + | \sigma_s, b | \text{ (Should be } \leq \text{SPS, s)}$$

$$= 145.005 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= Dc^2 / (4 \cdot tc (Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\Theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0.005 \cdot 425 / 2 \cdot 1) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 329.926 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_c, m | + | \sigma_c, b | \text{ (Should be } \leq \text{SPS, c)}$$

$$= 387.737 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, ks, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, kc, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E * (\lambda_s + \lambda_c + E * \ln(K)) \\
 &= (1 - 0.43) / 33773356 * (0 + 0 + 189341056 * \ln(1.06)) \\
 &= 0.1818
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 * [(p_s - 1) * (p_s^2 + 1) * P_s - (p_c - 1) * (p_c^2 + 1) * P_t] \\
 &= 2441^2 / 16 * [(1.02 - 1) * (1.02^2 + 1) * -100 - (1.02 - 1) * (1.02^2 + 1) * 4950] \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + \omega_c * P_t - \omega_s * P_s \\
 &= -61688608 + 0 * 4950 - 0 * -100 \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 * (3 + \nu) * (P_s - P_t) \\
 &= 92387032 + 2441^2 / 64 * (3 + 0.43) * (-100 - 4950) \\
 &= -1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |M_o| \\
 &= |-1520731776| \\
 &= 1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520731776 / (0.22 * (425 - 0)^2) \\
 &= 230.4476 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 * S \\
 &= 2 * 136 \\
 &= 272.40 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520722048 / (0.22 * (390 - 0)^2) \\
 &= 272.3968 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 390 + 0 + 0 \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(390; 141) \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |-100 - 4950| \\
 &= 5050.003 \text{ KPa}
 \end{aligned}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 28^2 \cdot [0 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0 / 2) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS}, s)$$

$$= 2.190 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0 \cdot 425 / 2 \cdot 1) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS}, c)$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 1.65 (25.4 - 1.65) 104 \\
 &= 1301.823 \text{ Kgf}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104 \\
 &= 1234.9329 \text{ Kgf}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 104/104 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104 \\
 &= 799.0742 \text{ Kgf}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right) \\
 &= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1) - 0.75 \cdot 25.4} \right) \\
 &= 2.141 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	230.4 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	33.7 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.2 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	390.216	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:
The Tubesheet stress reduction option was Simply Supported. Performing



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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1208 + 25.4 \\ &= 2440.600 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2482 / 2441 \\ &= 1.0168 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 399 \\ &= 0.125 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot \\ &\quad (104 / 136) \cdot (0.13); 25.4 - 2 \cdot 1.65) \\ &= 25.1015 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ 1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\ &= 32.1726 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.2 - 25.1) / 32.2 \\ &= 0.2198 \end{aligned}$$

$h/p = 12.566931$; $\mu^* = 0.219787$; $E^* = 33904192 \text{ KPa}$.

$E^*/E = 0.179064$; $v^* = 0.431382$

Note: As h/p (12.567) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2482 + 27.2) 27.2 } \\ &= 0.0070 \text{ 1/mm} \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.007 \cdot 194566320 \cdot 27.2^3 / (6(1 - 0.3^2)) \end{aligned}$$



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$$= 4990222848.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2482 + 27.2) \cdot 4.99E+09 / 399^3 \cdot (1 + 399 \cdot 0.007 + (399 \cdot 0.007)^2 / 2)$$

$$= 9024370.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2482^2 / (4 \cdot 194566320 \cdot 27.2) \cdot (1 - 2482 \cdot 0.3 / (2(2482 + 27.2)))$$

$$= 0.24774E+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 4.99E+09 \cdot 0.007 \cdot 0.24773699 (1 + 399 \cdot 0.007)$$

$$= 33037.6133 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 52) 52 }$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09 / 399^3 \cdot (1 + 399 \cdot 0.005 \cdot (1 + 1) / 2 + (399 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 29192318.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 399) - 0.000 (1 / 399 + 1 \cdot 0.005 / 2)]$$

$$= 49683.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584 / 2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (9024370 + 29192318 + 189341056 \cdot \ln(1.06))$$

$$= 0.8222$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 -$
 $(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$
 $= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$

Moment acting on unperforated tubesheet rim [M*]
 $= Mts + wc \cdot Pt - ws \cdot Ps$
 $= -61739052 + 49683 \cdot 4950 - 33038 \cdot -100$
 $= 187497808.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((187497808) - 2441^2/32 \cdot 0.82 (-100 - 4950)) / (1 + 0.82)$
 $= 527054432.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$
 $= 527054432 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$
 $= -1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 527054432 | ; | -1085721088 |)$
 $= 1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1085721088 / (0.22 (399 - 0)^2)$
 $= 186.1859 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S$
 $= 2 \cdot 136$
 $= 272.40 \text{ N./mm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1041836800 / (0.22 (322 - 0)^2)$
 $= 272.1585 \text{ N./mm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 322 + 13 + 13$
 $= 347.6495 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(348 ; 158)$
 $= 347.6495 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:
 $= | Ps - Pt |$
 $= | -100 - 4950 |$
 $= 5050.003 \text{ KPa}$

Shear Stress check [τ_{limit}]:



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$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205) \\
 &= 14.250 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 26 \\
 &= 157.6121 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\
 &= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100 \\
 &= -2.2563 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 1121872 / 27.2^2 \cdot [0.007 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot 2441 / 399^3 \cdot (1 + 399 \cdot 0.007 / 2) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= -161.308 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned}
 &= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS, s)} \\
 &= 163.564 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)
 \end{aligned}$$

Axial Channel Membrane Stress [σ_c, m]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt \\
 &= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950 \\
 &= 57.811 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σ_c, b]:

$$\begin{aligned}
 &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\Theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 + 0.005 \cdot 399 / 2 \cdot 1) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= 352.138 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 &= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS, c)} \\
 &= 409.949 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)
 \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$\beta_s, ks, \lambda_s, \delta_s$ are all = 0



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Intermediate Channel side parameters for the Simply Supported condition:
 β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (0 + 0 + 189341056 \cdot \ln(1.06))$$

$$= 0.1813$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2441^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -61739052 + 0 \cdot 4950 - 0 \cdot -100$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 92008104 + 2441^2 / 64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= | -1520767360 |$$

$$= 1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520767360 / (0.22 (399 - 0)^2)$$

$$= 260.7902 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520764928 / (0.22 (390 - 0)^2)$$

$$= 272.3629 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 390 + 13 + 13$$

$$= 416.2215 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(416; 167)$$

$$= 416.2215 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | -100 - 4950 |$$

$$= 5050.003 \text{ KPa.}$$

Shear Stress check [rlimit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 14.250 \text{ N./mm}^2$$

| Ps - Pt | is ≤ the Shear Stress limit (rlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 26$$

$$= 157.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100$$

$$= -2.2563 \text{ N./mm}^2$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot$$

$$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 27.2^2 \cdot [0 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot$$

$$2441 / 399^3 \cdot (1 + 399 \cdot 0 / 2) \cdot (92008104 + 2441^2 / 32 \cdot$$

$$(-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)}$$

$$= 2.256 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σc,m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 +$$

$$0 \cdot 399 / 2 \cdot 1) \cdot (92008104 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σc]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)}$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$



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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	260.8 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	35.9 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.3 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	416.221	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	390.216 ...	230	272 34 109	Ps+Pt D1	Ok



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D2uc	119.509	...	24	272	3	109	Ps+Pt D2	Ok
D3uc	364.932	...	202	272	30	109	Ps+Pt D3	Ok
D4uc	48.212	...	5	272	1	109	Ps+Pt D4	Ok
D1c	416.221	...	261	272	36	109	Ps+Pt-c D1	Ok
D2c	145.600	...	27	272	4	109	Ps+Pt-c D2	Ok
D3c	390.901	...	229	272	32	109	Ps+Pt-c D3	Ok
D4c	74.248	...	5	272	1	109	Ps+Pt-c D4	Ok

Max: 416.2215 ... mm 0.957 0.330 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.			Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	2	461	58	434	Ok
D2uc	...	...	11	461	...	434	Ok
D3uc	...	...	11	461	58	434	Ok
D4uc	...	...	2	461	...	434	Ok
D1c	...	...	2	461	58	434	Ok
D2c	...	...	12	461	...	434	Ok
D3c	...	...	12	461	58	434	Ok
D4c	...	...	2	461	...	434	Ok
Max	...	...	0.025		0.133		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	416.221	425.000	Ok
Tube thickness :	0.607	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 52.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5274.89	1.000	5267.42	1.000
Tubesheet Shear Stress	15309.77	1.000	15309.77	1.000
Shell Stress (Axial, Junction)	15113.33	1.000	3647.20	1.000



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ASME TS Calc: CASE-3 -Q-7

Case: 3 8:25a Apr 21, 2026

Tubesheet-Channel Junction	5289.86	1.000	11381.71	1.000
Tube Pressure Stress	14210.69	1.000	4065.19	1.000
Minimum MAWP	5274.89		3647.20	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6074.90	1.000	6074.90	1.000
Tubesheet Shear Stress	16522.81	1.000	16522.81	1.000
Shell Stress (Axial, Junction)	16522.81	1.000	4379.35	1.000
Tubesheet-Channel Junction	6696.49	1.000	15673.79	1.000
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	6074.90		4379.35	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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ASME TS Calc: CASE-4 -Q-7

Case: 4 8:25a Apr 21,2026

Input Echo, Tubesheet Number 4, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	216.99 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	52.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	245.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5080
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6075.00 mm
Design Temperature of the Tubes		245.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		103.70 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	114.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.600 mm
Largest Center-to-Center Tube Distance	Ul	76.2000 mm



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ASME TS Calc: CASE-4 -Q-7

Case: 4 8:25a Apr 21,2026

Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	44322.1 cm ²
Perimeter of Tube Layout	Cp	7804.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	245.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

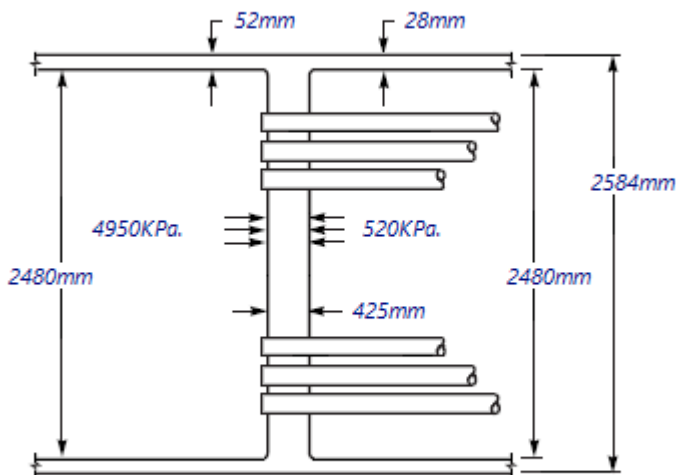
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	136.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	425.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2584.000 mm

Area of the Untubed Lanes	AL	1221.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 100.0002 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: CASE-4 -Q-7

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Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	245.0 °C	0.17950E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5050 \cdot (25.4/2 - 0)) / (104 \cdot 1 + 0.4 \cdot 5050)$
 $= 0.6067 + 0.0000 = 0.6067 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 245.00 °C
 Elastic Modulus for Material 175048832.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7282.60	15.39	50.0000	0.0046419	46.94

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.9) / (3 \cdot 15.4) = 4065 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4049	25.40	7282.60	62.73	50.0000	0.0002795	24.47

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.5) / (3 \cdot 62.7) = 520 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6067 mm
Allowable Internal Pressure at Corroded thickness	14210.70 KPa.
Required Internal Design Pressure	5050.00 KPa.
Allowable External Pressure at Corroded thickness	4065.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.4049 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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ASME TS Calc: CASE-4 -Q-7

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Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 1208 + 25.4 \\
 &= 2440.600 \text{ mm}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.200
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2480/2441 \\
 &= 1.0161
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 2480/2441 \\
 &= 1.0161
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 50/425 \\
 &= 0.118 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t_t(E_t T/E) \cdot (S_t T/S) \cdot (\rho); dt - 2t_t) \\
 &= \max(25.4 - 2 \cdot 1.65 (179502208/189341056) \cdot \\
 &\quad (104/136) \cdot (0.12); 25.4 - 2 \cdot 1.65) \\
 &= 25.1198 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\
 &= 31.8 / \sqrt{ (1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\
 &= 32.1726 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (32.2 - 25.1) / 32.2 \\
 &= 0.2192
 \end{aligned}$$

$h/p = 13.385828$; $\mu^* = 0.219219$; $E^* = 33773356 \text{ KPa}$.

$E^*/E = 0.178373$; $v^* = 0.432112$

Note: As h/p (13.386) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((2480 + 28) 28) } \\
 &= 0.0069 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.0069 \cdot 194566320 \cdot 28^3 / (6(1 - 0.3^2)) \\
 &= 5366140416.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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$$= 6(2480 + 28) \cdot 5.37E+09/425^3 \cdot (1 + 425 \cdot 0.0069 + (425 \cdot 0.0069)^2/2)$$

$$= 8589042.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2480^2 / (4 \cdot 194566320 \cdot 28) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 28)))$$

$$= 0.24036E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 5.37E+09 \cdot 0.0069 \cdot 0.2403622 (1 + 425 \cdot 0.0069)$$

$$= 35204.6680 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2480 + 52)52)$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09/425^3 \cdot (1 + 425 \cdot 0.005 \cdot (1 + 1)/2 + (425 \cdot 0.005)^2/2 \cdot 1)$$

$$= 26085156.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 425) - 0.000 (1/425 + 1 \cdot 0.005/2)]$$

$$= 51841.3750 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584/2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33773356 \cdot (8589042 + 26085156 + 189341056 \cdot \ln(1.06))$$

$$= 0.7648$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$



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ASME TS Calc: CASE-4 -Q-7

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$$= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61688608.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -61688608 + 51841 \cdot 4950 - 35205 \cdot -100$$

$$= 198446832.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M_{ts} - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= (198446832 - 2441^2/32 \cdot 0.76(-100 - 4950)) / (1 + 0.76)$$

$$= 519817472.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 519817472 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|519817472|; |-1093301376|)$$

$$= 1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1093301376 / (0.22(425 - 0)^2)$$

$$= 165.6760 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1039441152 / (0.22(321 - 0)^2)$$

$$= 272.1014 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 321 + 0 + 0$$

$$= 321.3042 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(321; 132)$$

$$= 321.3042 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$= |P_s - P_t|$$

$$= |-100 - 4950|$$

$$= 5050.003 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$



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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 1206383 / 28^2 \cdot [0.0069 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0.0069 / 2) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -142.815 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_{s, m} | + | \sigma_{s, b} | \text{ (Should be } \leq \text{SPS, s)}$$

$$= 145.005 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + Do^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0.005 \cdot 425 / 2 \cdot 1) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 329.926 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c, m} | + | \sigma_{c, b} | \text{ (Should be } \leq \text{SPS, c)}$$

$$= 387.737 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, k_s, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, k_c, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E * (\lambda_s + \lambda_c + E * \ln(K)) \\
 &= (1 - 0.43) / 33773356 * (0 + 0 + 189341056 * \ln(1.06)) \\
 &= 0.1818
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 * [(p_s - 1) * (p_s^2 + 1) * P_s - (p_c - 1) * (p_c^2 + 1) * P_t] \\
 &= 2441^2 / 16 * [(1.02 - 1) * (1.02^2 + 1) * -100 - (1.02 - 1) * (1.02^2 + 1) * 4950] \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + \omega_c * P_t - \omega_s * P_s \\
 &= -61688608 + 0 * 4950 - 0 * -100 \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 * (3 + \nu) * (P_s - P_t) \\
 &= 92387032 + 2441^2 / 64 * (3 + 0.43) * (-100 - 4950) \\
 &= -1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |M_o| \\
 &= |-1520731776| \\
 &= 1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520731776 / (0.22 * (425 - 0)^2) \\
 &= 230.4476 \text{ N/mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 * S \\
 &= 2 * 136 \\
 &= 272.40 \text{ N/mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520722048 / (0.22 * (390 - 0)^2) \\
 &= 272.3968 \text{ N/mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 390 + 0 + 0 \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}, H_{reqS}) \\
 &= \max(390; 141) \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |-100 - 4950| \\
 &= 5050.003 \text{ KPa}
 \end{aligned}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 28^2 \cdot [0 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0 / 2) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS}_s)$$

$$= 2.190 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0 \cdot 425 / 2 \cdot 1) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS}_c)$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1) - 0.75 \cdot 25.4} \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	230.4 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	33.7 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.2 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	390.216	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:
The Tubesheet stress reduction option was Simply Supported. Performing



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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1208 + 25.4 \\ &= 2440.600 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2482 / 2441 \\ &= 1.0168 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 399 \\ &= 0.125 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot \\ &\quad (104 / 136) \cdot (0.13); 25.4 - 2 \cdot 1.65) \\ &= 25.1015 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ 1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ 1 - 4 \cdot \min(1221 ; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\ &= 32.1726 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.2 - 25.1) / 32.2 \\ &= 0.2198 \end{aligned}$$

$h/p = 12.566931$; $\mu^* = 0.219787$; $E^* = 33904192 \text{ KPa}$.

$E^*/E = 0.179064$; $v^* = 0.431382$

Note: As h/p (12.567) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2482 + 27.2) 27.2 } \\ &= 0.0070 \text{ 1/mm} \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.007 \cdot 194566320 \cdot 27.2^3 / (6(1 - 0.3^2)) \end{aligned}$$



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$$= 4990222848.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2482 + 27.2) \cdot 4.99E+09 / 399^3 \cdot (1 + 399 \cdot 0.007 + (399 \cdot 0.007)^2 / 2)$$

$$= 9024370.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2482^2 / (4 \cdot 194566320 \cdot 27.2) \cdot (1 - 2482 \cdot 0.3 / (2(2482 + 27.2)))$$

$$= 0.24774E+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 4.99E+09 \cdot 0.007 \cdot 0.24773699 (1 + 399 \cdot 0.007)$$

$$= 33037.6133 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 52) 52 }$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09 / 399^3 \cdot (1 + 399 \cdot 0.005 \cdot (1 + 1) / 2 + (399 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 29192318.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 399) - 0.000 (1 / 399 + 1 \cdot 0.005 / 2)]$$

$$= 49683.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584 / 2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (9024370 + 29192318 + 189341056 \cdot \ln(1.06))$$

$$= 0.8222$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 -$
 $(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$
 $= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$

Moment acting on unperforated tubesheet rim [M*]
 $= Mts + wc \cdot Pt - ws \cdot Ps$
 $= -61739052 + 49683 \cdot 4950 - 33038 \cdot -100$
 $= 187497808.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((187497808) - 2441^2/32 \cdot 0.82 (-100 - 4950)) / (1 + 0.82)$
 $= 527054432.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$
 $= 527054432 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$
 $= -1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 527054432 | ; | -1085721088 |)$
 $= 1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1085721088 / (0.22 (399 - 0)^2)$
 $= 186.1859 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S$
 $= 2 \cdot 136$
 $= 272.40 \text{ N./mm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1041836800 / (0.22 (322 - 0)^2)$
 $= 272.1585 \text{ N./mm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 322 + 13 + 13$
 $= 347.6495 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(348 ; 158)$
 $= 347.6495 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:
 $= | Ps - Pt |$
 $= | -100 - 4950 |$
 $= 5050.003 \text{ KPa}$

Shear Stress check [τ_{limit}]:



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$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205) \\
 &= 14.250 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 26 \\
 &= 157.6121 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\
 &= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100 \\
 &= -2.2563 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 1121872 / 27.2^2 \cdot [0.007 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot 2441 / 399^3 \cdot (1 + 399 \cdot 0.007 / 2) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= -161.308 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned}
 &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)} \\
 &= 163.564 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)
 \end{aligned}$$

Axial Channel Membrane Stress [σ_c, m]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt \\
 &= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950 \\
 &= 57.811 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σ_c, b]:

$$\begin{aligned}
 &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 + 0.005 \cdot 399 / 2 \cdot 1) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= 352.138 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)} \\
 &= 409.949 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)
 \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$\beta_s, ks, \lambda_s, \delta_s$ are all = 0



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Intermediate Channel side parameters for the Simply Supported condition:
 β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (0 + 0 + 189341056 \cdot \ln(1.06))$$

$$= 0.1813$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 2441^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -61739052 + 0 \cdot 4950 - 0 \cdot -100$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 92008104 + 2441^2 / 64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= | -1520767360 |$$

$$= 1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520767360 / (0.22 (399 - 0)^2)$$

$$= 260.7902 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520764928 / (0.22 (390 - 0)^2)$$

$$= 272.3629 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 390 + 13 + 13$$

$$= 416.2215 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(416; 167)$$

$$= 416.2215 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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ASME TS Calc: CASE-4 -Q-7

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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | -100 - 4950 |$$

$$= 5050.003 \text{ KPa.}$$

Shear Stress check [rlimit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 14.250 \text{ N./mm}^2$$

| Ps - Pt | is ≤ the Shear Stress limit (rlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 26$$

$$= 157.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100$$

$$= -2.2563 \text{ N./mm}^2$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot$$

$$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 27.2^2 \cdot [0 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot$$

$$2441 / 399^3 \cdot (1 + 399 \cdot 0 / 2) \cdot (92008104 + 2441^2 / 32 \cdot$$

$$(-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)}$$

$$= 2.256 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σc,m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 +$$

$$0 \cdot 399 / 2 \cdot 1) \cdot (92008104 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σc]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)}$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$



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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	260.8 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	35.9 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.3 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	416.221	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	390.216 ...	230	272	34 109	Ps+Pt D1	Ok



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ASME TS Calc: CASE-4 -Q-7

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D2uc	119.509	...	24	272	3	109	Ps+Pt D2	Ok
D3uc	364.932	...	202	272	30	109	Ps+Pt D3	Ok
D4uc	48.212	...	5	272	1	109	Ps+Pt D4	Ok
D1c	416.221	...	261	272	36	109	Ps+Pt-c D1	Ok
D2c	145.600	...	27	272	4	109	Ps+Pt-c D2	Ok
D3c	390.901	...	229	272	32	109	Ps+Pt-c D3	Ok
D4c	74.248	...	5	272	1	109	Ps+Pt-c D4	Ok

Max: 416.2215 ... mm 0.957 0.330 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.			Shell Stress		Channel Stress		Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	2	461	58	434	Ok
D2uc	...	...	11	461	...	434	Ok
D3uc	...	...	11	461	58	434	Ok
D4uc	...	...	2	461	...	434	Ok
D1c	...	...	2	461	58	434	Ok
D2c	...	...	12	461	...	434	Ok
D3c	...	...	12	461	58	434	Ok
D4c	...	...	2	461	...	434	Ok
Max	...	...		0.025		0.133	

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	416.221	425.000	Ok
Tube thickness :	0.607	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 52.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5274.89	1.000	5267.42	1.000
Tubesheet Shear Stress	15309.77	1.000	15309.77	1.000
Shell Stress (Axial, Junction)	15113.33	1.000	3647.20	1.000



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ASME TS Calc: CASE-4 -Q-7

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Tubesheet-Channel Junction	5289.86	1.000	11381.71	1.000
Tube Pressure Stress	14210.69	1.000	4065.19	1.000
Minimum MAWP	5274.89		3647.20	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6074.90	1.000	6074.90	1.000
Tubesheet Shear Stress	16522.81	1.000	16522.81	1.000
Shell Stress (Axial, Junction)	16522.81	1.000	4379.35	1.000
Tubesheet-Channel Junction	6696.49	1.000	15673.79	1.000
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	6074.90		4379.35	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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ASME TS Calc: CASE 5 -Q-7

Case: 5 8:25a Apr 21,2026

Input Echo, Tubesheet Number 5, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	235.99 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	52.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	245.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5080
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6075.00 mm
Design Temperature of the Tubes		245.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		103.70 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	114.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.600 mm
Largest Center-to-Center Tube Distance	Ul	76.2000 mm



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ASME TS Calc: CASE 5 -Q-7

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Length of Expanded Portion of Tube	ltx	50.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	44322.1 cm ²
Perimeter of Tube Layout	Cp	7804.6 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	245.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

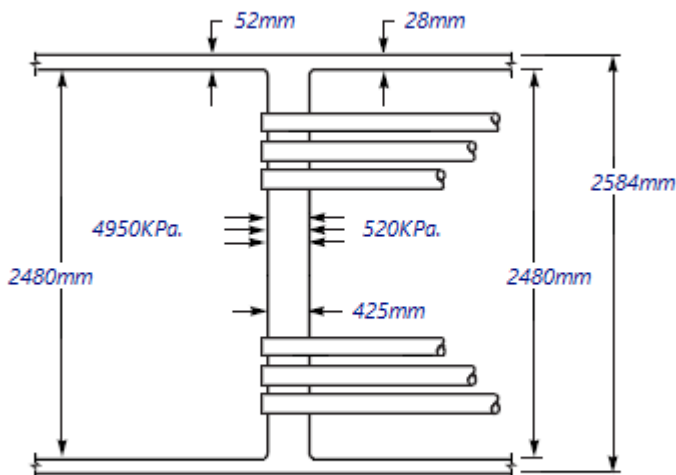
Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	136.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	425.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2584.000 mm

Area of the Untubed Lanes	AL	1221.0 cm ²
---------------------------	----	------------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Shell Side Min. Design Pressure	Psd,min 100.0002 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: CASE 5 -Q-7

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Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	245.0 °C	0.17950E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	245.0 °C	0.18934E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (5050 \cdot (25.4/2 - 0)) / (104 \cdot 1 + 0.4 \cdot 5050)$$

$$= 0.6067 + 0.0000 = 0.6067 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-3	at	245.00 °C
Elastic Modulus for Material			175048832.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	25.40	7282.60	15.39	50.0000	0.0046419	46.94

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.9) / (3 \cdot 15.4) = 4065 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4049	25.40	7282.60	62.73	50.0000	0.0002795	24.47

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 24.5) / (3 \cdot 62.7) = 520 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6067 mm
Allowable Internal Pressure at Corroded thickness	14210.70 KPa.
Required Internal Design Pressure	5050.00 KPa.
Allowable External Pressure at Corroded thickness	4065.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.4049 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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ASME TS Calc: CASE 5 -Q-7

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Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 1208 + 25.4 \\
 &= 2440.600 \text{ mm}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.200
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 2480 / 2441 \\
 &= 1.0161
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 2480 / 2441 \\
 &= 1.0161
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 50 / 425 \\
 &= 0.118 \text{ (must be } 0 \leq \rho \leq 1)
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\
 &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot (104 / 136) \cdot (0.12); 25.4 - 2 \cdot 1.65) \\
 &= 25.1198 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2) } \\
 &= 31.8 / \sqrt{ 1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2) } \\
 &= 32.1726 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (32.2 - 25.1) / 32.2 \\
 &= 0.2192
 \end{aligned}$$

$h/p = 13.385828$; $\mu^* = 0.219219$; $E^* = 33773356 \text{ KPa}$.

$E^*/E = 0.178373$; $\nu^* = 0.432112$

Note: As h/p (13.386) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 28) 28 } \\
 &= 0.0069 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\
 &= 0.0069 \cdot 194566320 \cdot 28^3 / (6(1 - 0.3^2)) \\
 &= 5366140416.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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ASME TS Calc: CASE 5 -Q-7

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$$= 6(2480 + 28) \cdot 5.37E+09/425^3 \cdot (1 + 425 \cdot 0.0069 + (425 \cdot 0.0069)^2/2)$$

$$= 8589042.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2480^2 / (4 \cdot 194566320 \cdot 28) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 28)))$$

$$= 0.24036E+00 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 5.37E+09 \cdot 0.0069 \cdot 0.2403622 (1 + 425 \cdot 0.0069)$$

$$= 35204.6680 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2480 + 52)52)$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43E+09/425^3 \cdot (1 + 425 \cdot 0.005 \cdot (1 + 1)/2 + (425 \cdot 0.005)^2/2 \cdot 1)$$

$$= 26085156.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43E+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 425) - 0.000 (1/425 + 1 \cdot 0.005/2)]$$

$$= 51841.3750 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584/2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33773356 \cdot (8589042 + 26085156 + 189341056 \cdot \ln(1.06))$$

$$= 0.7648$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$



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$$= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61688608.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps$$

$$= -61688608 + 51841 \cdot 4950 - 35205 \cdot -100$$

$$= 198446832.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((198446832) - 2441^2/32 \cdot 0.76 (-100 - 4950)) / (1 + 0.76)$$

$$= 519817472.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu*) \cdot (Ps - Pt)$$

$$= 519817472 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| 519817472 | ; | -1093301376 |)$$

$$= 1093301376.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 1093301376 / (0.22 (425 - 0)^2)$$

$$= 165.6760 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu* (h - hg')^2)$$

$$= 6 \cdot 1039441152 / (0.22 (321 - 0)^2)$$

$$= 272.1014 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 321 + 0 + 0$$

$$= 321.3042 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(321 ; 132)$$

$$= 321.3042 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$= | Ps - Pt |$$

$$= | -100 - 4950 |$$

$$= 5050.003 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.2 \cdot 425/2441 \cdot \min(0.8 \cdot 136 ; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$



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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 1206383 / 28^2 \cdot [0.0069 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0.0069 / 2) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -142.815 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_s, m | + | \sigma_s, b | \text{ (Should be } \leq \text{SPS, s)}$$

$$= 145.005 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= Dc^2 / (4 \cdot tc (Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\Theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0.005 \cdot 425 / 2 \cdot 1) \cdot (519817472 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 329.926 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_c, m | + | \sigma_c, b | \text{ (Should be } \leq \text{SPS, c)}$$

$$= 387.737 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, ks, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, kc, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu) / E * (\lambda_s + \lambda_c + E * \ln(K)) \\
 &= (1 - 0.43) / 33773356 * (0 + 0 + 189341056 * \ln(1.06)) \\
 &= 0.1818
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= D_o^2 / 16 * [(p_s - 1) * (p_s^2 + 1) * P_s - (p_c - 1) * (p_c^2 + 1) * P_t] \\
 &= 2441^2 / 16 * [(1.02 - 1) * (1.02^2 + 1) * -100 - (1.02 - 1) * (1.02^2 + 1) * 4950] \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= M_{ts} + \omega_c * P_t - \omega_s * P_s \\
 &= -61688608 + 0 * 4950 - 0 * -100 \\
 &= -61688608.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= M_p + D_o^2 / 64 * (3 + \nu) * (P_s - P_t) \\
 &= 92387032 + 2441^2 / 64 * (3 + 0.43) * (-100 - 4950) \\
 &= -1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |M_o| \\
 &= |-1520731776| \\
 &= 1520731776.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520731776 / (0.22 * (425 - 0)^2) \\
 &= 230.4476 \text{ N/mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 * S \\
 &= 2 * 136 \\
 &= 272.40 \text{ N/mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 * M / (\mu * (h - h_g')^2) \\
 &= 6 * 1520722048 / (0.22 * (390 - 0)^2) \\
 &= 272.3968 \text{ N/mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 390 + 0 + 0 \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(390; 141) \\
 &= 390.2159 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |-100 - 4950| \\
 &= 5050.003 \text{ KPa}
 \end{aligned}$$



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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 425 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 15.179 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 0$$

$$= 131.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2480^2 / (4 \cdot 28 (2480 + 28)) \cdot -100$$

$$= -2.1897 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 28^2 \cdot [0 \cdot -0.024 + 6(1 - 0.43) / 33773356 \cdot 2441 / 425^3 \cdot (1 + 425 \cdot 0 / 2) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq \text{SPS}_s)$$

$$= 2.190 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot \cos(\theta) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33773356) \cdot 2441 / 425^3 \cdot (1 + 0 \cdot 425 / 2 \cdot 1) \cdot (92387032 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq \text{SPS}_c)$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1) - 0.75 \cdot 25.4} \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm
Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	230.4 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	33.7 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.2 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	390.216	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:
The Tubesheet stress reduction option was Simply Supported. Performing



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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 1208 + 25.4 \\ &= 2440.600 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.8 - 25.4) / 31.8 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 2482 / 2441 \\ &= 1.0168 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 2480 / 2441 \\ &= 1.0161 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$\begin{aligned} &= l_{tx} / h \\ &= 50 / 399 \\ &= 0.125 \text{ (must be } 0 \leq p \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (p); dt - 2tt) \\ &= \max(25.4 - 2 \cdot 1.65 (179502208 / 189341056) \cdot \\ &\quad (104 / 136) \cdot (0.13); 25.4 - 2 \cdot 1.65) \\ &= 25.1015 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ 1 - 4 \cdot \min(A1; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) } \\ &= 31.8 / \sqrt{ 1 - 4 \cdot \min(1221; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) } \\ &= 32.1726 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (32.2 - 25.1) / 32.2 \\ &= 0.2198 \end{aligned}$$

$h/p = 12.566931$; $\mu^* = 0.219787$; $E^* = 33904192 \text{ KPa}$.

$E^*/E = 0.179064$; $v^* = 0.431382$

Note: As h/p (12.567) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2482 + 27.2) 27.2 } \\ &= 0.0070 \text{ 1/mm} \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned} &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\ &= 0.007 \cdot 194566320 \cdot 27.2^3 / (6(1 - 0.3^2)) \end{aligned}$$



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$$= 4990222848.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2482 + 27.2) \cdot 4.99\text{E}+09 / 399^3 \cdot (1 + 399 \cdot 0.007 + (399 \cdot 0.007)^2 / 2)$$

$$= 9024370.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2482^2 / (4 \cdot 194566320 \cdot 27.2) \cdot (1 - 2482 \cdot 0.3 / (2(2482 + 27.2)))$$

$$= 0.24774\text{E}+00 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 4.99\text{E}+09 \cdot 0.007 \cdot 0.24773699 (1 + 399 \cdot 0.007)$$

$$= 33037.6133 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (2480 + 52) 52 }$$

$$= 0.0050 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.005 \cdot 189341056 \cdot 52^3 / (6(1 - 0.3^2))$$

$$= 24427798528.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2480 + 52 / \cos(0)) \cdot 24.43\text{E}+09 / 399^3 \cdot (1 + 399 \cdot 0.005 \cdot (1 + 1) / 2 + (399 \cdot 0.005)^2 / 2 \cdot 1)$$

$$= 29192318.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2480^2 / (4 \cdot 189341056 \cdot 52) \cdot (1 - 2480 \cdot 0.3 / (2(2480 + 52)))$$

$$= 0.13322\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.02 \cdot 24.43\text{E}+09 [0.13321713 (1 + 0.005 \cdot 1 \cdot 399) - 0.000 (1 / 399 + 1 \cdot 0.005 / 2)]$$

$$= 49683.4219 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2584 / 2441$$

$$= 1.0588$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (9024370 + 29192318 + 189341056 \cdot \ln(1.06))$$

$$= 0.8222$$



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ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 $= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 $= 2441^2/16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 -$
 $(1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$
 $= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$

Moment acting on unperforated tubesheet rim [M*]
 $= Mts + wc \cdot Pt - ws \cdot Ps$
 $= -61739052 + 49683 \cdot 4950 - 33038 \cdot -100$
 $= 187497808.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:
 $= ((M*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$
 $= ((187497808) - 2441^2/32 \cdot 0.82 (-100 - 4950)) / (1 + 0.82)$
 $= 527054432.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on center of tubesheet [Mo]:
 $= Mp + Do^2/64 \cdot (3 + v*) \cdot (Ps - Pt)$
 $= 527054432 + 2441^2/64 \cdot (3 + 0.43) \cdot (-100 - 4950)$
 $= -1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

Maximum bending moment acting on tubesheet [M]:
 $= \max(| Mp | ; | Mo |)$
 $= \max(| 527054432 | ; | -1085721088 |)$
 $= 1085721088.0000 \text{ KPa} \cdot \text{mm}^2$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1085721088 / (0.22 (399 - 0)^2)$
 $= 186.1859 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ_{All}]:
 $= 2 \cdot S$
 $= 2 \cdot 136$
 $= 272.40 \text{ N./mm}^2$

Tubesheet bending stress at final thickness [σ]:
 $= 6 \cdot M / (\mu* (h - hg')^2)$
 $= 6 \cdot 1041836800 / (0.22 (322 - 0)^2)$
 $= 272.1585 \text{ N./mm}^2$

Required tubesheet thickness for Bending Stress [HreqB]:
 $= h + Cats + Catc$
 $= 322 + 13 + 13$
 $= 347.6495 \text{ mm}$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:
 $= \max(HreqB; HreqS)$
 $= \max(348 ; 158)$
 $= 347.6495 \text{ mm}$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:
 $= | Ps - Pt |$
 $= | -100 - 4950 |$
 $= 5050.003 \text{ KPa}$

Shear Stress check [τ_{limit}]:



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$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy) \\
 &= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205) \\
 &= 14.250 \text{ N./mm}^2
 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot |Ps - Pt| + Cats + Catc \\
 &= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot |-100 - 4950| + 26 \\
 &= 157.6121 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps \\
 &= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100 \\
 &= -2.2563 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned}
 &= 6 \cdot ks / ts^2 \cdot [\beta S \cdot Ps + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 1121872 / 27.2^2 \cdot [0.007 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot 2441 / 399^3 \cdot (1 + 399 \cdot 0.007 / 2) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= -161.308 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned}
 &= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)} \\
 &= 163.564 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)
 \end{aligned}$$

Axial Channel Membrane Stress [σ_c, m]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt \\
 &= 2480^2 / (4 \cdot 52 (2480 + 52)) \cdot \cos(0) \cdot 4950 \\
 &= 57.811 \text{ N./mm}^2
 \end{aligned}$$

Axial Channel Bending Stress [σ_c, b]:

$$\begin{aligned}
 &= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))] \\
 &= 6 \cdot 5491713 / 52^2 \cdot [(0.13321713 \cdot 1 \cdot 0.005 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 + 0.005 \cdot 399 / 2 \cdot 1) \cdot (527054432 + 2441^2 / 32 \cdot (-100 - 4950))] \\
 &= 352.138 \text{ N./mm}^2
 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned}
 &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)} \\
 &= 409.949 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)
 \end{aligned}$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$\beta_s, ks, \lambda_s, \delta_s$ are all = 0



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Intermediate Channel side parameters for the Simply Supported condition:
 β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.43) / 33904192 \cdot (0 + 0 + 189341056 \cdot \ln(1.06))$$

$$= 0.1813$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 2441^2 / 16 \cdot [(1.02 - 1) \cdot (1.02^2 + 1) \cdot -100 - (1.02 - 1) \cdot (1.02^2 + 1) \cdot 4950]$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -61739052 + 0 \cdot 4950 - 0 \cdot -100$$

$$= -61739052.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 92008104 + 2441^2 / 64 \cdot (3 + 0.43) \cdot (-100 - 4950)$$

$$= -1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= |-1520767360|$$

$$= 1520767360.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520767360 / (0.22 (399 - 0)^2)$$

$$= 260.7902 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 136$$

$$= 272.40 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 1520764928 / (0.22 (390 - 0)^2)$$

$$= 272.3629 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$= h + C_{ats} + C_{atc}$$

$$= 390 + 13 + 13$$

$$= 416.2215 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(416; 167)$$

$$= 416.2215 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:



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Absolute pressure differential [| Ps - Pt |]:

$$= | P_s - P_t |$$

$$= | -100 - 4950 |$$

$$= 5050.003 \text{ KPa.}$$

Shear Stress check [rlimit]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 399 / 2441 \cdot \min(0.8 \cdot 136; 0.533 \cdot 205)$$

$$= 14.250 \text{ N./mm}^2$$

| Ps - Pt | is ≤ the Shear Stress limit (rlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 109) \cdot (4 \cdot 44322 / 7805) \cdot | -100 - 4950 | + 26$$

$$= 157.6121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σs,m]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 2482^2 / (4 \cdot 27.2 (2482 + 27.2)) \cdot -100$$

$$= -2.2563 \text{ N./mm}^2$$

Axial Shell Bending Stress [σs,b]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P'_s + 6(1 - \nu^*) / E^* \cdot$$

$$D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 27.2^2 \cdot [0 \cdot -0.025 + 6(1 - 0.43) / 33904192 \cdot$$

$$2441 / 399^3 \cdot (1 + 399 \cdot 0 / 2) \cdot (92008104 + 2441^2 / 32 \cdot$$

$$(-100 - 4950))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σs]:

$$= | \sigma_{s,m} | + | \sigma_{s,b} | \text{ (Should be } \leq \text{SPS,s)}$$

$$= 2.256 \text{ N./mm}^2 \text{ (Should be } \leq 460.82)$$

Axial Channel Membrane Stress [σc,m]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 2480^2 / (4 \cdot 52 (2480 + 52) \cdot \cos(0)) \cdot 4950$$

$$= 57.811 \text{ N./mm}^2$$

Axial Channel Bending Stress [σc,b]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t -$$

$$6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta +$$

$$h \cdot \beta C / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 0 / 52^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 4950 -$$

$$6((1 - 0.43) / 33904192) \cdot 2441 / 399^3 \cdot (1 +$$

$$0 \cdot 399 / 2 \cdot 1) \cdot (92008104 + 2441^2 / 32 \cdot (-100 - 4950))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σc]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS,c)}$$

$$= 57.811 \text{ N./mm}^2 \text{ (Should be } \leq 434.02)$$

Analysis Finished.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.231 \text{ cm}^2$$



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Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 104$$

$$= 1301.823 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 1234.9329 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 104/104$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 104$$

$$= 799.0742 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.141 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0705 mm

Minimum Required Groove Weld Leg [agr]: 1.0705 mm

Tube-Tubesheet Jt allowable, 1302 is $\geq$ tube strength 1302 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	260.8 $\leq$	272.4 N./mm ²	Ok
Tubesheet shear stress	35.9 $\leq$	109.0 N./mm ²	Ok
Stress in Shell at Tubesheet	2.3 $\leq$	460.8 N./mm ²	Ok
Stress in Channel at Tubesheet	57.8 $\leq$	434.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	416.221	425.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.070	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.070	1.650	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc 390.216	...	230	272 34 109	Ps+Pt D1	Ok



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D2uc	119.509	...	24	272	3	109	Ps+Pt D2	Ok
D3uc	364.932	...	202	272	30	109	Ps+Pt D3	Ok
D4uc	48.212	...	5	272	1	109	Ps+Pt D4	Ok
D1c	416.221	...	261	272	36	109	Ps+Pt-c D1	Ok
D2c	145.600	...	27	272	4	109	Ps+Pt-c D2	Ok
D3c	390.901	...	229	272	32	109	Ps+Pt-c D3	Ok
D4c	74.248	...	5	272	1	109	Ps+Pt-c D4	Ok

Max: 416.2215 ... mm 0.957 0.330 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.			Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	2	461	58	434	Ok
D2uc	...	...	11	461	...	434	Ok
D3uc	...	...	11	461	58	434	Ok
D4uc	...	...	2	461	...	434	Ok
D1c	...	...	2	461	58	434	Ok
D2c	...	...	12	461	...	434	Ok
D3c	...	...	12	461	58	434	Ok
D4c	...	...	2	461	...	434	Ok
Max	...	...	0.025		0.133		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	416.221	425.000	Ok
Tube thickness :	0.607	1.650	Ok
Tube-tubesheet fillet weld leg :	1.070	1.650	Ok
Tube-tubesheet groove weld leg :	1.070	1.650	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 52.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	5274.89	1.000	5267.42	1.000
Tubesheet Shear Stress	15309.77	1.000	15309.77	1.000
Shell Stress (Axial, Junction)	15113.33	1.000	3647.20	1.000



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Tubesheet-Channel Junction	5289.86	1.000	11381.71	1.000
Tube Pressure Stress	14210.69	1.000	4065.19	1.000
Minimum MAWP	5274.89		3647.20	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6074.90	1.000	6074.90	1.000
Tubesheet Shear Stress	16522.81	1.000	16522.81	1.000
Shell Stress (Axial, Junction)	16522.81	1.000	4379.35	1.000
Tubesheet-Channel Junction	6696.49	1.000	15673.79	1.000
Tube Pressure Stress	15758.76	1.000	7090.00	1.000
Minimum MAPnc	6074.90		4379.35	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(425/4, \min(425, 28)) \\
 &= 106.250 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(520/4379, 1) \\
 &= 0.119
 \end{aligned}$$

Thickness Ratio = 0.12, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	33 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(425/4, \min(425, 52)) \\
 &= 106.250 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(4950/6075, 1) \\
 &= 0.815
 \end{aligned}$$

Thickness Ratio = 0.81, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	33 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	6 °C

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